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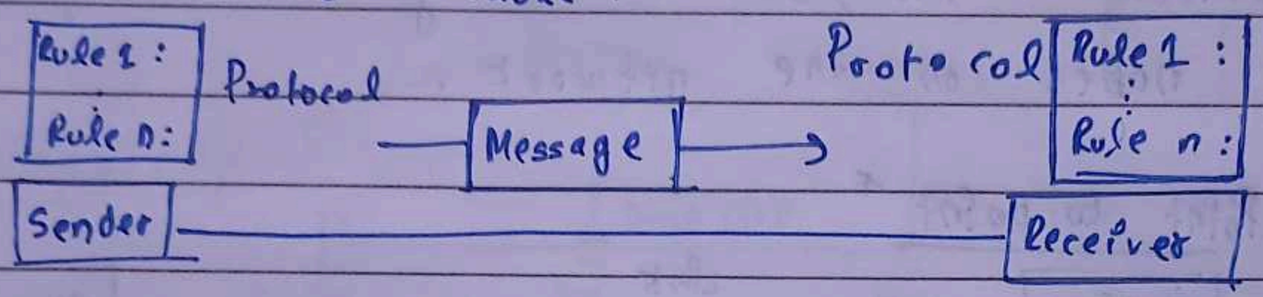
9056299166
5 AM to 10 PM helpline.

Data Communication. - term telecommunication means communication at a distance. The word data refers to information presented in whatever form is agreed upon by the parties creating and using the data. Data communications are the exchange of data b/w 2 devices via some form of transmission medium such as a wire cable.



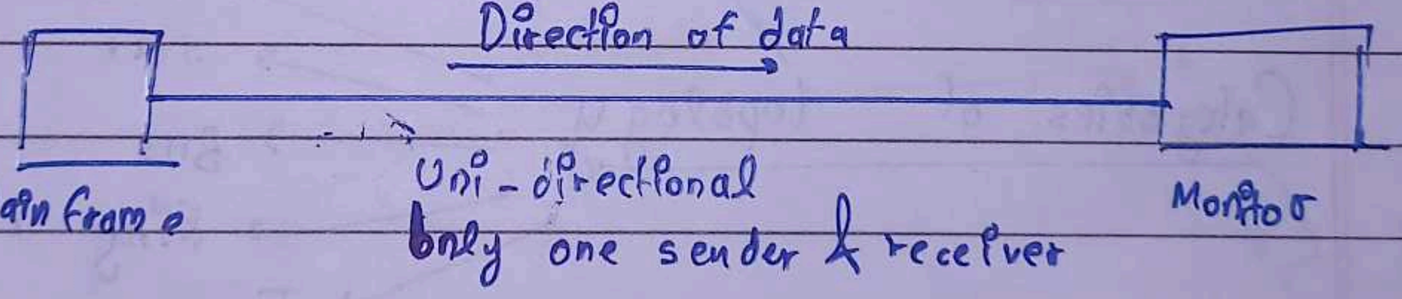
Message, Sender, Receiver, Transmission medium, Protocol.

Protocol : Set of rules that govern data communication.

Data flow

Simplex

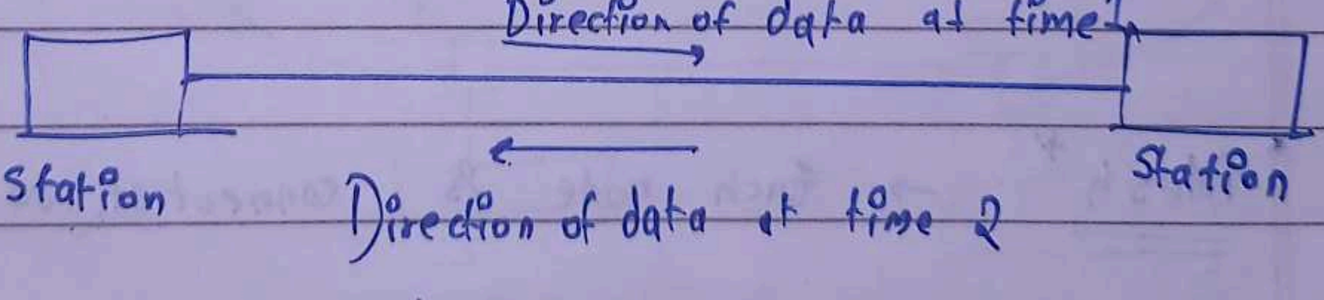
eg. Radio, TV



Uni-directional
only one sender & receiver

Half-duplex

eg. walkie-talkie

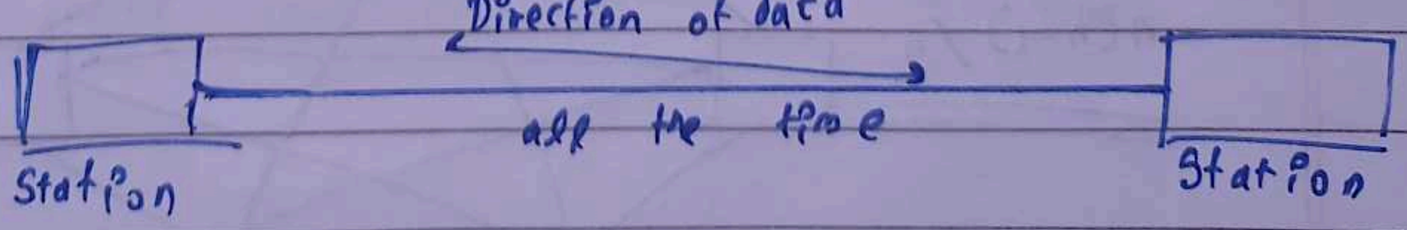


Direction of data at time 2

bidirectional, only one direction at a time

Full - duplex

eg. telephone, video conferencing.

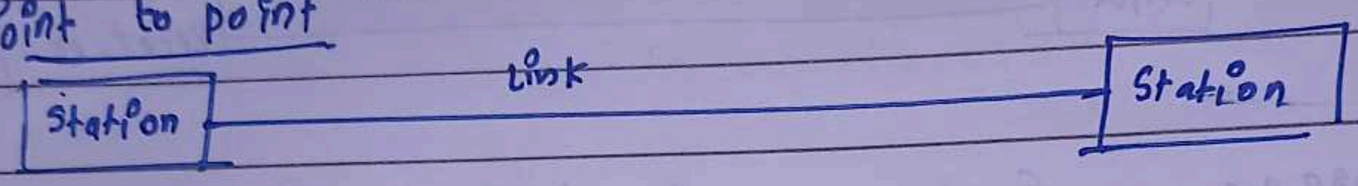


bidirectional, but both direction simultaneously.

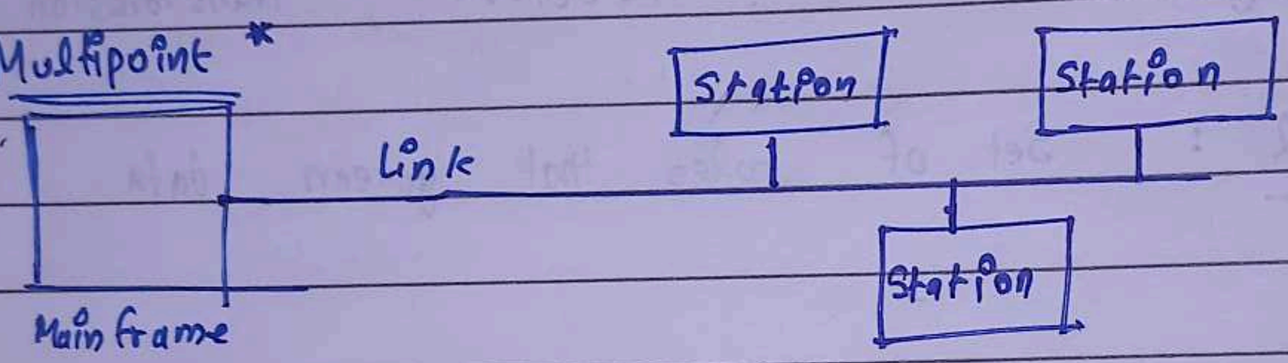
Networks.

A network is a set of devices (often referred to as nodes) connected by communication links. A node can be a computer, printer, or any other devices capable of sending and/or receiving data generated by other nodes on the network.

* Point to point *



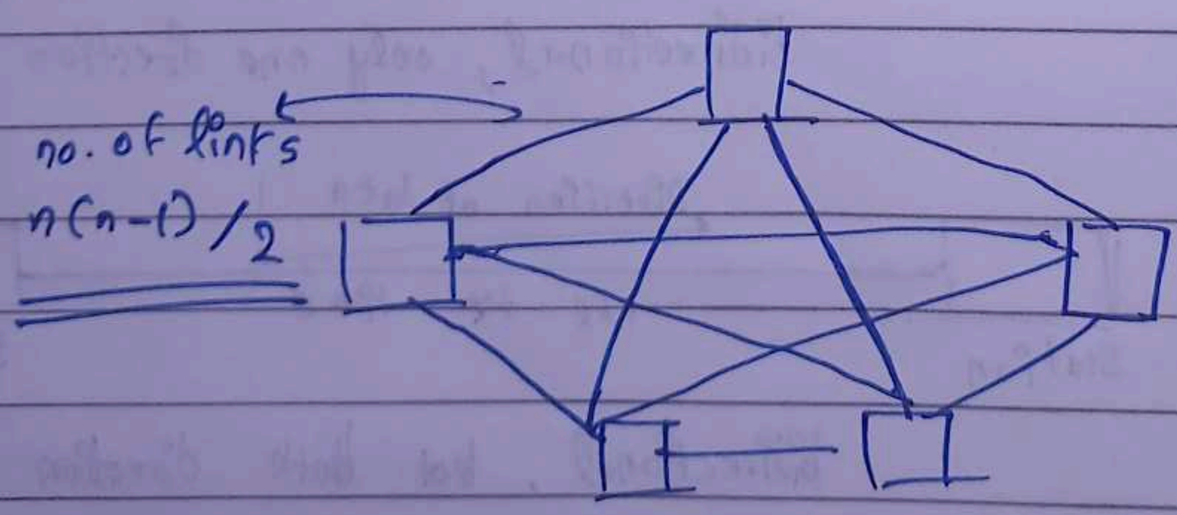
* Multipoint *



Categories of topology

- Mesh
- Star
- Bus
- Ring
- Tree

* Mesh * → Each node is connected to every other node.



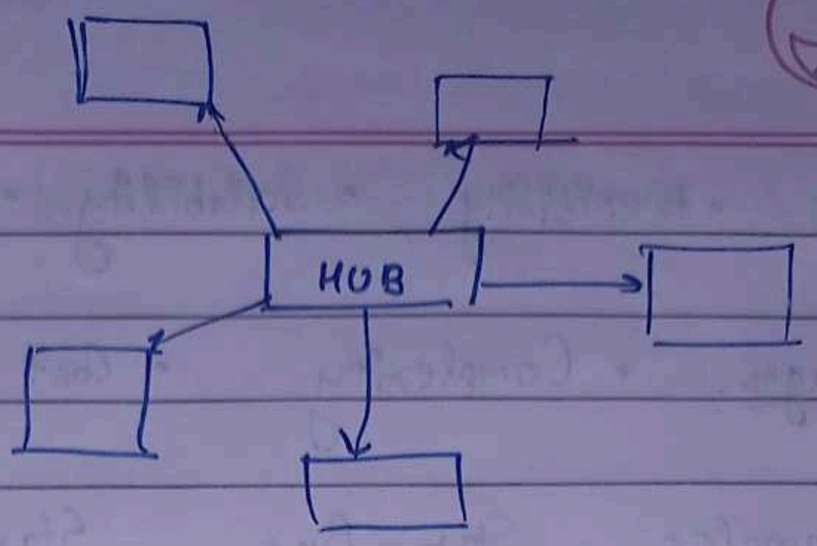
Advantages.

- Fault tolerant
- Scalable
- Self-configuring

Disadvantages

- Complex
- Expensive

Star topology.



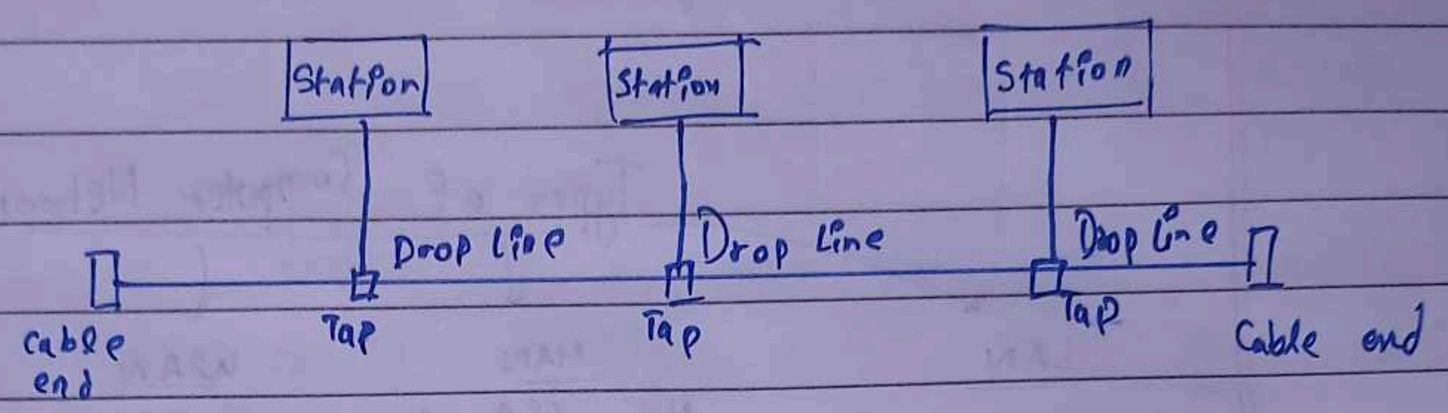
Disadvantages

- Central point failure
- Performance.

Advantages.

- Easy to install and manage
- Scalable
- Fault-tolerant

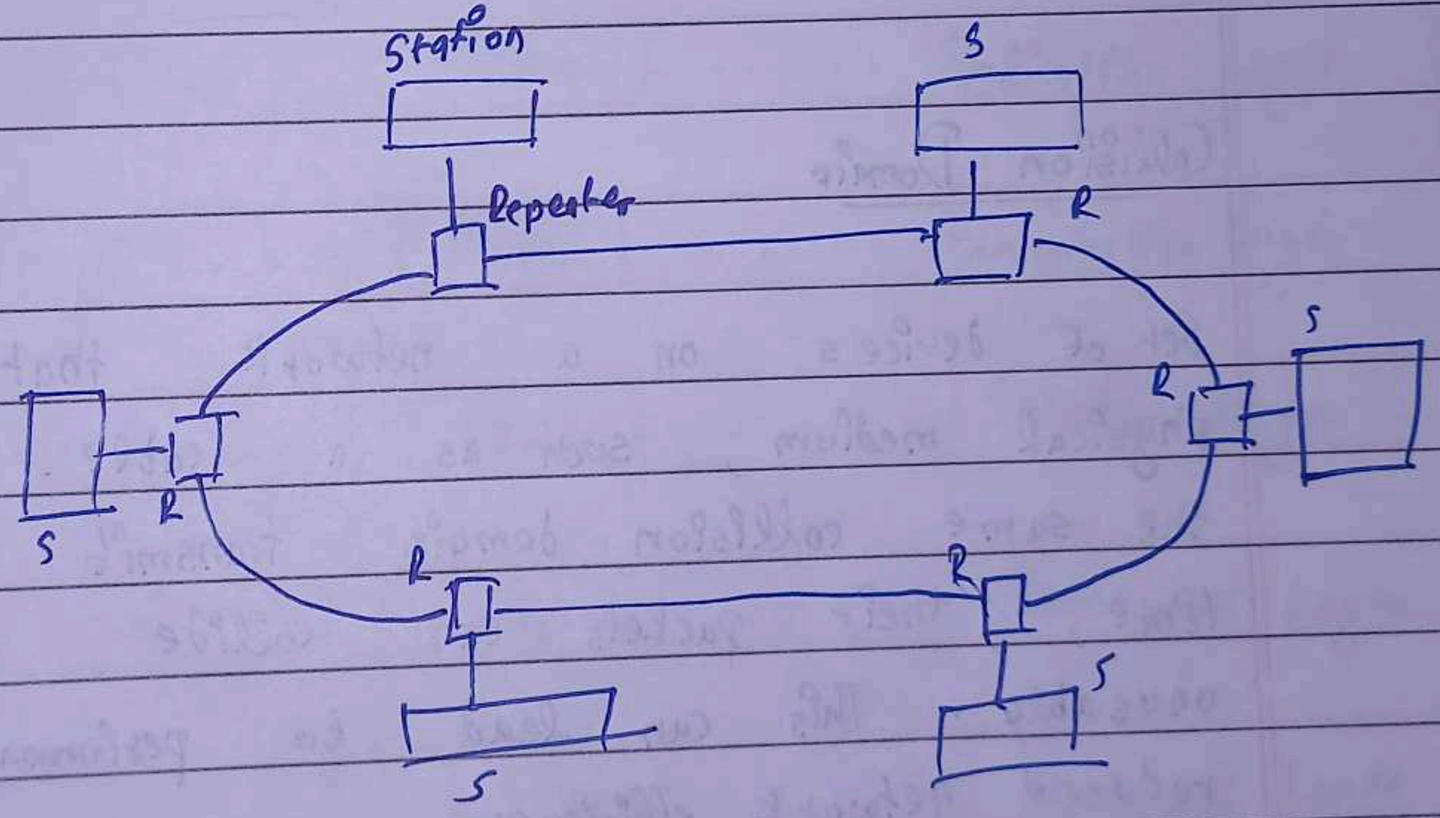
Bus Topology.



Advantages. • Easy to Install • Cost-Effective • Scalable

Disadvantages. • Performance • Fault tolerant • Collisions

Ring Topology.

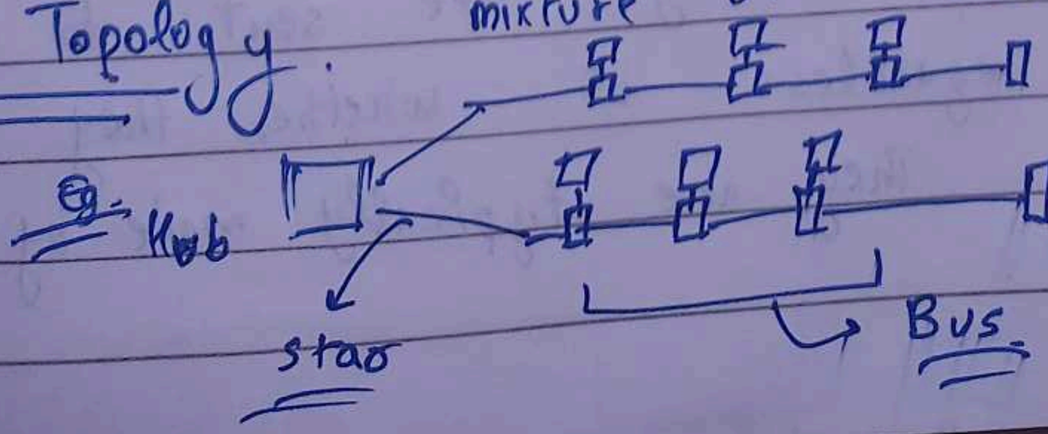


Advantages

- Fault-tolerant
- Performance
- Scalability

Disadvantages • Single point of failure • Performance • Complexity

Hybrid Topology.



Advantages • Flexibility • Scalability • Performance • Reliability

Disadvantages. • Complexity • Cost

Common Examples Star - Bus, Star - Ring, Mesh - Star

Types of Computer Networks.

LAN
Local Area Network
Small area
Buildings
and ale

MAN
Metropolitan Area Network
b/w cities

WAN
Wide Area Network
b/w countries

PAN
Personal Area Network
within 10 m. "our" network

Collision Domain

Set of devices on a network that share the same physical medium, such as a cable. When 2 devices on the same collision domain transmit data at the same time, their packets can collide, rendering them both unusable. This can lead to performance problems & reduced network efficiency.

Broadcast Domain

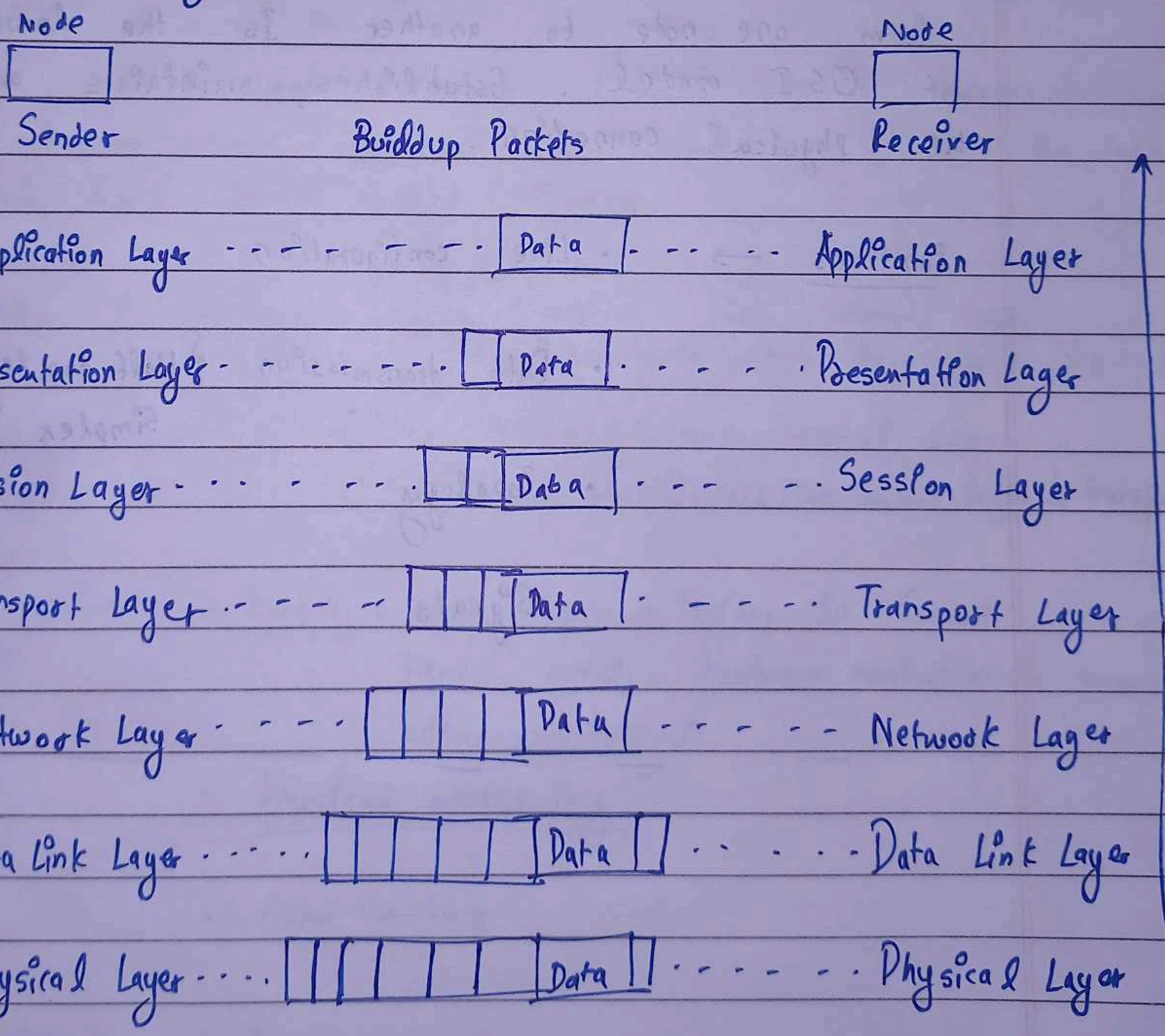
Set of devices on a network that can receive broadcast messages. Broadcast message are sent to all devices on a network, regardless of whether they are intended for a specific device. They are typically made by hubs & switches No collisions here.

Devices that can break collision domain are Bridge, switch & routers.

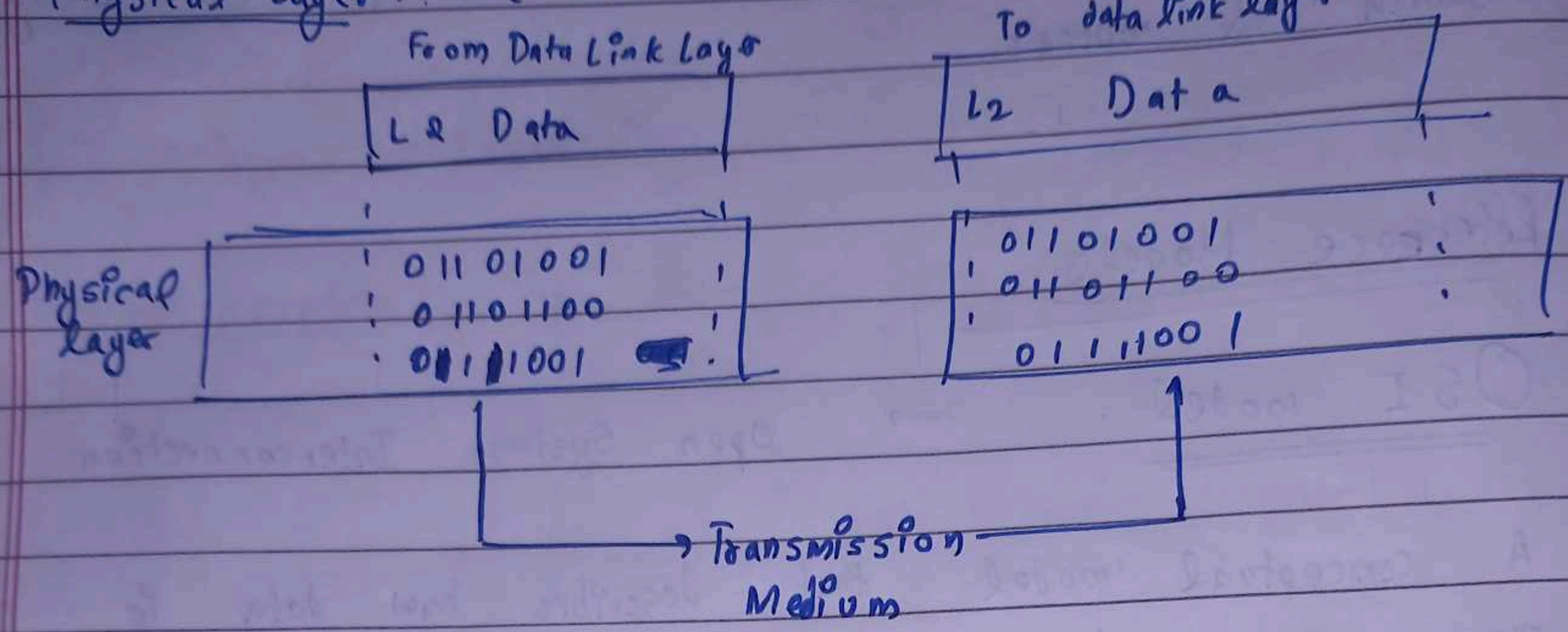
Reference Models.

OSI model. → Open Systems Interconnection model.

A conceptual model that describes how data is communicated over a network. It is a 7 layer model. First introduced in 1970s. Made by ISO.



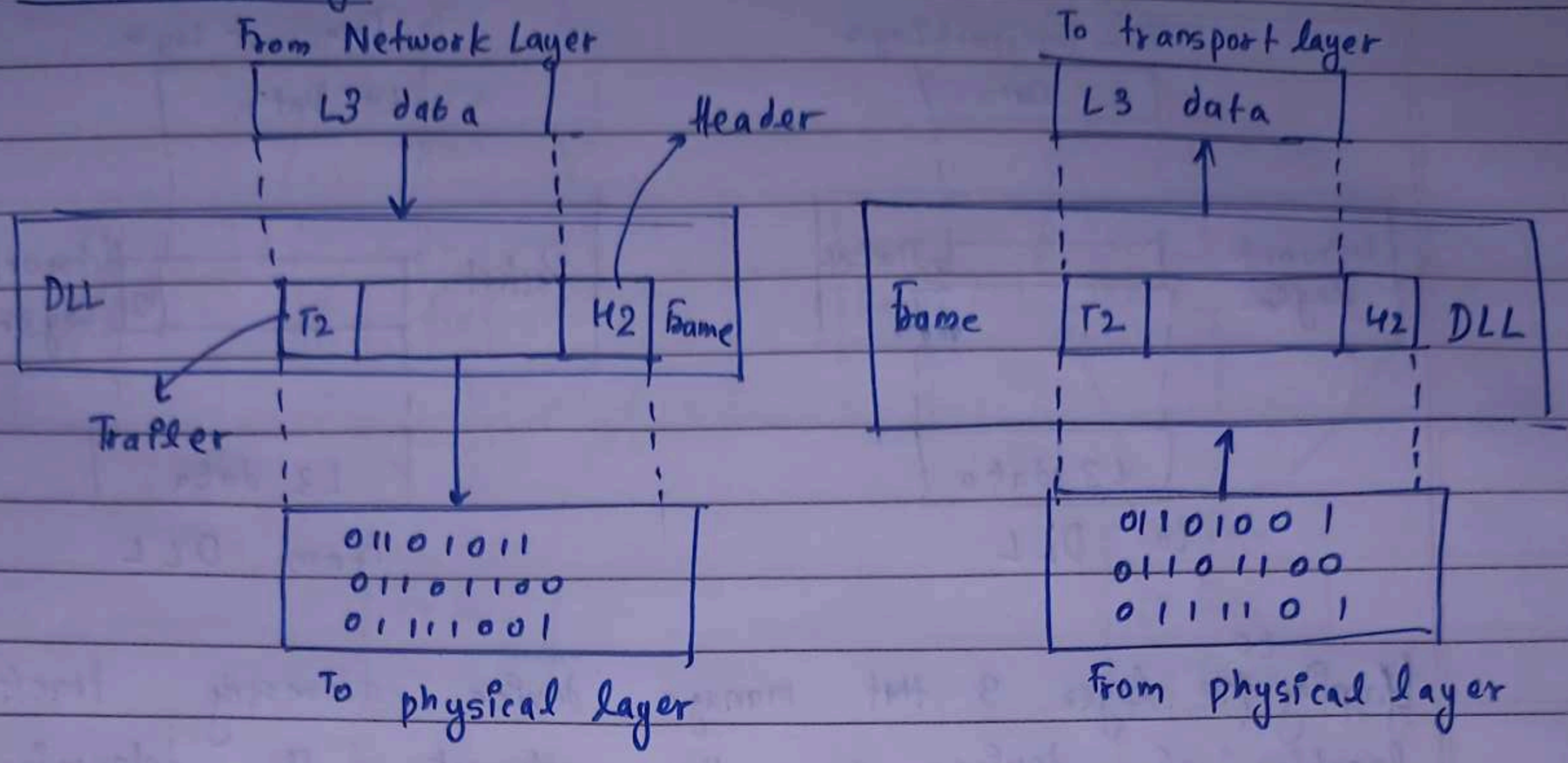
Physical Layer → {bits; hop to hop delivery}



Institutes the ~~transformatio~~ transmission of individual bits from one node to another. Is the lowest layer of OSI model. Establishes, maintains and deactivates the physical connection.

Functions →
- Line configuration
- Data transmission {Half duplex, Full duplex, simplex}
- Topology
- Signals

Data Link Layer → { frames, hop to hop }

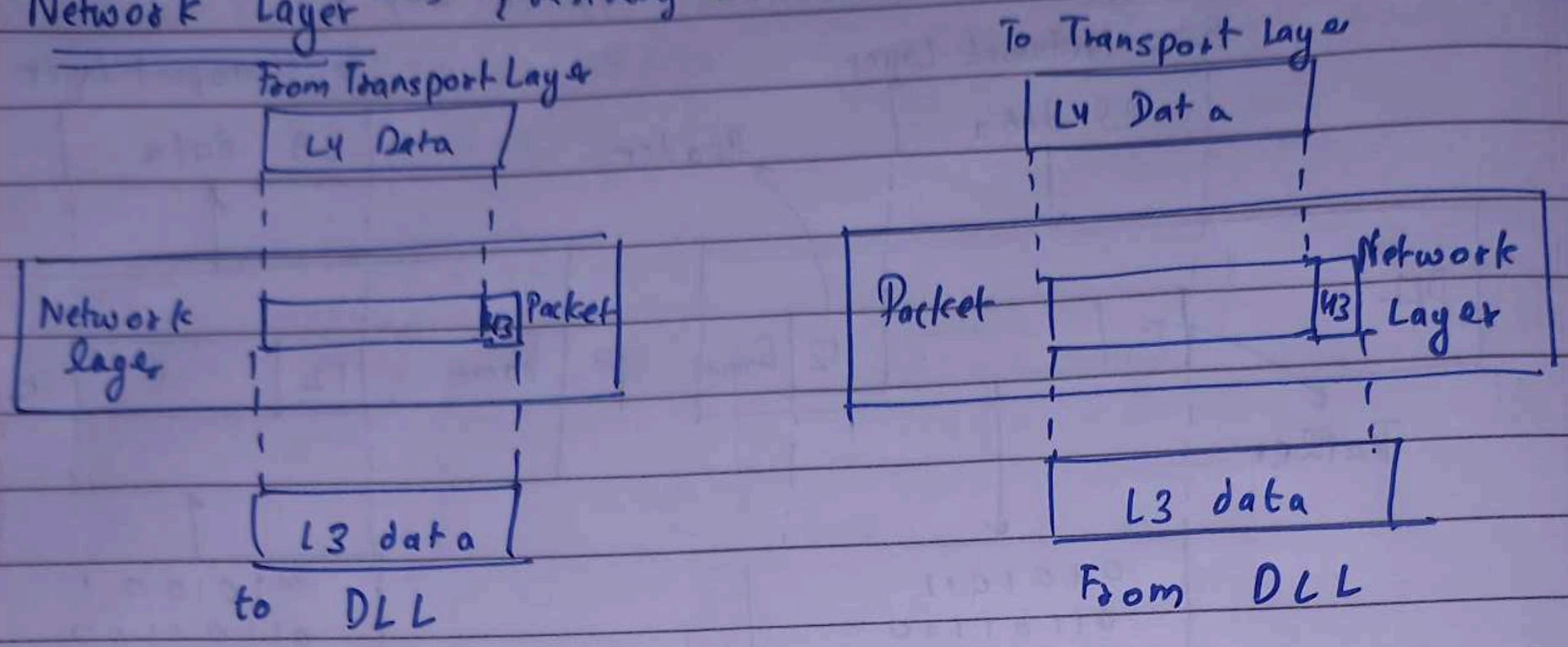


Secure, reliable and efficient { Error free } communication b/w 2 or more devices is done by DLL. The data in this layer is called **Frames**.

Contains 2 sub-layers →
- Logical Link Control layer { transferring packets to network layer }
- Media Access Control Layer { link b/w Logical Link Control layer & Transport }

Functions →
- **Framing**: adding Header & Trailer to frame. Header contains hardware destination & source address.
- **Physical addressing**
- Flow Control
- Error Control
- Access Control.

Network Layer → {delivery of individual packets from host to destination}



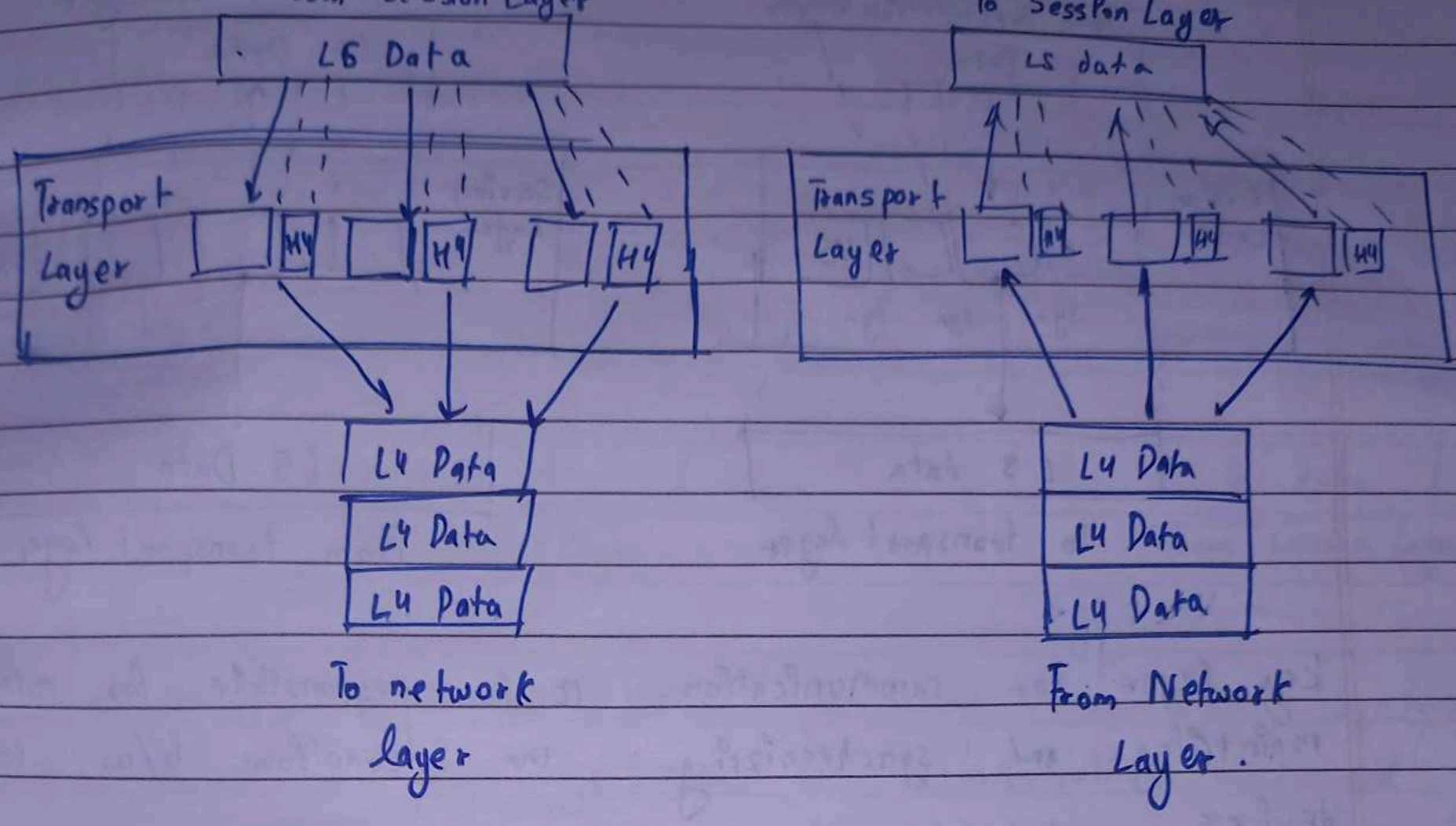
Mapping to the layer 3 that manage device addressing, tracks the location of devices on the network. It determines the optimal Route for transferring data & for forwarding packets. Routers work in this layer. Network layer protocols also work: here (IPv6 ..).

Functions

- Internetworking
- Addressing
- Routing
- Packetizing

↑ delivery of message from one process to another

Transport layer { Heart of OSI }

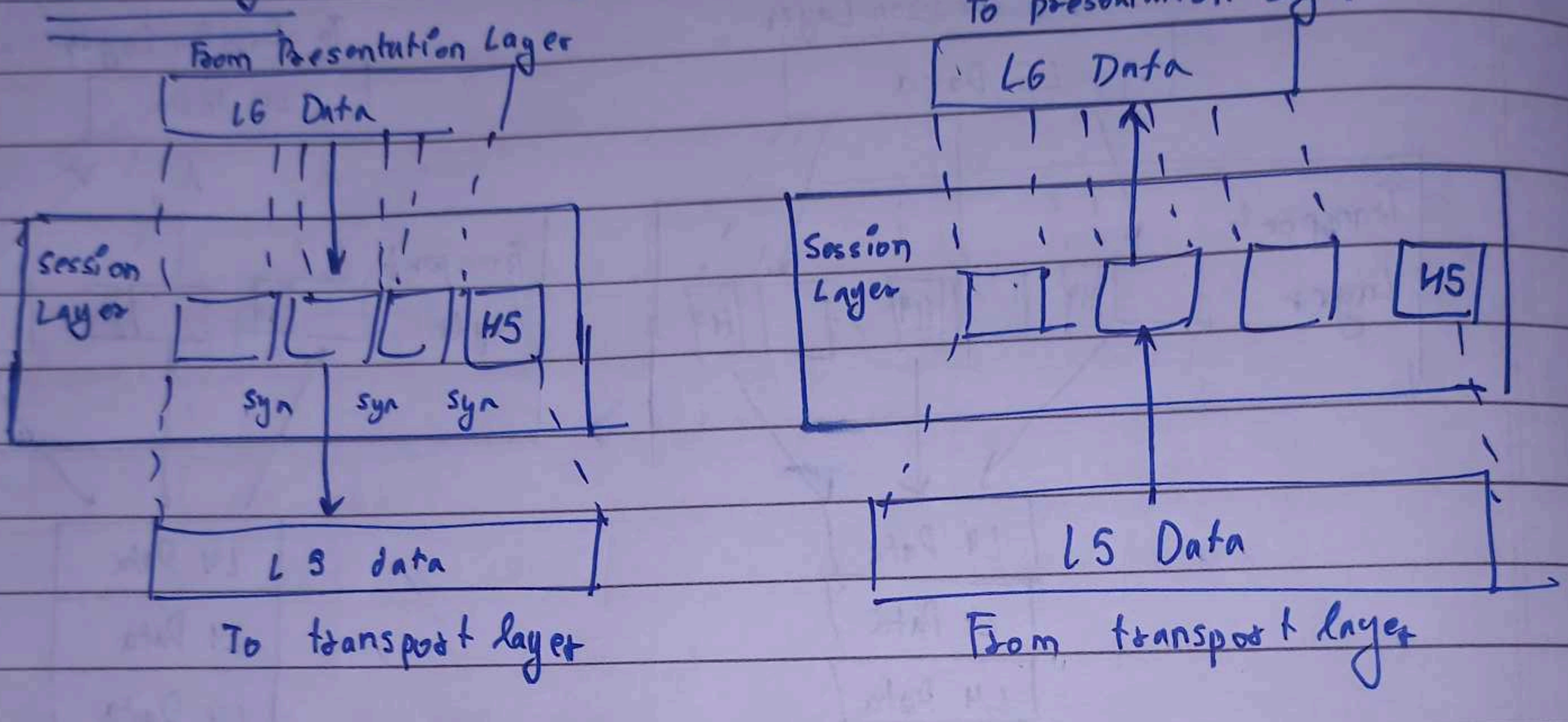


It receives data from upper layers & converts them into smaller units known as segments. It ensures that messages are transmitted in order in which they are sent & no data duplication occurs. It provides end to end / point to point connection b/w source and destination. It has TCP & UDP.

Functions →

- Service Point Addressing
- Segmentation and Reassembly.
- Connection control.
- Flow control.
- Error control.

Session Layer → { dialog control & Synchronization }

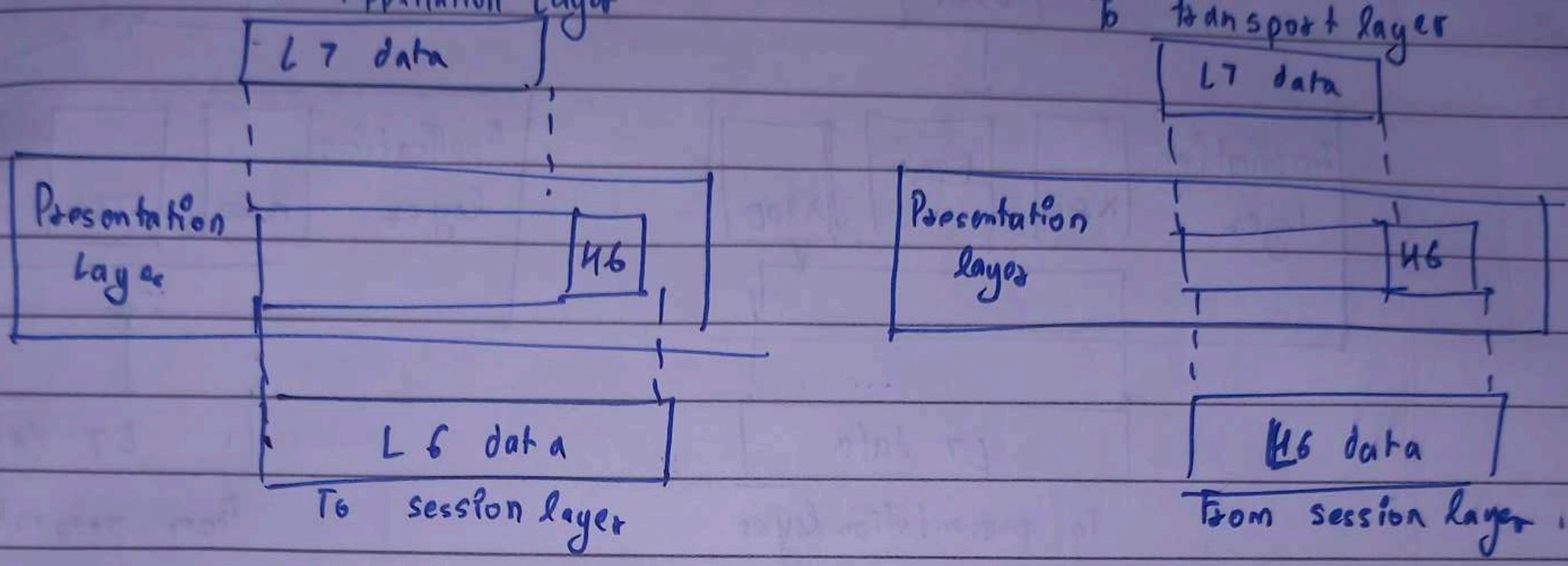


key layer for communication. It is responsible for establishing maintaing and synchronizing the interaction b/w communicating devices.

Function → • Dialog control

• Synchronization

Presentation layer → { compression, translation, encryption }

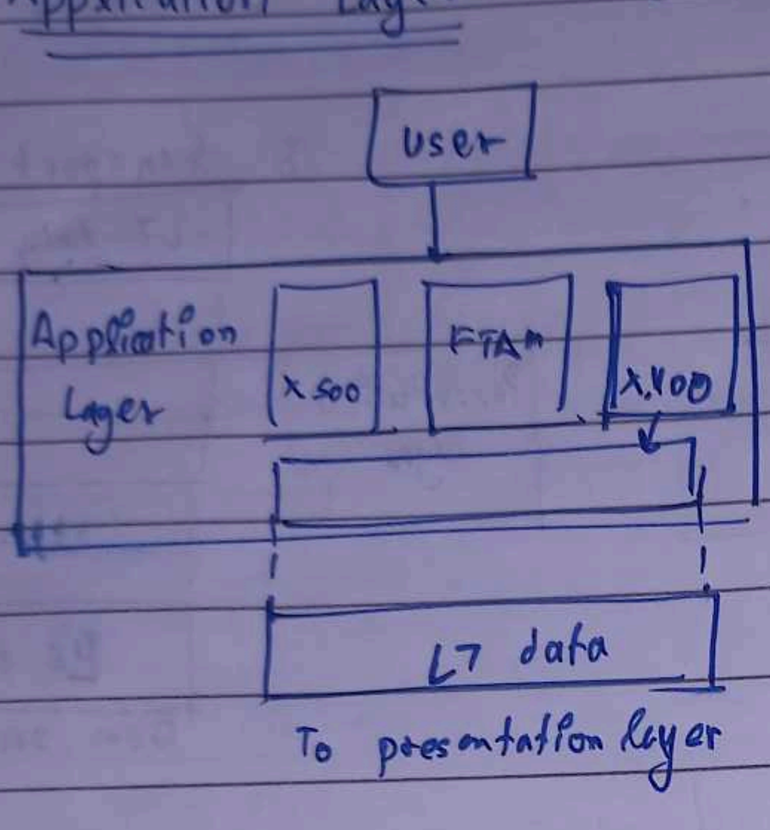


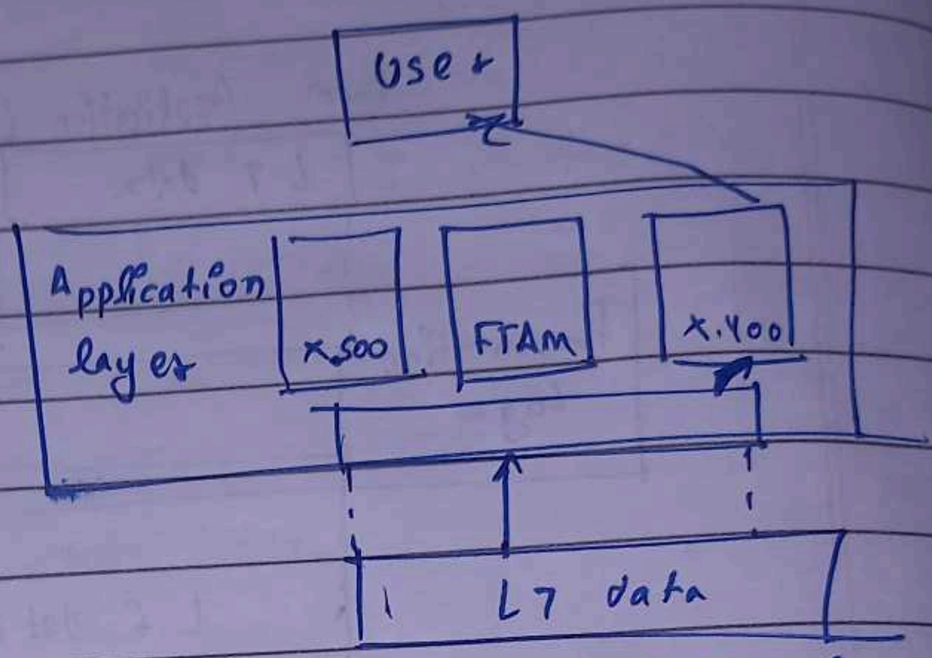
A presentation layer is mainly concerned with the syntax and ~~semantics~~ *Semantics* of the information exchanged b/w the 2 systems. Acts as a data translator for a network. Also known as *Syntax layer*. Converts from one presentation format to another.

Functions ⟶

- Translation
- Encryption & Decryption
- Compression.

Application Layer → { responsible for providing service to user }





Honestly, I am very tired now lol.

An application layer serves as a window for users an application processes to access network service. Handles issues such as network transparency, resource allocation, etc.

Functions → File transfer, access and management* (FTAM)

→ Mail service

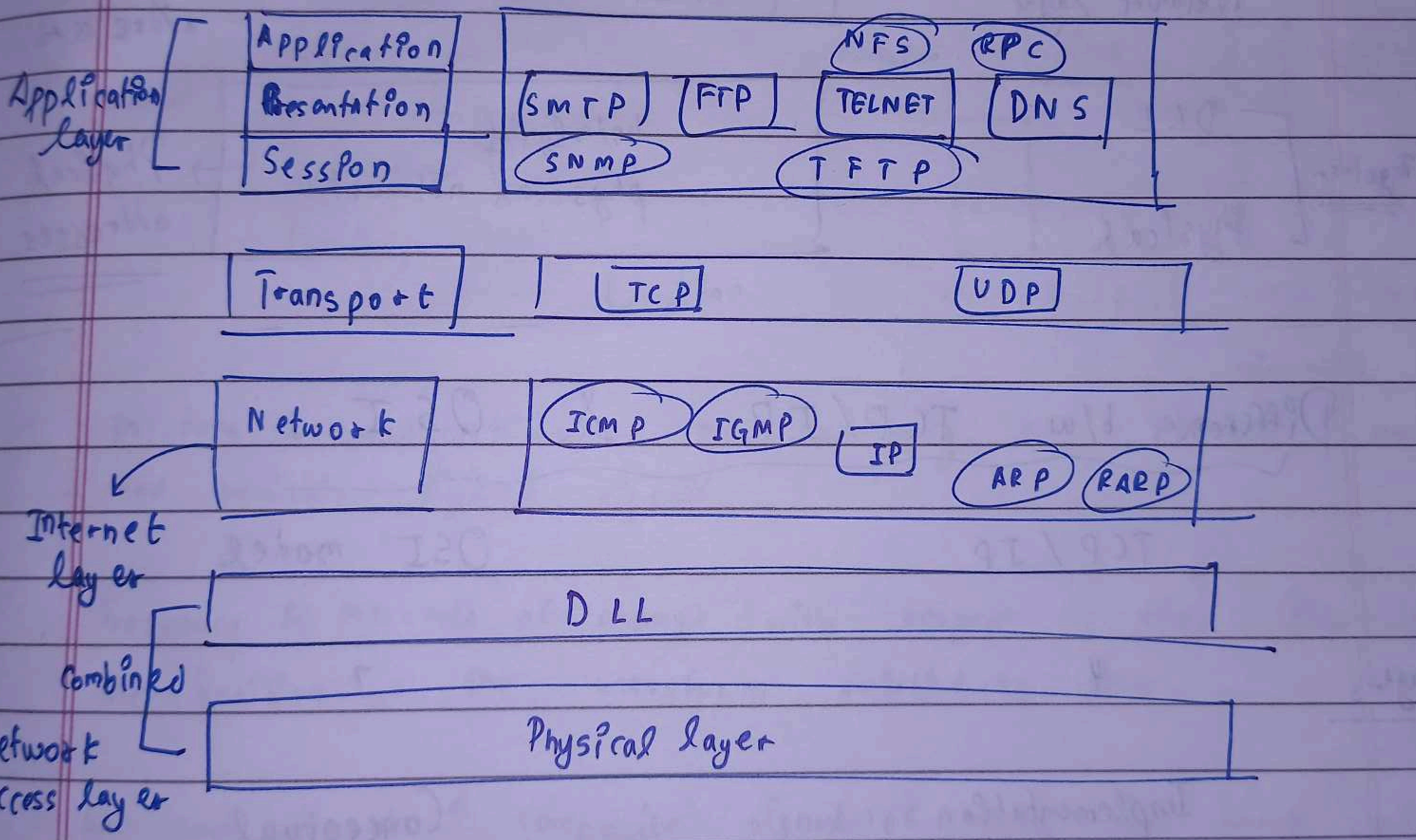
→ Directory services

header → payload

PDU → Protocol Data Unit.

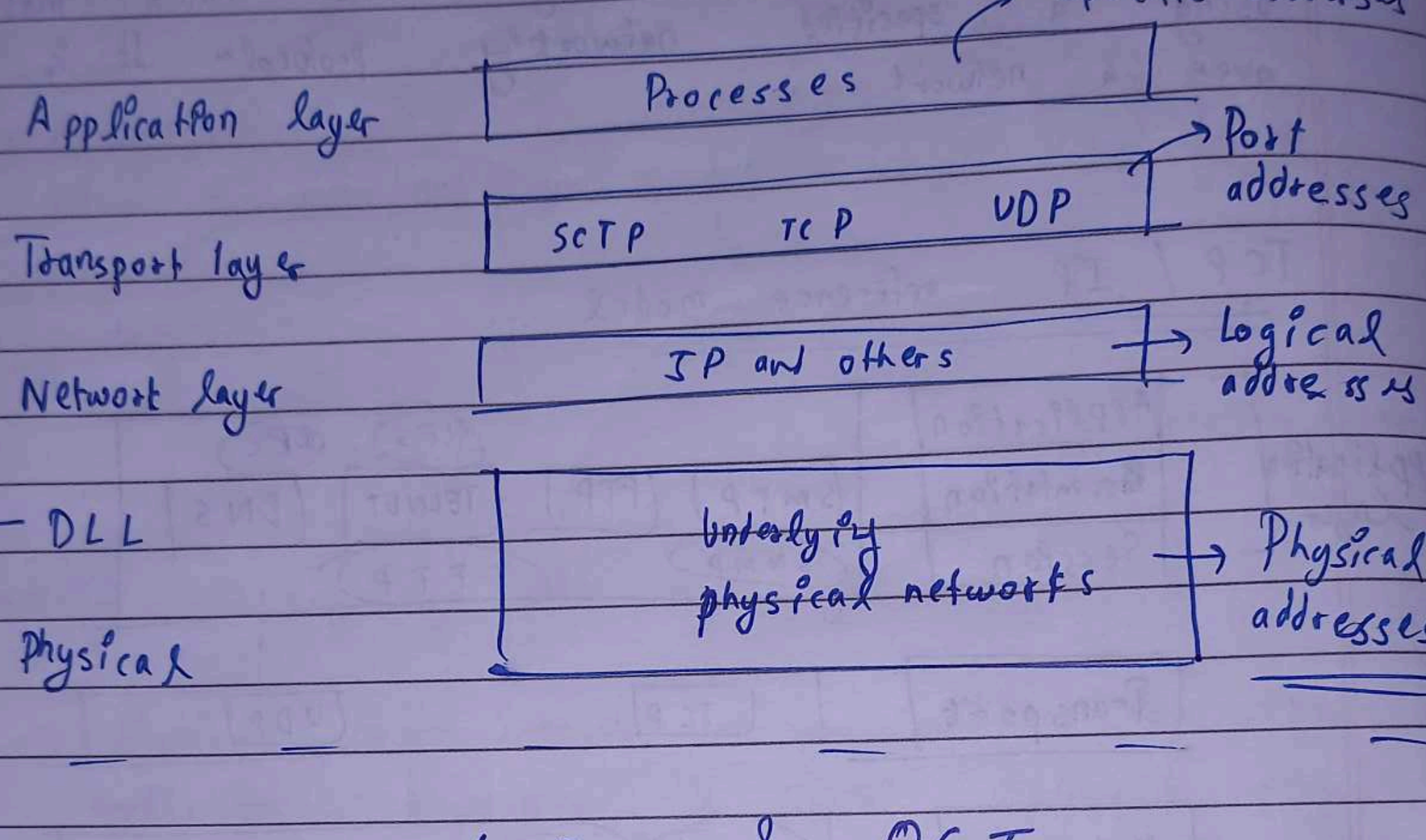
The basic unit of exchange b/w entities that communication using a specified networking protocol. It is transferred over a network.

TCP / IP reference model.



Sorry for the bad diagram!!

Addressing — 4 levels of addressing are used in an internet employing, physical, logical, port and specific.



Differences b/w TCP/IP & OSI.

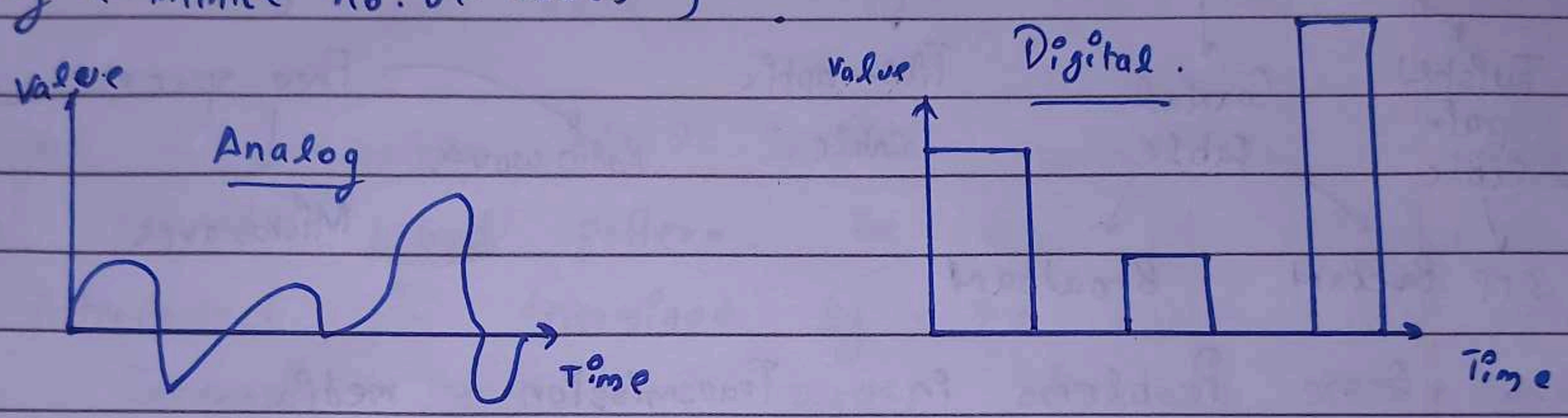
	TCP/IP	OSI model
No. of layers	4	7
Focus	Implementation	Conceptual
Usage	Widely used	less widly used
Approach	Horizontal	Vertical
Delivery guarantee	No	Yes
Reliable	No	Yes

* This page is kind of like good to know info. if you are in hurry skip. *

Physical layer.

To be transmitted, data must be transformed to electromagnetic signals. Data can be analog or Digital. Analog dat are continuous aN take continuous values. Digital data have discrete states and take discrete values.

Signals too, can be Digital (limited no. of values) & Analog (infinite no. of values).

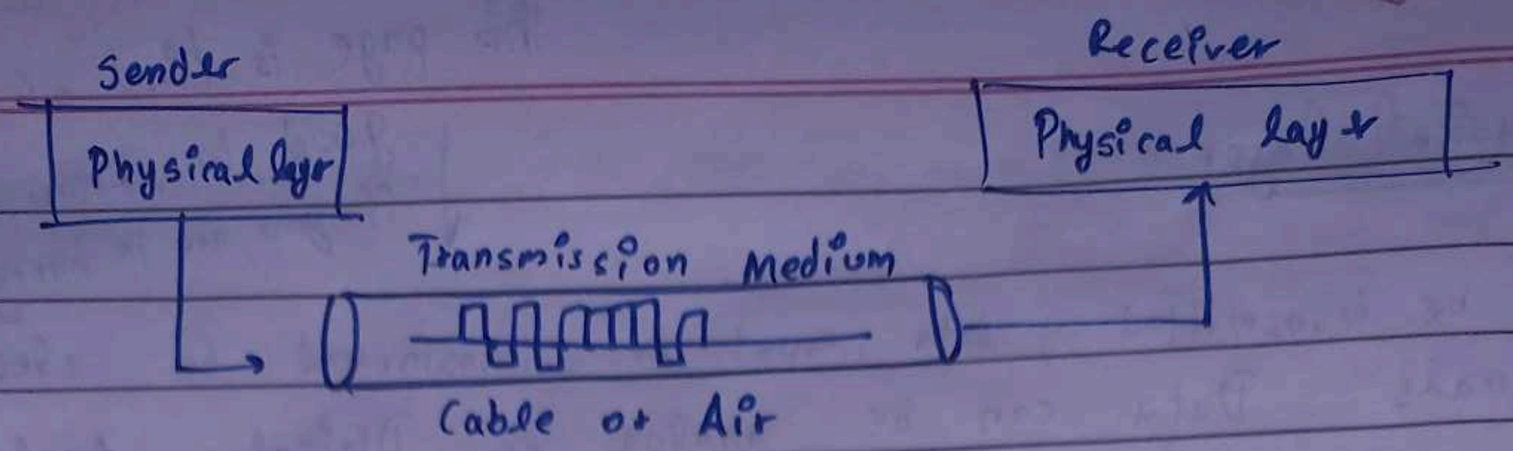


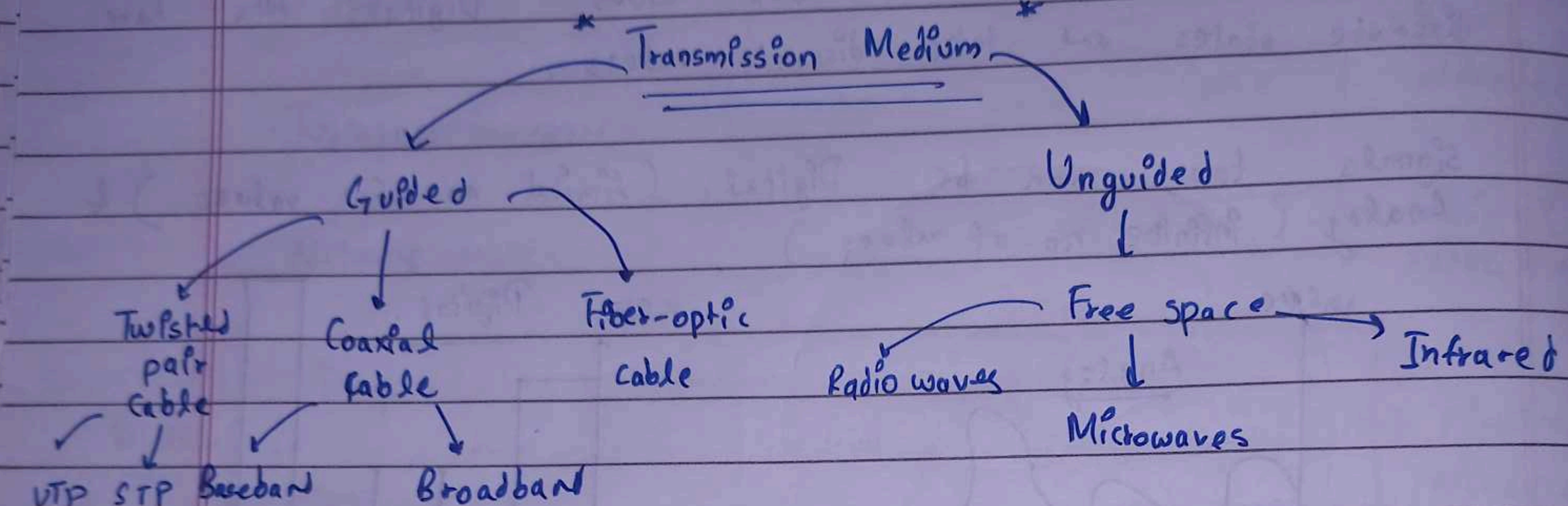
In data communications, we commonly use periodic analog signal (sine wave) and periodic digital signals.

Frequency is the rate of change with respect to time. Phase describes the position of the waveform relative to time.

We need to sed a composite signal of many sine wave not just one.

P.T.O.





Some Problems in Transmission media.

Attenuation → loss of signal / energy with increase in distance

Distortion → change in shape of signal.

Noise → unwanted signal which gets added to main signal.

key factors in making Transmission media.

- Bandwidth
- Transmission impairment
- Interference.

Guided Media

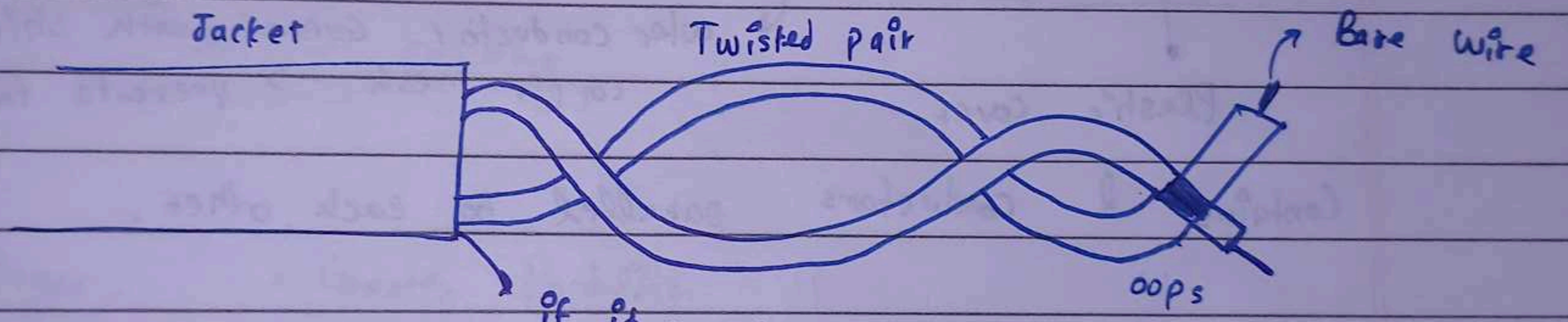
also called Bounded media.

The Physical medium through which the signals are transmitted.

Twisted Pair Cable

Made up of a pair of cables twisted with each other
A twisted pair cable is cheappp. Installation is easy and a lightweight cable. Frequency range → 0 - 3.5 kHz

A twisted pair consists of 2 insulated copper wires arranged in a regular spiral pattern. The degree of reduction in noise interference is determined by the no. of turns per foot. Increasing the no. of turns decreases noise interference & cross-talk.



if it would have in STP then there would be a Metal shield here.

Unshielded Twisted Pair

Cat 1 : telephone lines low speed
Cat 2 : 4 Mbps
Cat 3 : 16 Mbps
Cat 4 : 20 Mbps
Cat 5 : 200 Mbps.

Advantages

- Cheap
- Easy installation
- can be used for high speed LAN.

Disadvantages

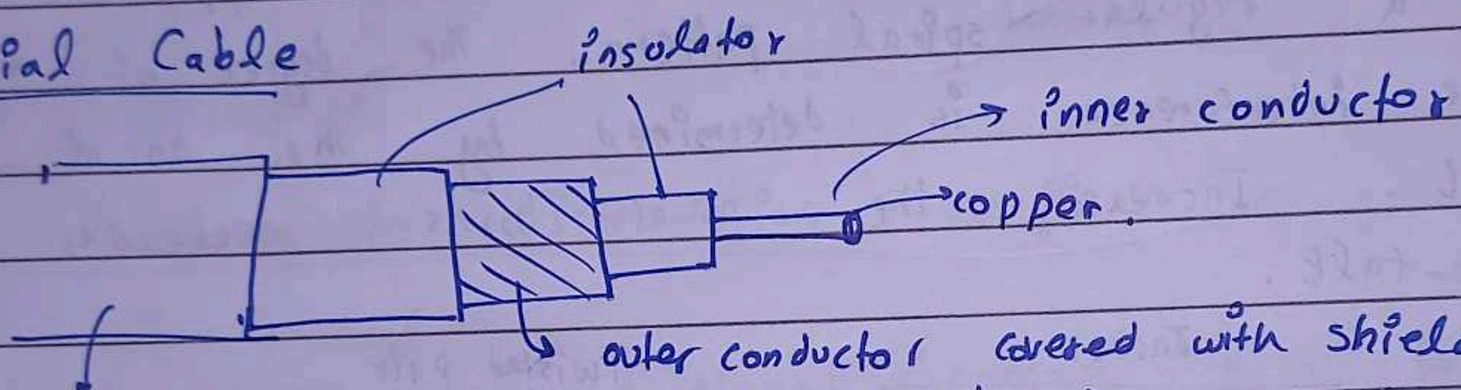
- Only used for shorter distances.

Shielded Twisted Pair → has a mesh surrounding wires.
Can provide much higher speeds. 600 is 4 Mbps

Disadvantages
- Higher attenuation
- More expensive than UTP.

- Medium cost
- Easy installation
- Higher capacity
- ~~Higher~~ attenuation
- Provides High data transmission rate

Coaxial Cable



Contains 2 conductors parallel to each other.

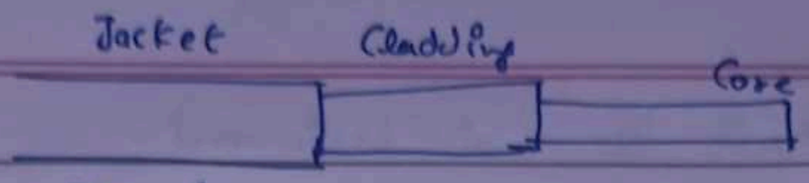
Coaxial
- Baseband → signal signal at high speed
- Broadband → transmitting multiple signals.

Advantages
- Higher speeds
- Better shielding
- Provides higher bandwidth

Disadvantages
- More expensive
- if any fault occurs, failure in ~~eth~~ entire network.

RG-59 → Cable TV RG-58 → Thin Ethernet RG-11 → Thick

Fibre Optic.



Side View

Uses Electrical signals for communication. Sends data by pulses of light.

Core: The optical fibre consists of a narrow strand of glass or plastic known as a core. A core is a light transmission area of the fibre. The more the area of the core, the more light will be transmitted into fibre.

Cladding: The concentric layer of glass is known as cladding. The main functionality of the cladding is to provide to lower refractive index at the core interface as to cause the TIR.

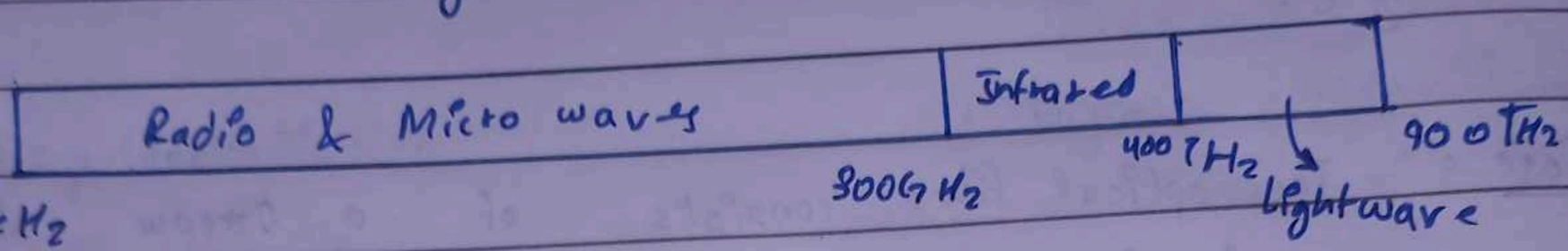
Jacket: Protective coating.

Advantages.

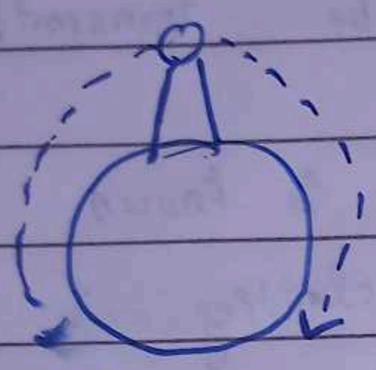
- Greater Bandwidth
- Faster Speed
- Longer distances
- Better reliability
- Thinner and Sturdier

Unguided Media

Transmits electromagnetic waves with physical medium.

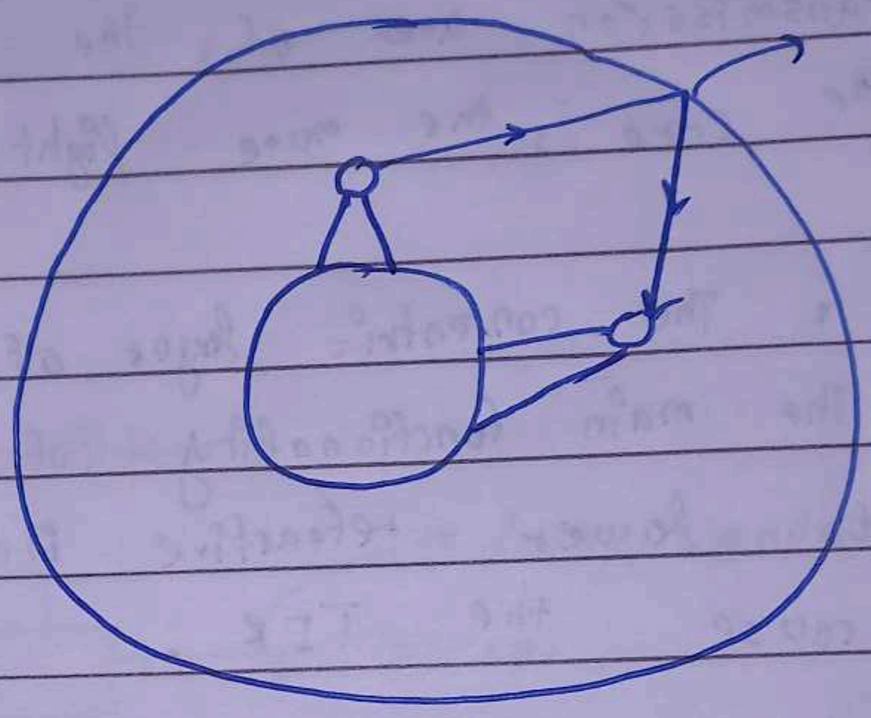


Propagation methods



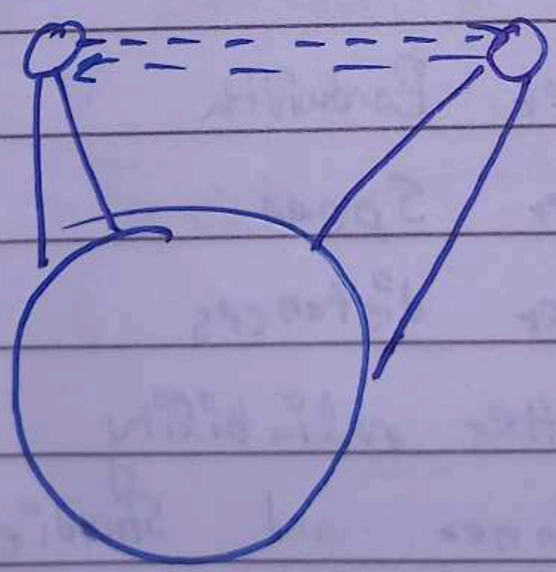
Ground Propagation

below 2MHz



Sky propagation

(2-30 MHz)



Line of Sight propagation.

Above 30MHz.

Radio Waves → transmitted in all direction of free space.
Omnidirectional & range from 3kHz to 1kHz.
Eg FM.

Application → useful for multicasting, FM radio, TV ...

Advantages → large coverage area, can penetrate through walls.
→ wide area network
→ high transmisspon rate.

Microwares Line of sight and uni directional. Ranges from long range.
1 GHz to 1000 GHz.

Advantages → High bandwidth
→ long range
→ Immunity to interference

Applications → Terrestial relay systems, satellite communication, Wireless Lans.

Infrared. Line of sight, short range. 300 GHz to 400 THz.

- Supports very high bandwidth.
- cannot penetrate walls
- better security
- unreliable outdoors. cause of Sun rays.

Packet switching → A data transmission in which data is divided into smaller units called packets and then transmitted independently over the network.

Circuit switching → In which a dedicated communication path is established b/w sender and receiver before transmission begins.

Differences

	Packet switching	Circuit Switching
Connection	Connectionless	Connection-oriented
Efficiency	More	less
Bandwidth	No guaranteed	Guaranteed,
Cost	Low	High.
Eg.	Internet	Telephone
Advantages	Efficient use of network resources, lower cost.	Guaranteed bandwidth, guaranteed latency.

Types of connecting devices

Hub : connects multiple devices to a single network segment. Sends a broadcast to all connected devices.

Switch : Smart version of hub, sends the packet to the devices that it is itended to.

Router : Connects 2 or more networks together. Determines the optimal routing path b/w 2 nodes.

Bridge : Connects 2 similar networks together.

Gateway : Joins 2 networks together. Eg connectiy LAN to internet. They provide many security features.

There are more devices though ... Repeaters, Modem, Access Point, Firewall.

Data Link Layer. Services of DLL,

- Framing & Link Access
- Flow Control
- Error Correction
- Reliable Delivery
- Error Detection

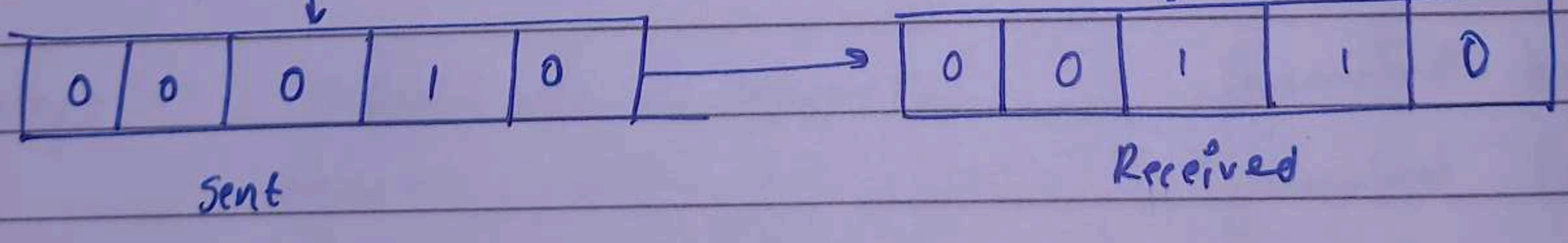
* Error Detection *

When data is transmitted from one device to another, the system does not guarantee whether the data received by the device is identical to the data transmitted by another device.

Types of Errors

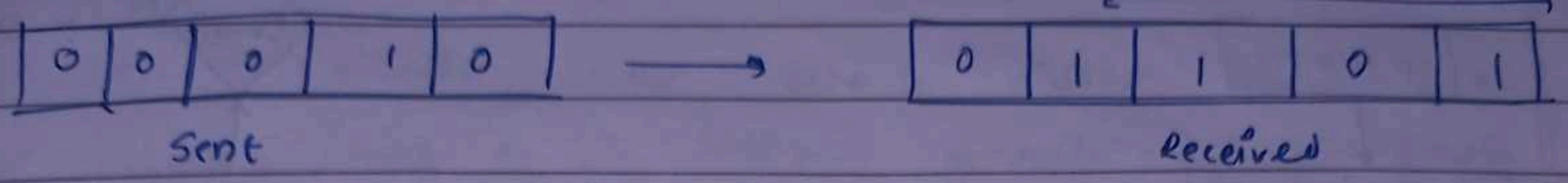
- Single Bit
- Burst

Single bit.



mainly occurs in Parallel data transmission.

Burst error



mainly occurs in Serial Data transmission.

Error Detection Techiques.

- Single Parity Check
- Two-dimentional Parity Check
- Checksum
- Cyclic Redundancy Check (CRC)

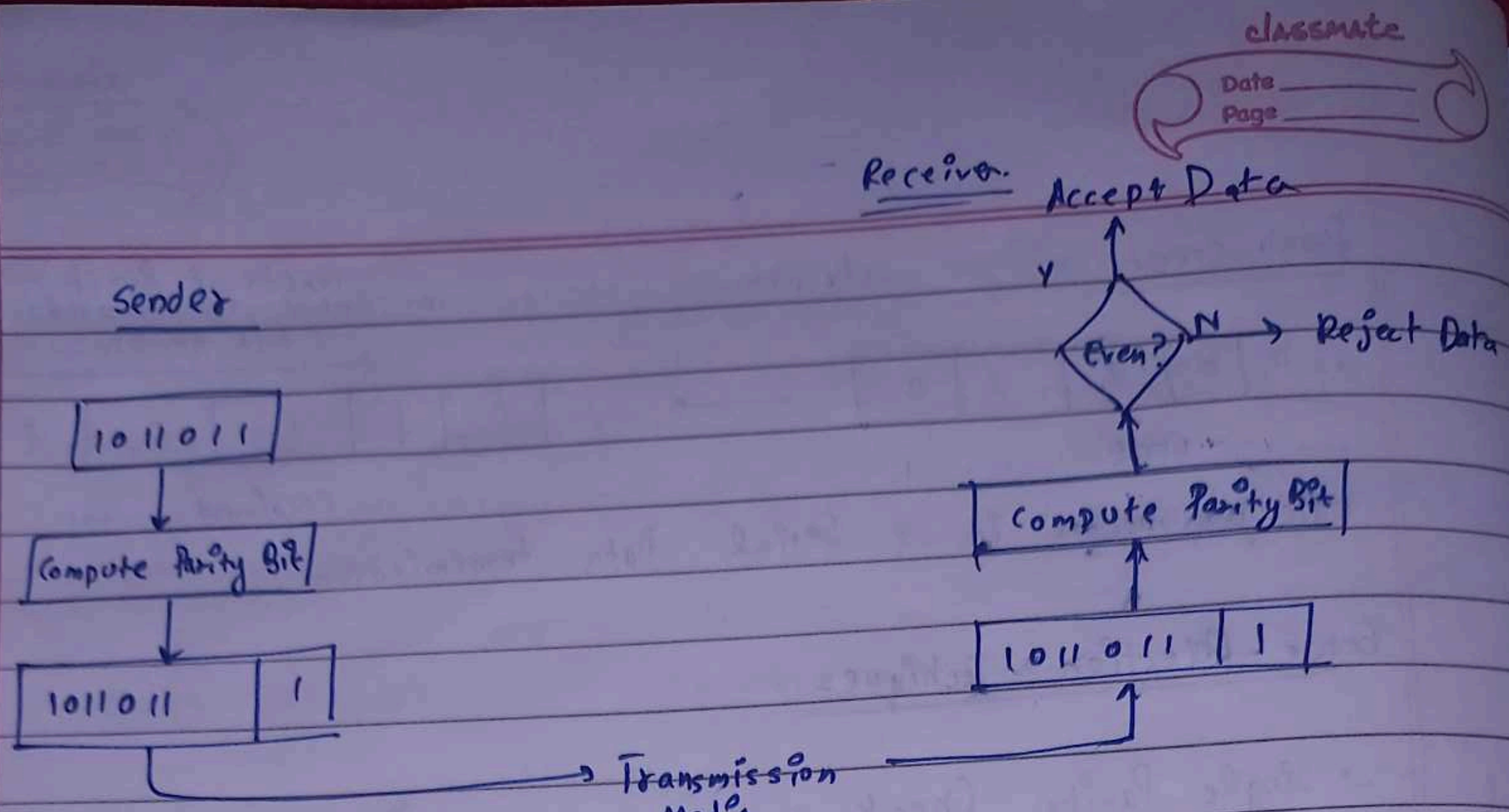
Single Parity Check.

A redundant bit also known as a parity bit which is appended at the end of the data unit so that the no. of 1s become even.

This is Even-parity checking. There is Odd parity checkiy opposite to this.

If the no. of 1s is odd the parity bit 1 is appended and if the no. of bit is even the parity bit 0 is appended.

At the receiving end, the parity bit is calculated from the received data bits and compared with the received parity bit.



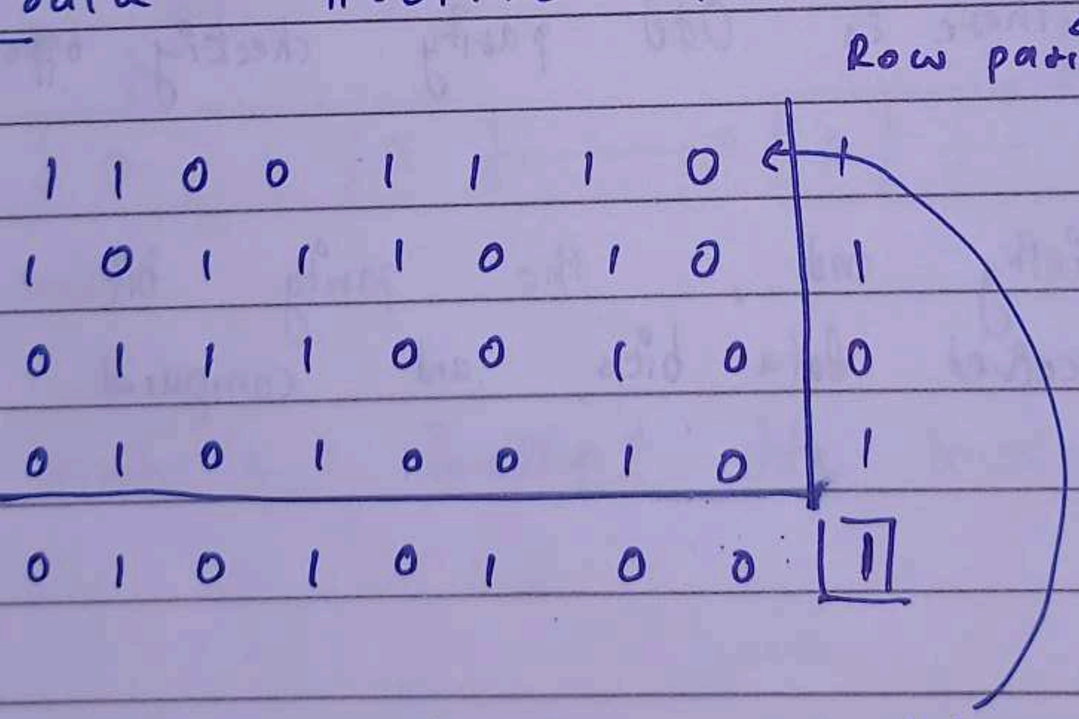
Drawbacks of Single Parity.

- Can only detect single bit error. Which is extremely rare.

2-D Parity Check.

Parity bits are computed for each row & column.
At the receiving end parity bits are compared.

Original data 11001110 10111010 01110010 01010010

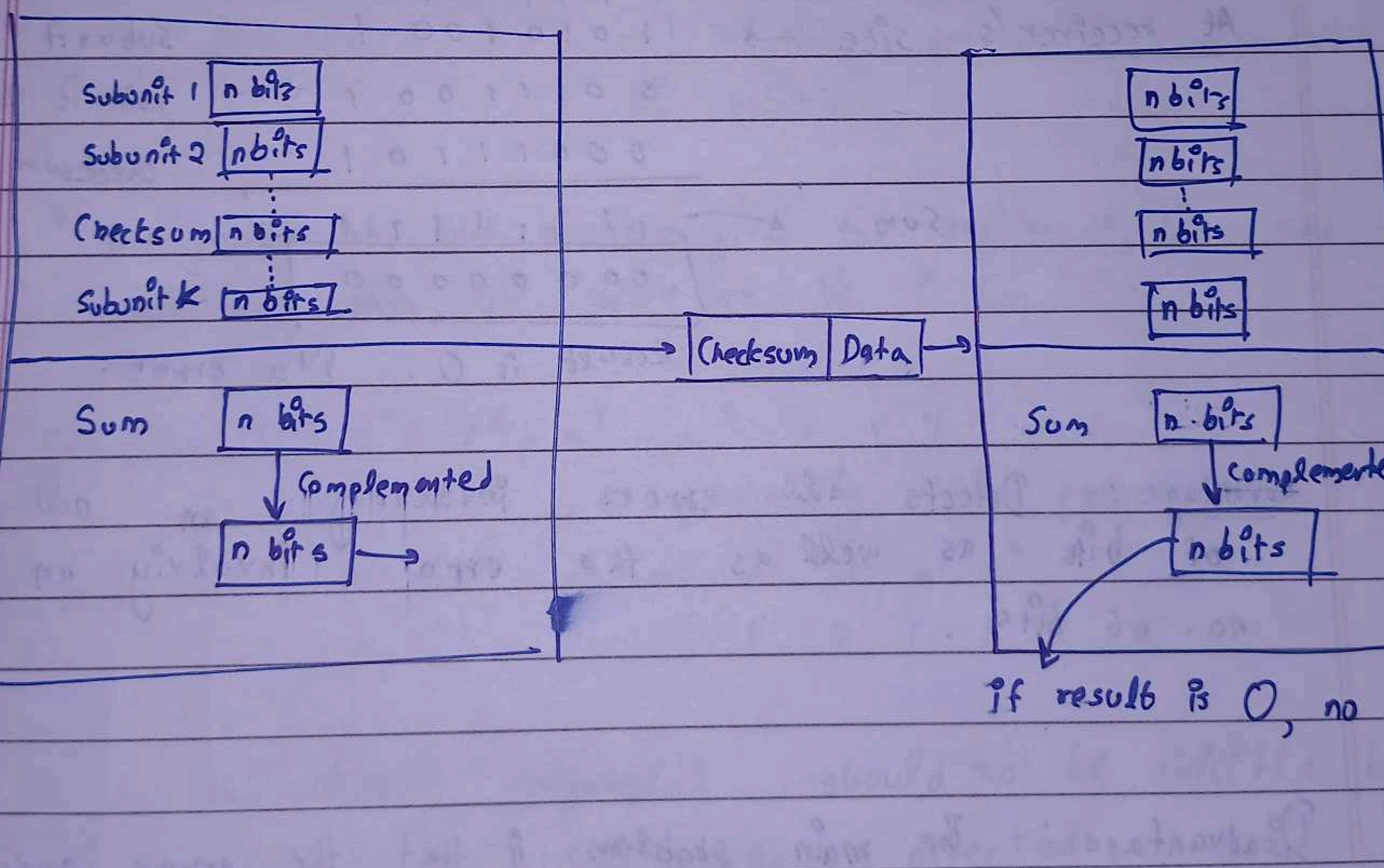


Drawbacks.
- If 2 bits in one data unit are corrupted and 2 bits exactly the same position in another data unit are also corrupted, then 2D will not be able to detect errors.
- Cannot be used to detect 4 or more bit errors.

Checksum error detection based on concept of redundancy.

At the sender side, the data is divided into equal subunits of n bit length by checksum generator. 16-bit length.

These subunits are then added togethor using one's complement method. This is the sum of n bits. The resultant bit is the complemented. This complemented sum is called checksum and the is apprehended to the end of original data unit and is then transmitted to receiver.



P.T.O .

Eg. If the data unit to be transmitted is 10101001 00111001, the following procedure is used at Sender & Receiver Site.

Sender's Site →

```
        1 1 1   1
  1 0 1 0 1 0 0 1    subunit 1
  0 0 1 1 1 0 0 1    subunit 2
-------------------
  1 1 1 0 0 0 1 0    Sum
  0 0 0 1 1 1 0 1    → Checksum (complement of sum)
```

→ no wrapping ∴ no extra bits

10101001	00111001	00011101
Data		Checksum

At receiver's site →

```
        1 1 1   1
  1 0 1 0 1 0 0 1    subunit 1
  0 0 1 1 1 0 0 1    subunit 2
  0 0 0 1 1 1 0 1    checksum
-------------------
  1 1 1 1 1 1 1 1    Sum
  0 0 0 0 0 0 0 0
```

Result is 0. No error.

Advantage: Detects all errors involving an odd no. of bits as well as the error involving an even no. of bits.

Disadvantage: The main problem is that the error goes undetected if one or more bits of a subunit is damaged and the corresponding bit or bits of a subunit are damaged and the corresponding bit or bits of opposite value in second subunit are also damaged. This is because sum remains unchanged.

Last eg's data checksum

Example of disadvantage: | 10101001 00111001 | 00011101 | . But the data received at destination is 00101001 10111001 00011101

```
  1 1 1   1
  0 0 1 0 1 0 0 1
  1 0 1 1 1 0 0 1
  0 0 0 1 1 1 0 1
  ---------------
  1 1 1 1 1 1 1 1
  0 0 0 0 0 0 0 0   → Error went undetected.
```

Cyclic Redundancy Check. (CRC)

Based on Binary Division.

CRC generator → is an algebraic polynomial represented as a bit pattern.

* The power of each bit term gives the position of bit and coefficient gives the value of bit. *

Eg: $x^7 + x^6 + x^4 + x^3 + x + 1$

↓ Correspondig bit pattern.

x^7	x^6	x^5	x^4	x^3	x^2	x^1	x^0
1	1	0	1	1	0	1	1

- The algebric polynomial should not be divisible by x. So that burst errors are detected.
- It should be divisible by x+1. So that burst errors affecting odd no. of bits are detected.

Let's understand by example only.

Eg: A bit stream 1101011011 is transmitted using CRC. The generator polynomial is x^4+x+1. What is the actual bit string?

5 bits

$x^4+x+1 \rightarrow 10011$

a)

We add 5-1 bits 0 to stream 1101011011 0000

```
1 0 0 1 1 | 1 1 0 1 0 1 1 0 1 1 0 0 0 0
            1 0 0 1 1
            -----------
              1 0 0 1 1
              1 0 0 1 1
              ---------
              0 0 0 0 0
                0 0 0 0 1
                1 0 0 1 1
                ---------
                0 0 1 0 1
```

leading should be always 1

b)

```
1 0 0 1 1 | 1 1 0 1 0 1 1 0 1 1 0 0 0 0
          | 1 0 0 1 1
          | ---------
          | 0 1 0 0 1 1
            1 0 0 1 1
            ---------
            0 0 0 0 0 1 0 1 1 0
                1 0 0 1 1
                ---------
                0 0 1 0 1 0 0
                  1 0 0 1 1
                  ---------
                  0 0 1 1 0
```

perform XOR.

the last 4 zeros that we added to the stream, replace by

Remainder.

Actual String. → 1101011011110

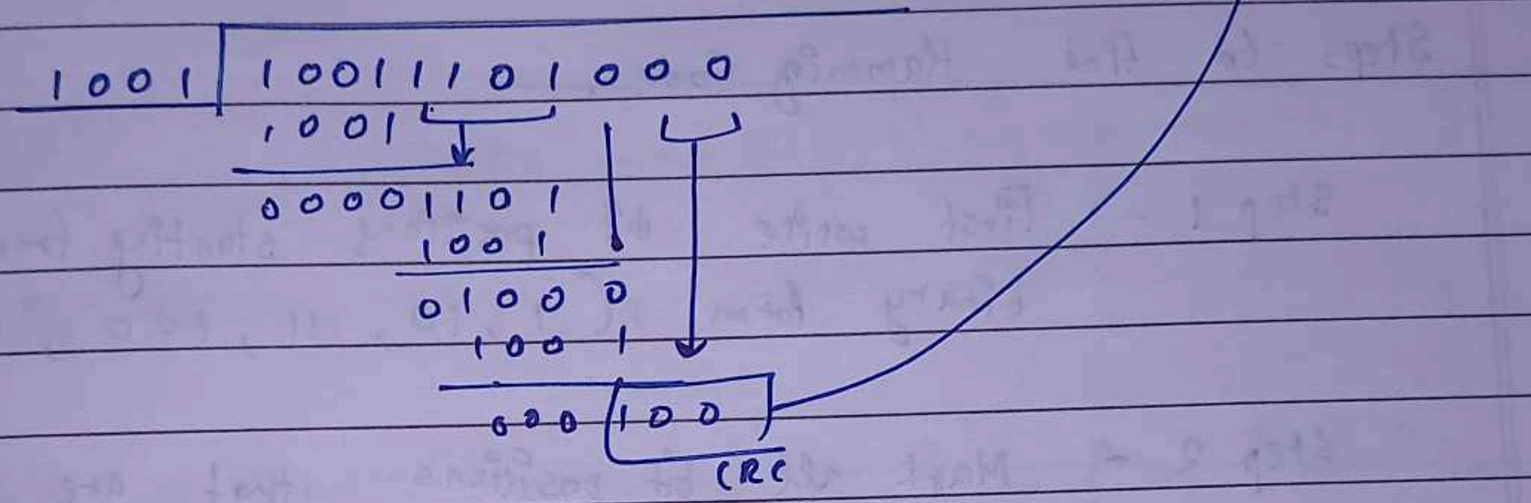
Eg. A bit stream 10011101 is transmitted. $x^3 + 1$.

a) What is actual string ?

b) Suppose 3rd bit from left is inverted. How will receiver detect error.

a) $x^3 + 1 \rightarrow 1001$

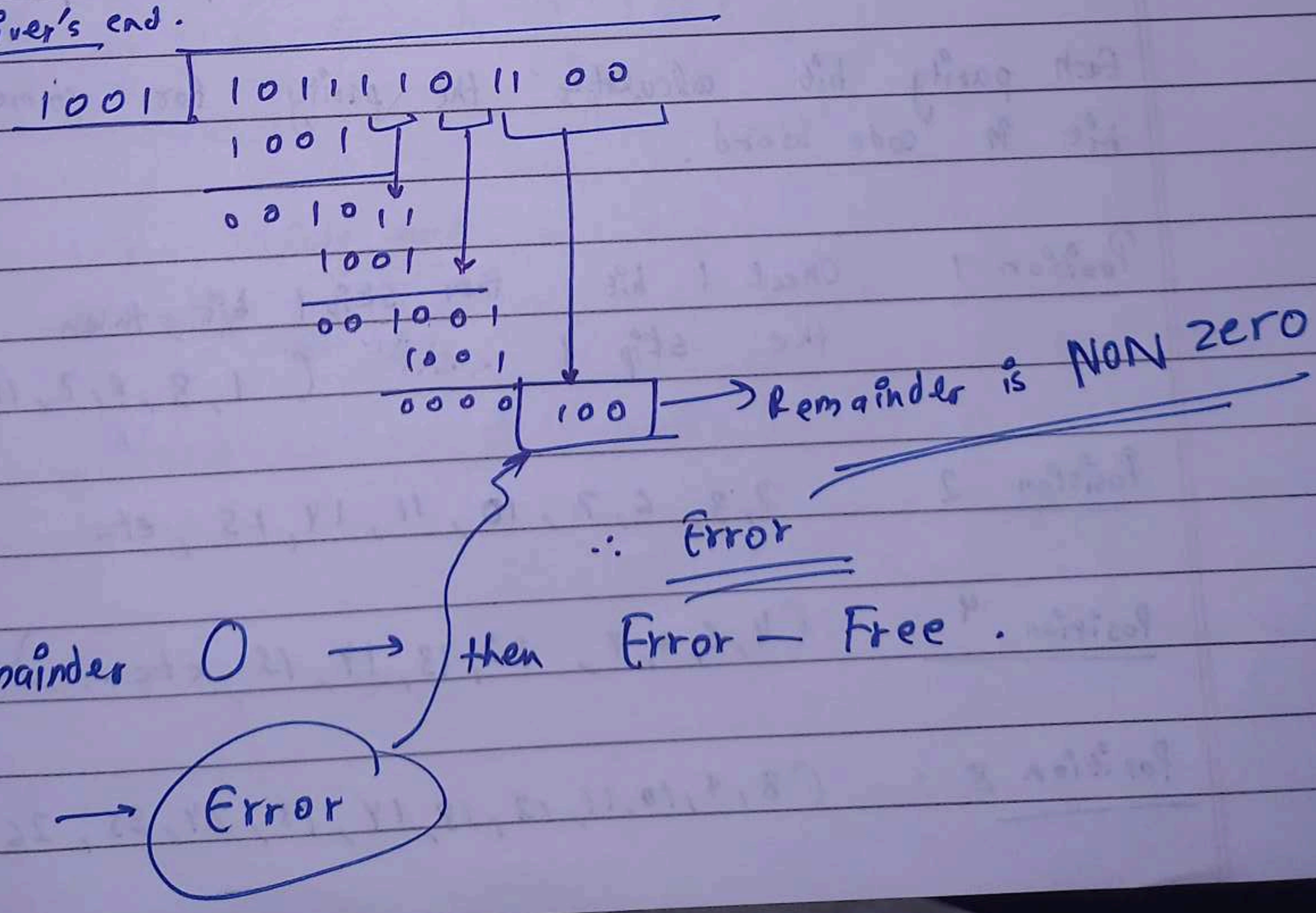
add 4-1 bits to stream 10011101 000



$\therefore$ Code word = 10011101100

b) 3rd bit flipped → 10111101100

At receiver's end.



Hamming Code → for error correction.

Here Total length of frame (N) = M + R → redundant bits
M → M data bits

No. of states of error = l

$$2^R \geq M + R + 1$$

$n + P + 1 > m$

Steps to find Hamming Code.

Step 1 - First write bit positions starting from 1 in binary form (1, 10, 11, 100, etc..)

Step 2 - Mark all bit positions that are powers of 2 as parity bits. (1, 2, 4, 8, 16, 32...)

Step 3 - All other bit positions are for data to be encoded. (3, 5, 6, 7, 9, 10, 11, etc..)

Each parity bit calculates the parity for some of the bit in code word.

Position 1 Check 1 bit, then skip 1 bit, then check 1 and the skip 1 .... (1, 3, 5, 7, 11, ...)

Position 2 2, 3, 6, 7, 10, 11, 14, 15, etc

Position 4 (4, 5, 6, 7, 12, 13, 14, 15, etc..)

Position 8 (8, 9, 10, 11, 12, 13, 14, 15, 24, 25, 26, 27, 28, 29, 30, 3

Eg.

Eg. Construct even parity Hammiy Code for 1001101
(7 bits)

$2^R \geq M + R + 1$

$2^4 \geq 7 + 4 + 1$ ← parity bits

$R = 4$ if $P = 2$ $2^P = 2^2 = 4$ & $n+P+1 = 7$ (4 2 1)

if $P = 3$ then $2^P = 2^3 = 8$ & $n+P+1 = 8$ (4 3 1) ✓.

Total bits in codeword = 4 + 3 = 7.

Position 1, checks, 1, 3, 5, 7, 9, 11 ...

Parity bits are 1, 2, 4.

Filling the hammiy code, ? ? ? ? ? ? ?

Parity bit 1 = XOR (data bit 1, data bit 3, data bit 5, data bit 7)

even parity, odd parity, odd parity, even parity

0 1 1 1 0 0 1 0 1 0 1

1st 2nd 4th 8th

Code Word

We will use this.

classmate
Date
Page

Multiple Access Protocols

- Random Access Protocols
  - ALOHA
  - CSMA
  - CSMA/CD
  - CSMA/CA
- Controlled-Access Protocols
  - Reservation
  - Polling
  - Token passing
  - Piggybacking
- Channelization Protocols
  - FDMA
  - TDMA
  - CDMA

Random Access Protocols

Allows devices to access a shared medium without taking turns or waiting for permission. No device is superior to another.

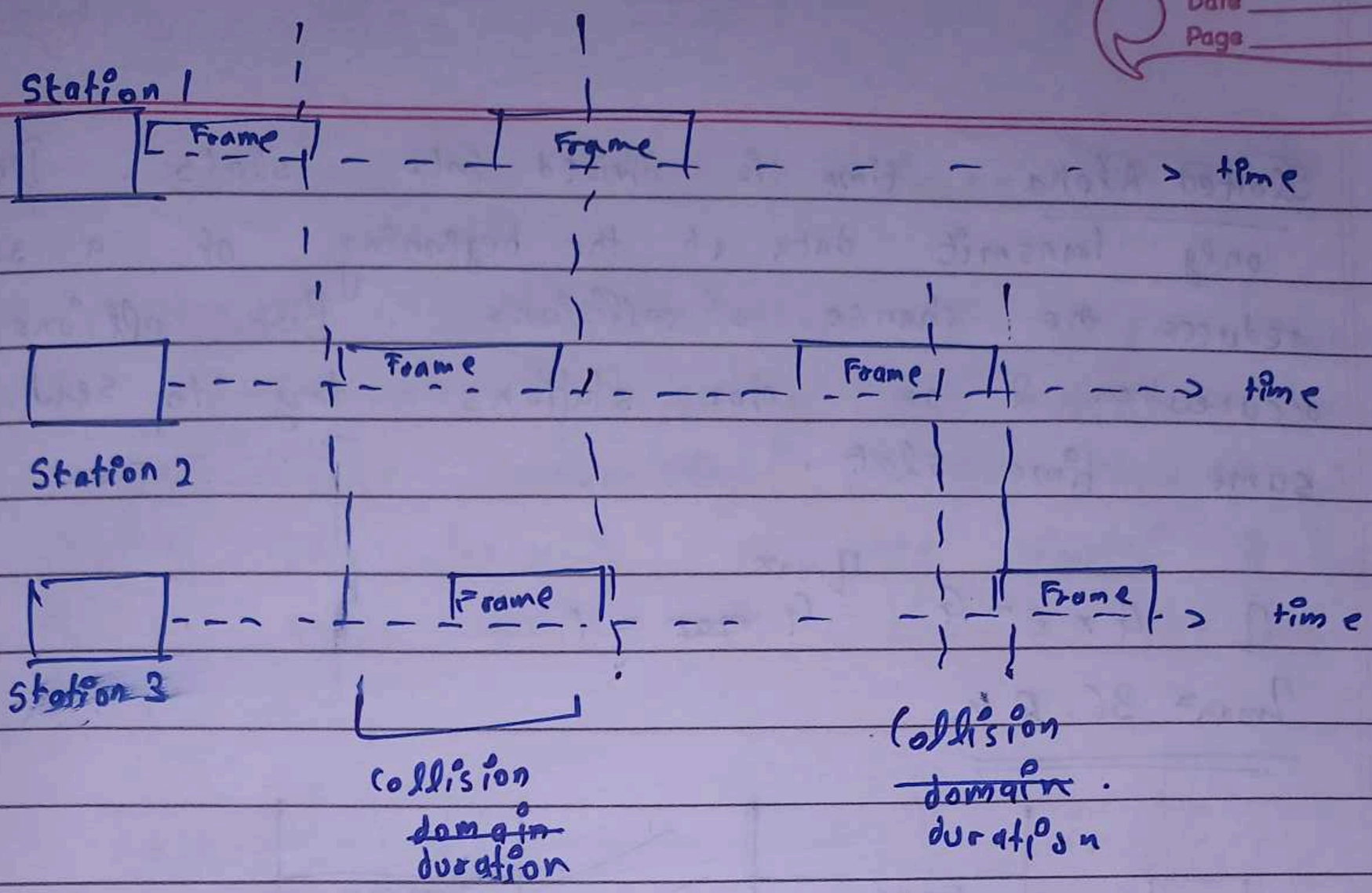
Aloha any device can transmit data at any time, without checking whether the channel is in use.

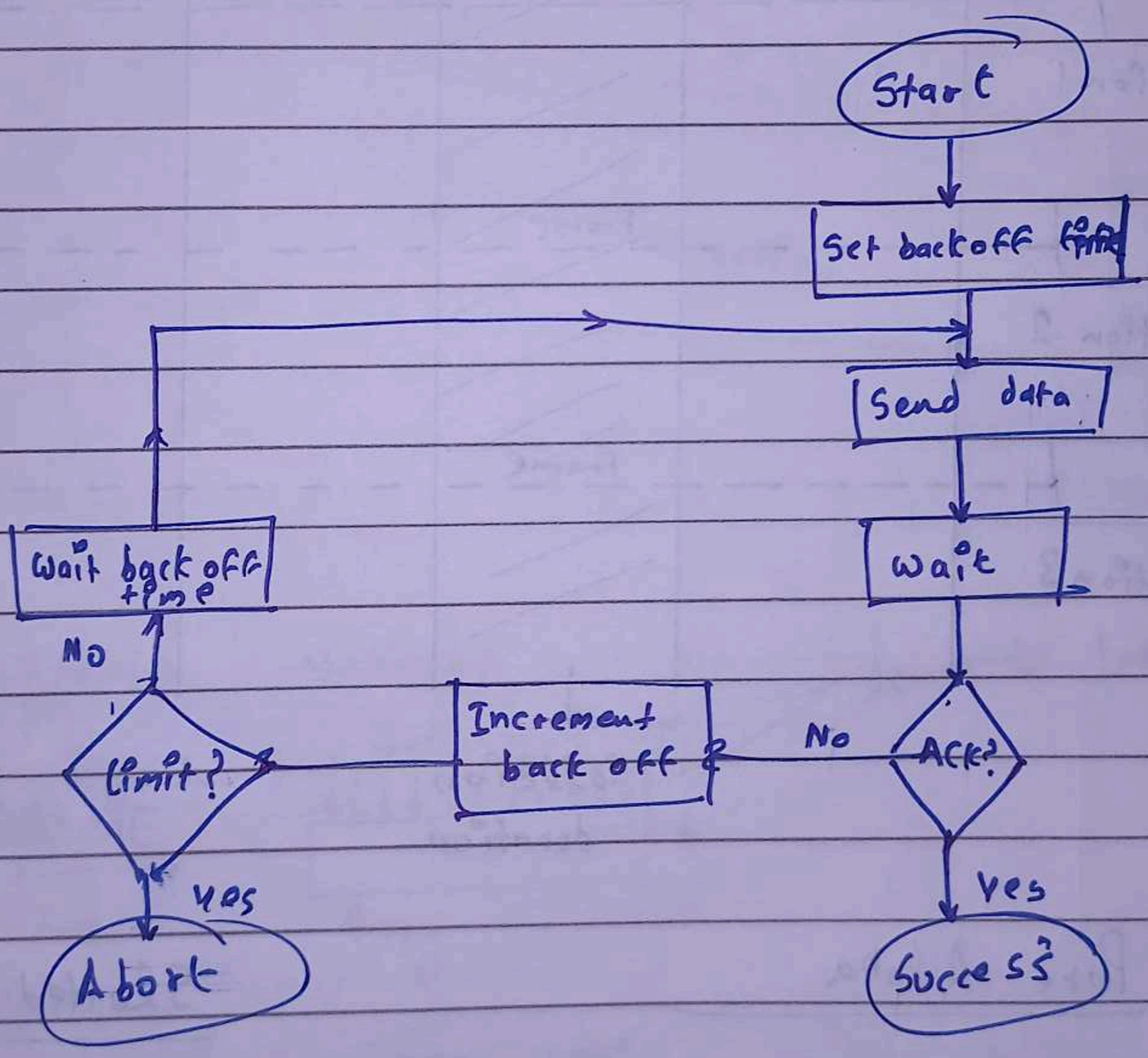
Pure Aloha simplest form of Aloha. There is no time synchronization b/w devices. Devices send whenever they have a packet to send and not wait anytime.

Efficiency of Pure Aloha (η) = $G \times e^{-2G}$ G = no. of stations

$G_{max} = 1/2$

$\therefore \eta_{max} = 18.4\%$


Station 1
Frame
Frame
time
Frame
Frame
time
Station 2
Frame
Frame
time
Station 3
Collision domain duration
Collision domain duration

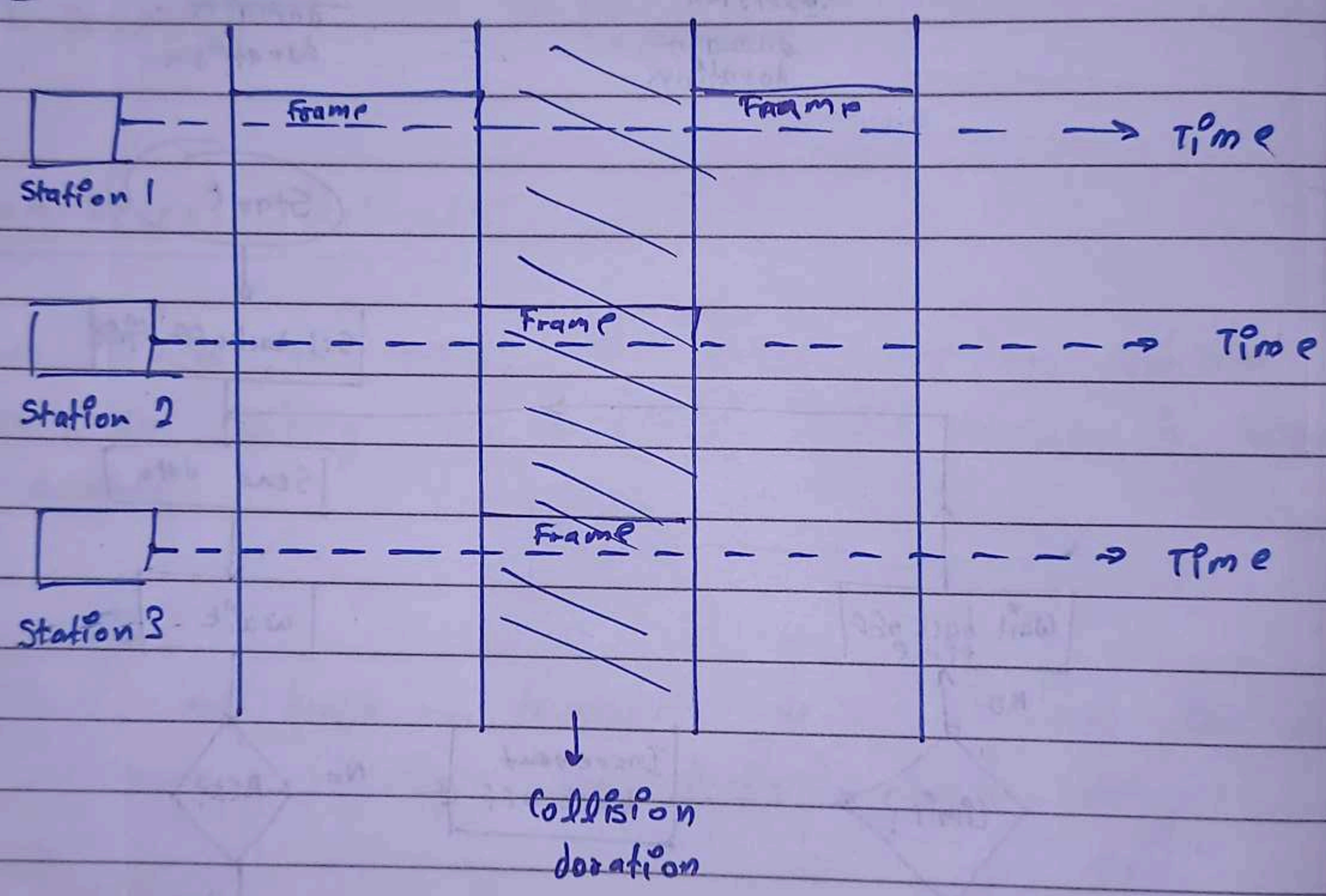

Start
Set backoff time
Send data
wait
Ack?
No
Increment back off
limit?
No
Wait back off time
Yes
Abort
Yes
Success

Slotted Aloha. time is divided into slots. Devices can only transmit data at the beginning of a slot. This reduces the chance of collisions. But collisions can still occur when 2 or mor stations try to send at the same time slot.

η_{max}

$\eta = G \times e^{-G}$ $G_{max} = 1$

$\eta_{max} = 36.8\%$.



Pure Aloha

Any station can transmit at any time.

Continuous time.

Vulnerable time = $2 \times T_t$

$\eta_{max} = 18.4\%$.

Slotted Aloha

Any station can transmit the data at the beginning of an time slot.

Discrete time.

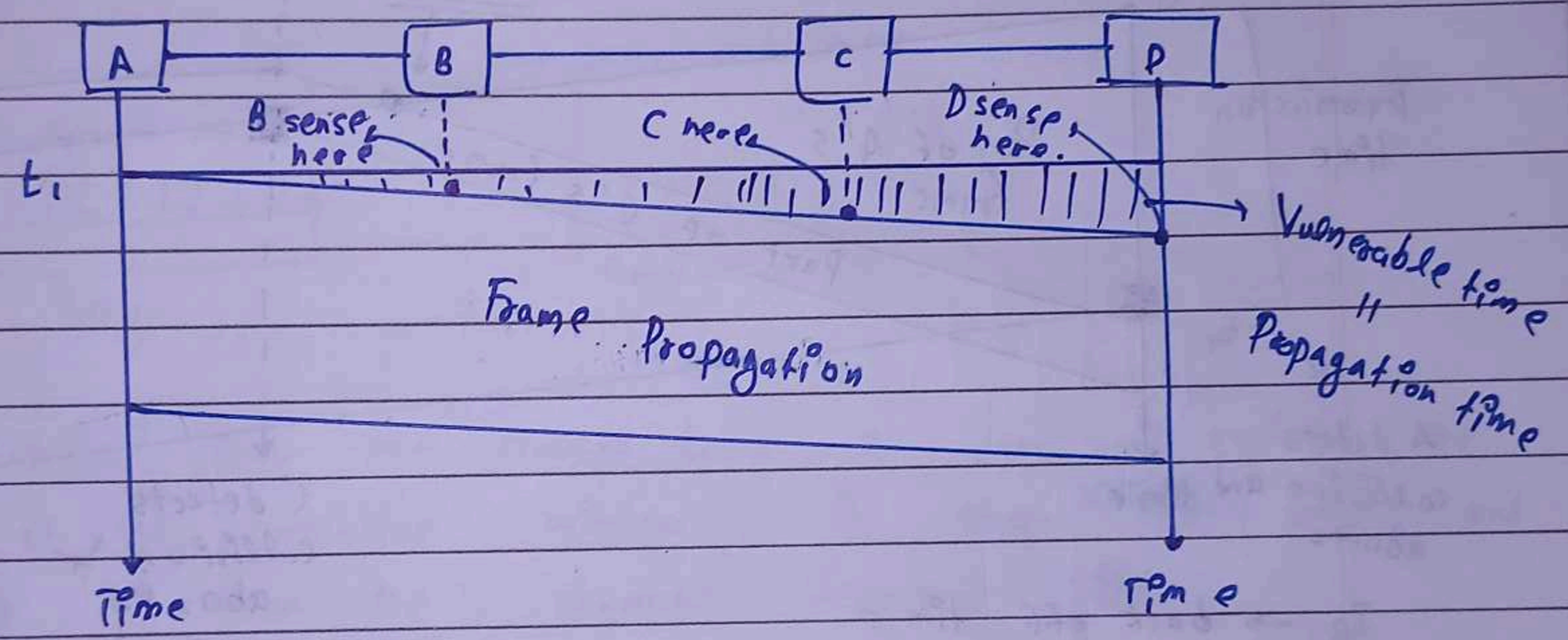
Vulnerable Time = T_t

$\eta_{max} = 36.8\%$.

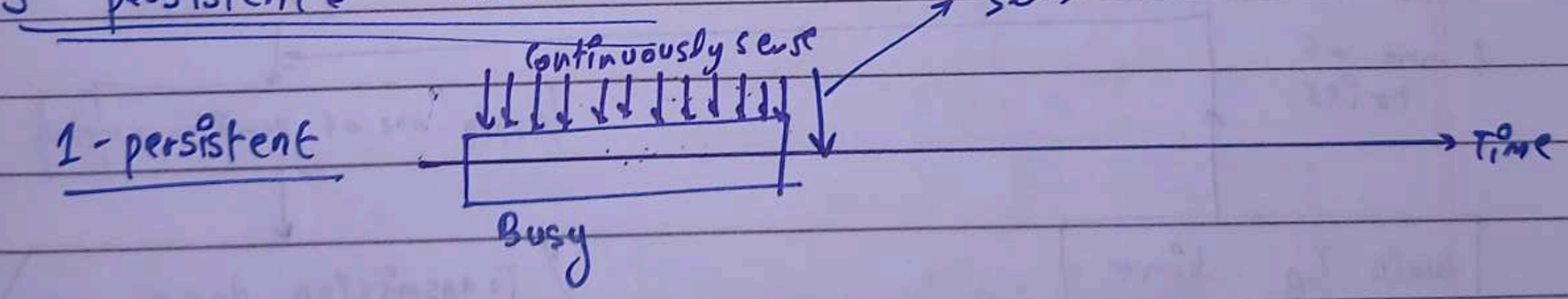
Vulnerable time is the period of time during which a collision can occur.

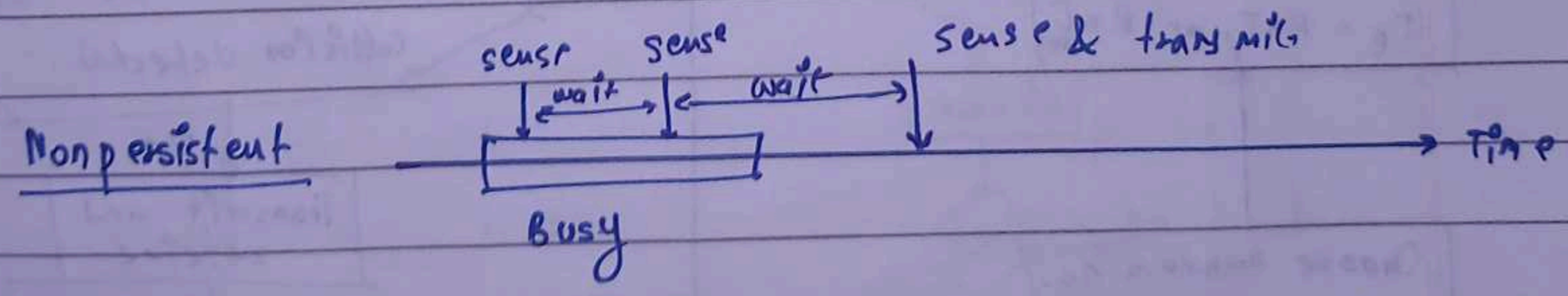
CSMA (Carrier Sense Multiple Access)

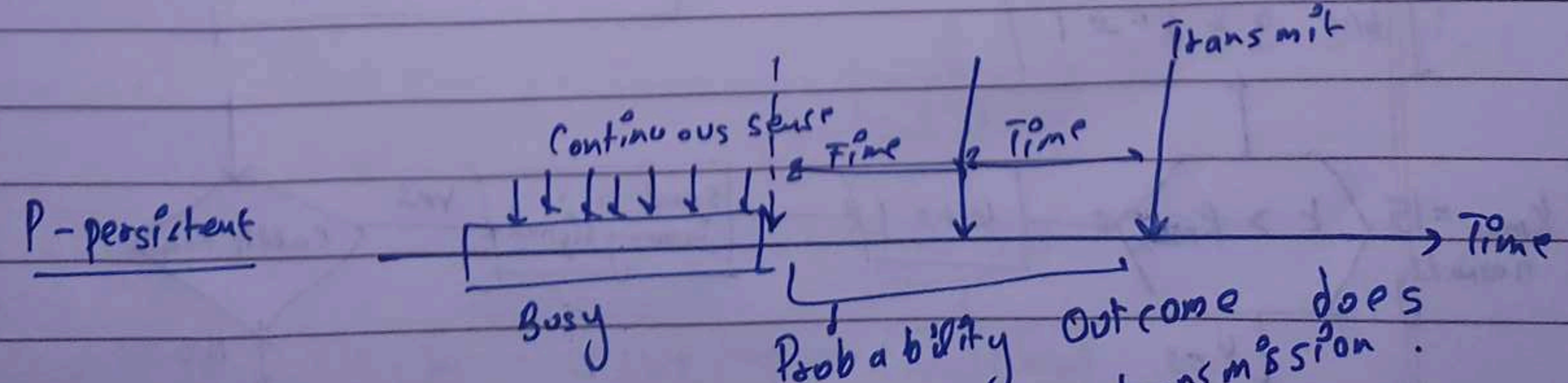
allows multiple devices to share a single communication channel. It works by having each device listen to the channel before transmitting. If the channel is idle, the device can transmit. If the channel is busy, the device waits until the channel is idle before transmitting.



3 persistence methods.

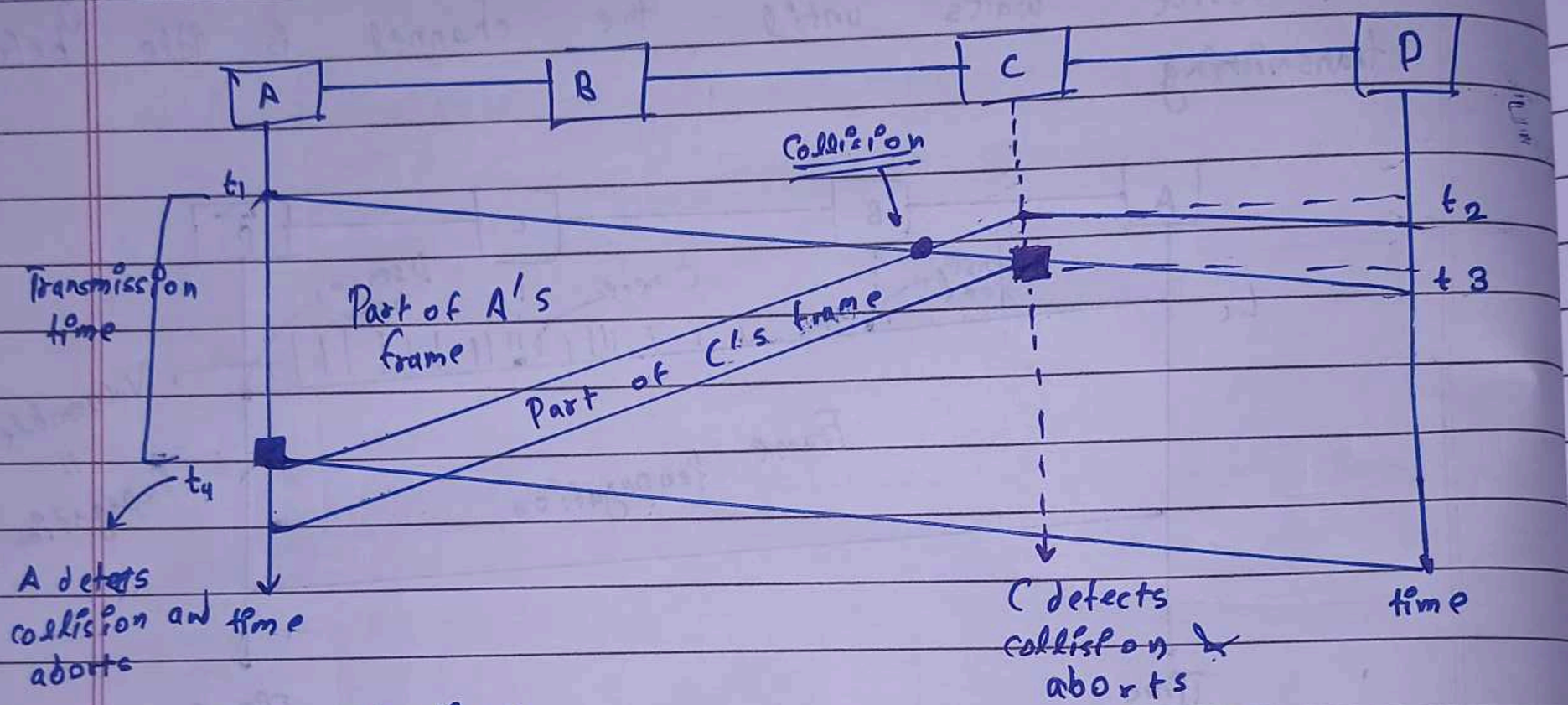




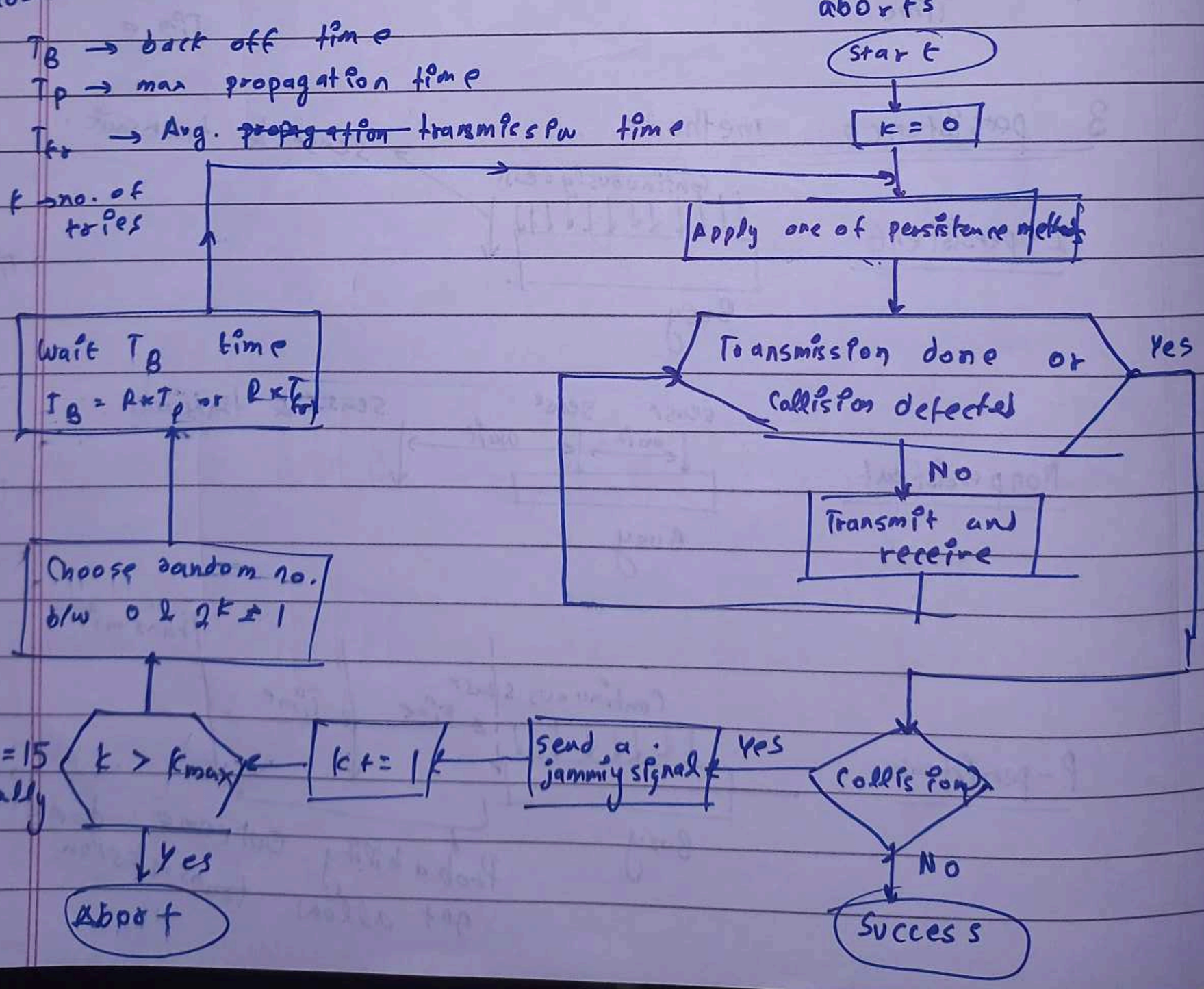


CSMA/CD (Collision Detection)

If 2 or more devices transmit at the same time, a collision will occur. Allows devices to detect collisions and stop transmitting immediately.



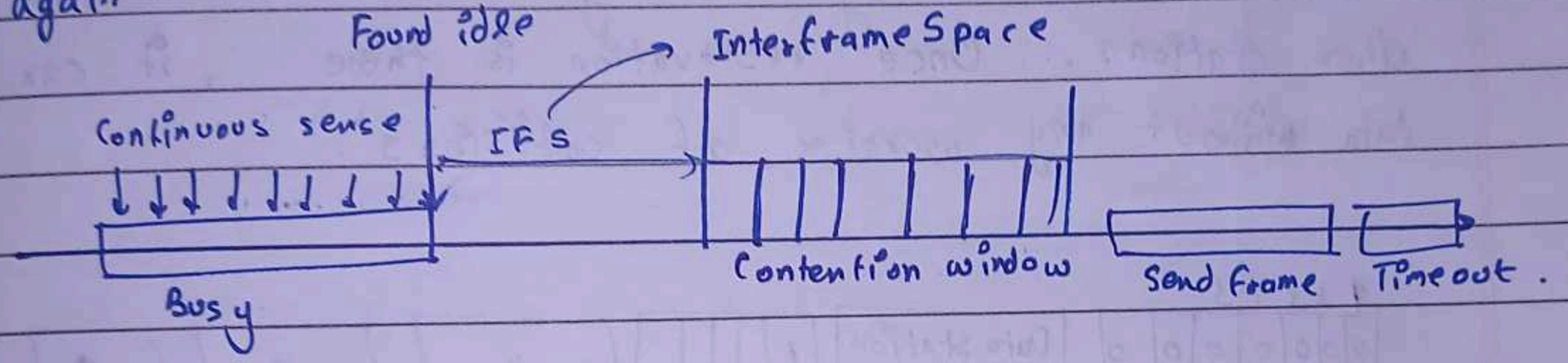
T_B → back off time
T_p → max propagation time
T_{tr} → Avg. ~~propagation~~ transmission time
k → no. of tries



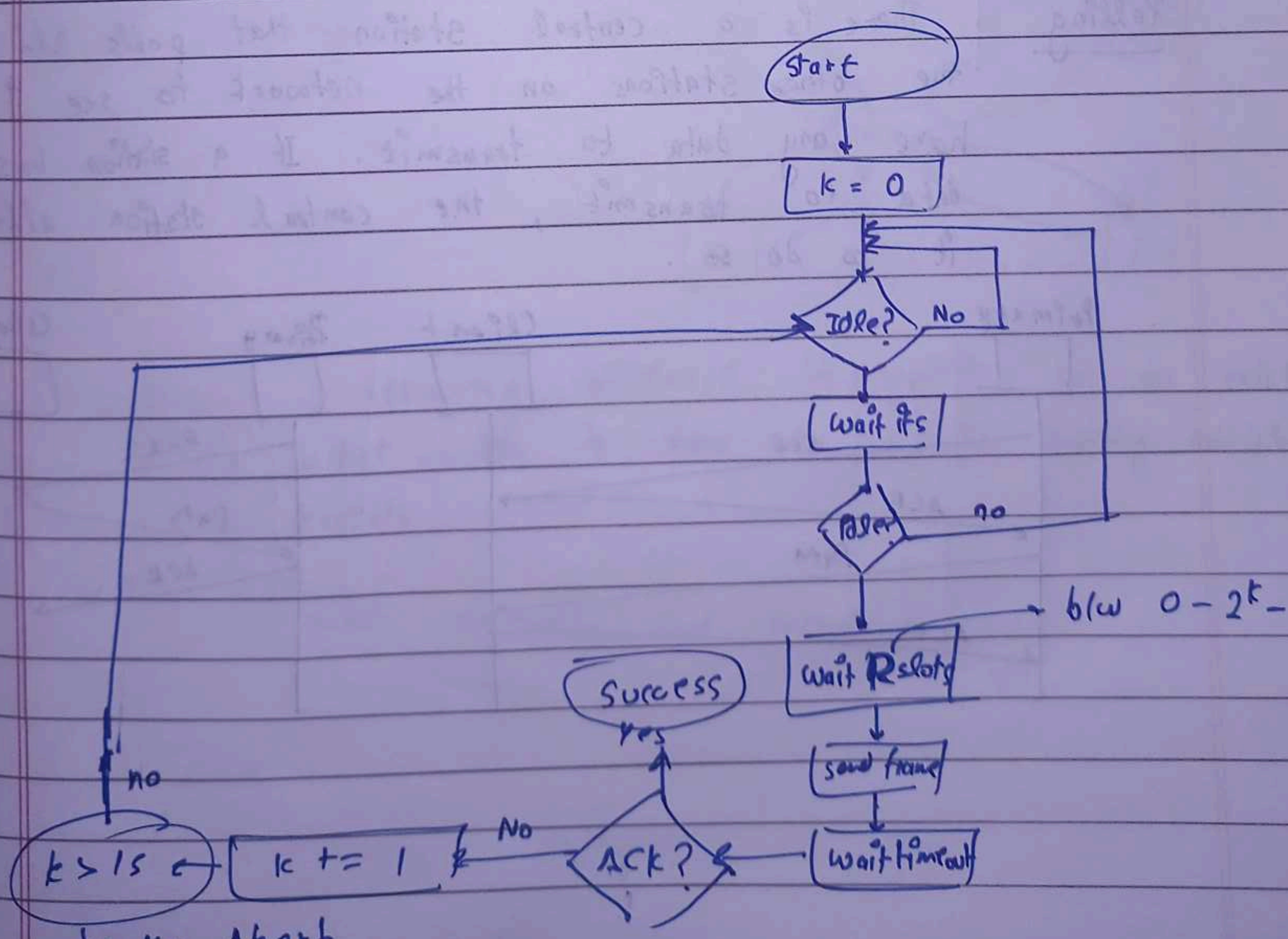
Collision avoidance

CSMA/CA

Allows devices to avoid collisions by sending a small request to send (RTS) packet to the intended recipient before transmitting data. If recipient is ready to receive data, it sends clear-to-send (CTS) packet back to sender. If a collision occurs, all devices that detected the collision wait a random time before trying again.



If station finds the channel busy, it does not restart the timer of the contention window; it stops the timer and restarts it when the channel becomes idle.

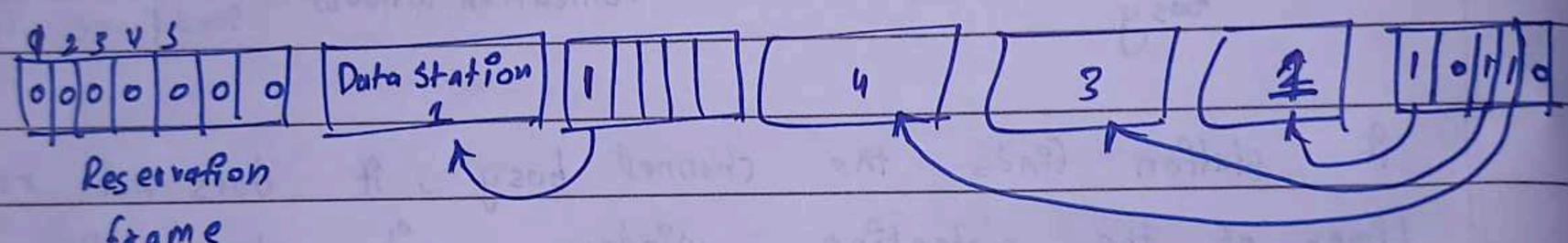


Controlled Access Protocols

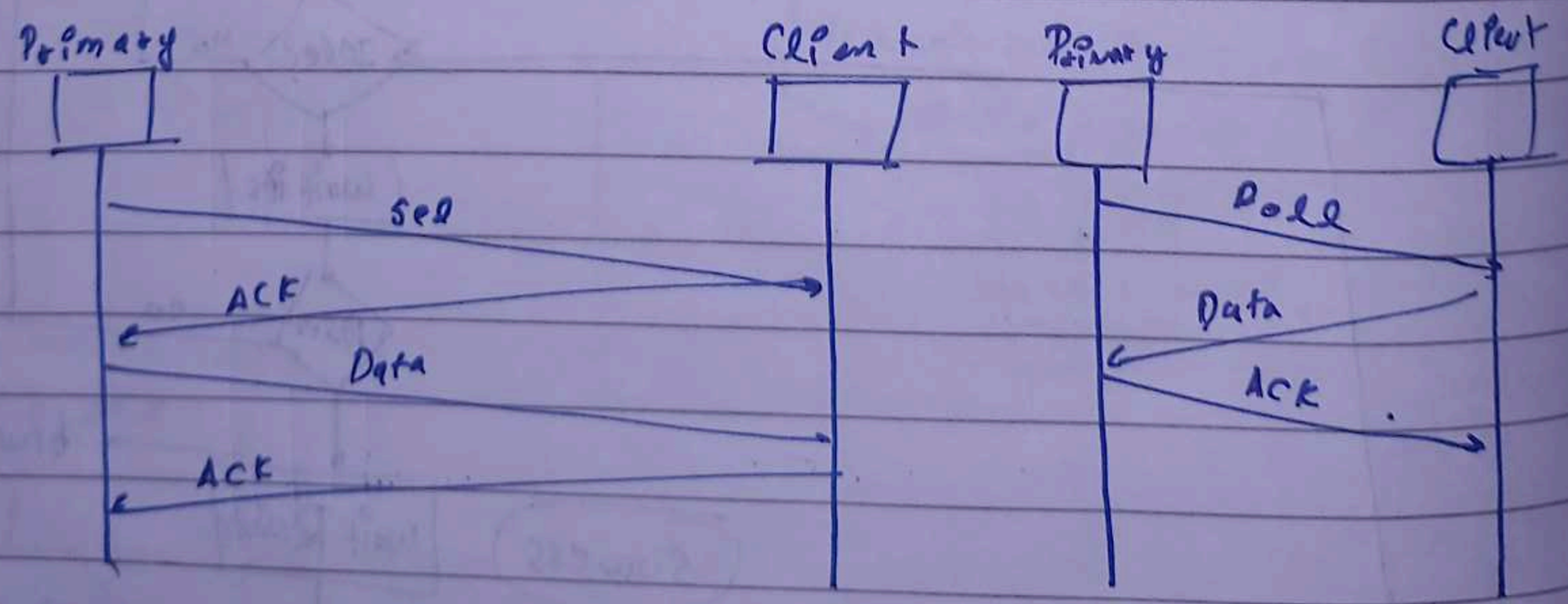
Type of network protocol that allow only one station to transmit data on a shared medium at a time.

Reservation Protocols

Stations reserve a slot of time in which to transmit data. This is done by sending reservation message to other stations. Once reservation is there, it can transmit data without any worries of collisions.

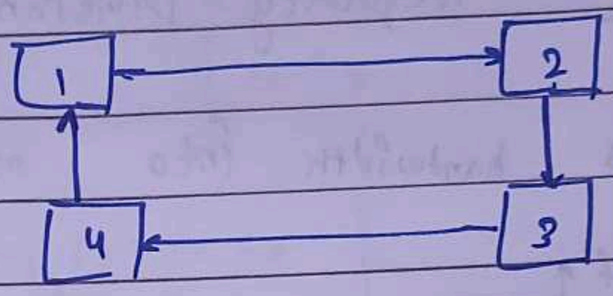


Polling There is a central station that polls all of the other stations on the network to see if they have any data to transmit. If a station has data to transmit, the central station allows it to do so.

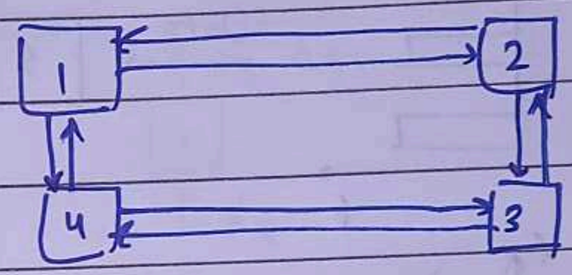


Token Passing. A special token is passed around. The station that has the token can pass data around. When station is finished transmitting, token is passed.

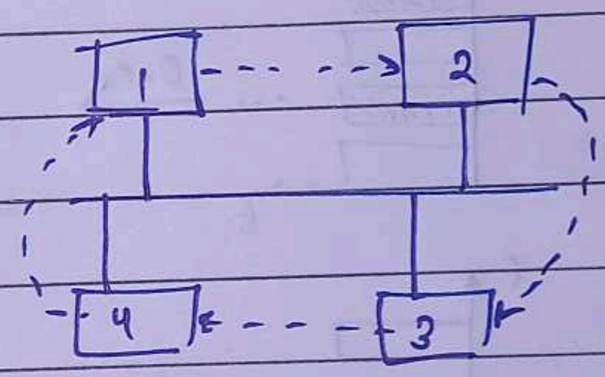
a) Physical ring



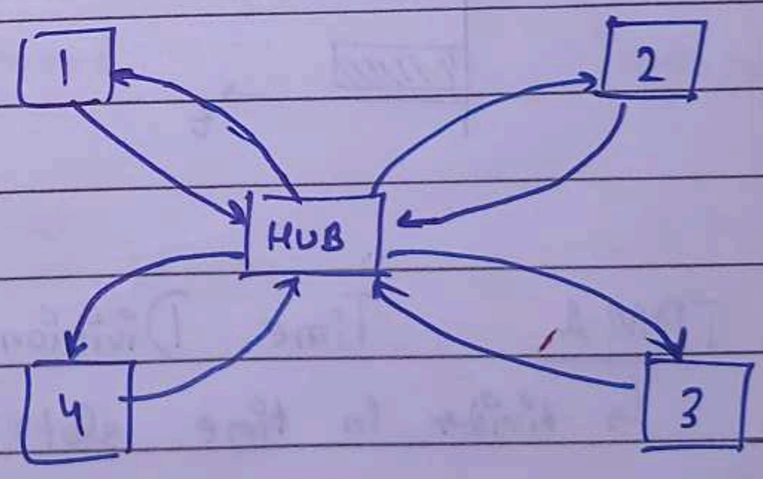
b) Dual ring



c) Bus ring



d) Star ring



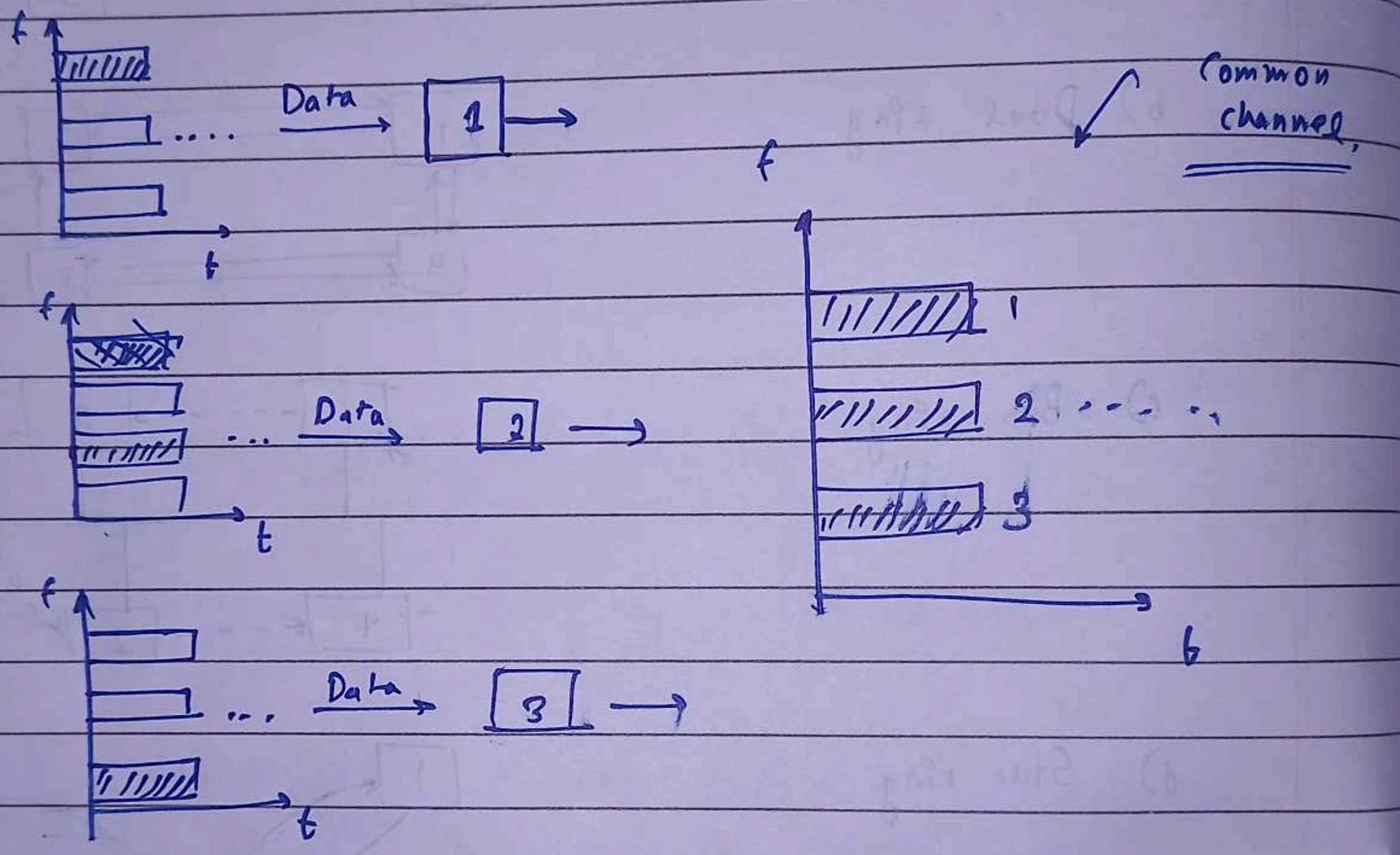
Piggybacking attaching additional information to an existing packet. This is done to avoid sending additional packets.

most commonly used technique.

Channelization protocols → type of network protocol that divides the available bandwidth into multiple channels.

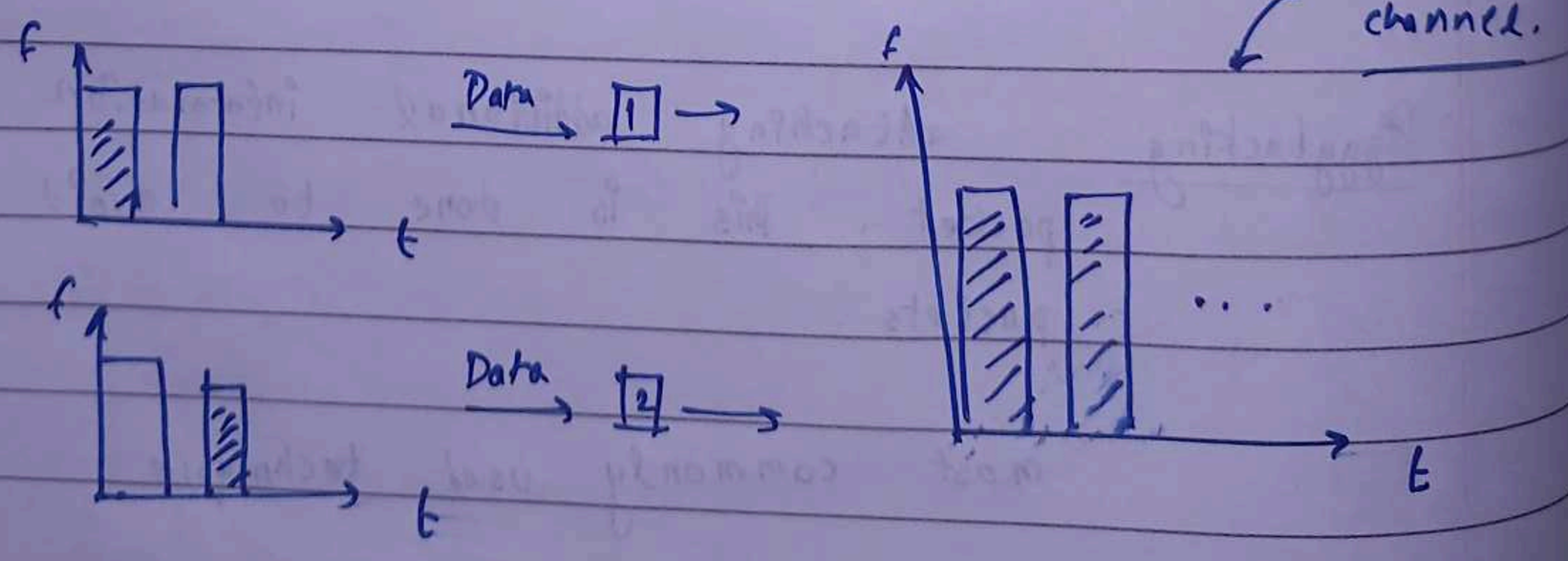
FDMA Frequency Division Multiple Access.

divides bandwidth into multiple frequency bands.



TDMA Time Division Multiple Access

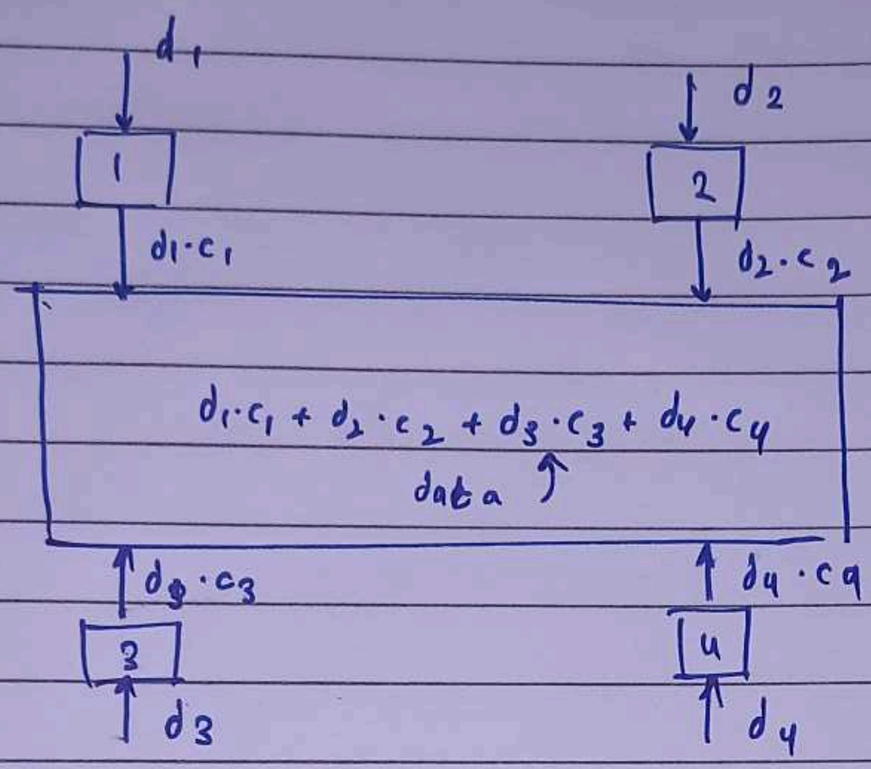
↳ divide in time slots.



Data bit 0 is -1
Data bit 1 is +1

Silence → 0

CDMA — Code Division Multiple Access.



FDMA	TDMA	CDMA
frequency	time	code
no codeword	no codeword	Codeword needed
Synchronization is not req.	Required	Not required
Rat of data is low	Medium	High
Continuous Signal	Bursts	Digital
little flexible	moderate	highly

Harshpreet Singh
2210990393 G-9

For any doubts contact
→ 9056299166
lol.

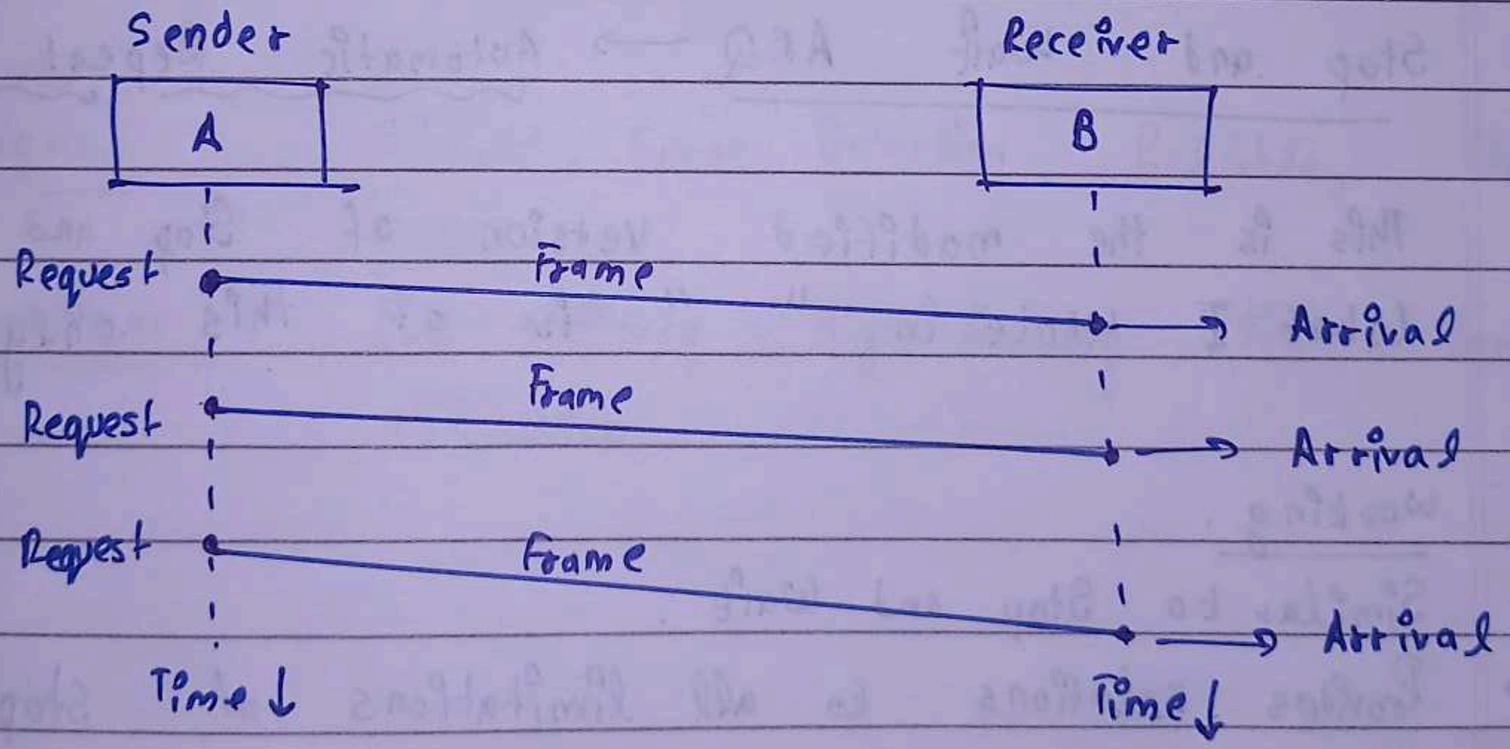
CN-ST-2.

Data link layer
Protocol.

- Noiseless channels
  - Simplest
  - Stop & Wait
- Noisy Channels
  - Stop & Wait ARQ
  - Go-Back-N ARQ
  - Selective Repeat ARQ.

Noiseless Channels. → ideal channel, no frames are lost, duplicated or corrupted.

Simplest. →
- Does not have any error or flow control.
- Unidirectional protocol where data frames are travelling in one direction from sender to receiver.

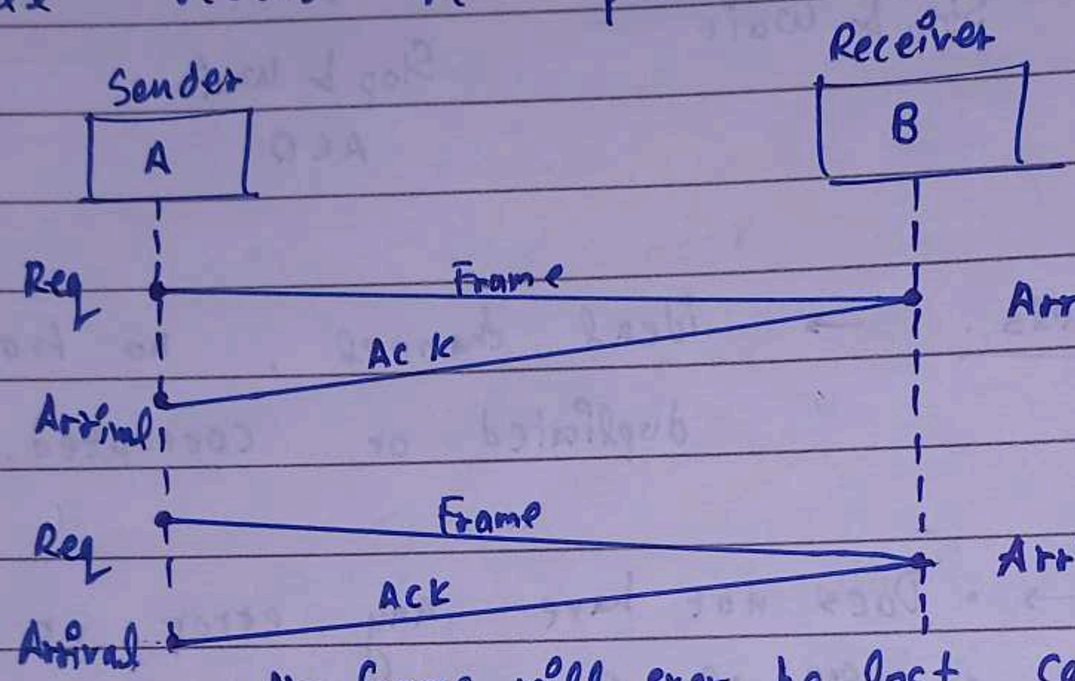


That's it.

Sender data ko bhej raha hai aur Receiver data ko accept karra hai.

Stop and Wait Protocol.

It basically works by having the sender send a single packet of data at a time, and the wait for an acknowledgement (Ack) from the receiver before sending another packet. If the sender does not receive an Ack within a certain time period, it will resend the packet." → for ARQ in Noisy



No frame will ever be lost cause "Noiseless"

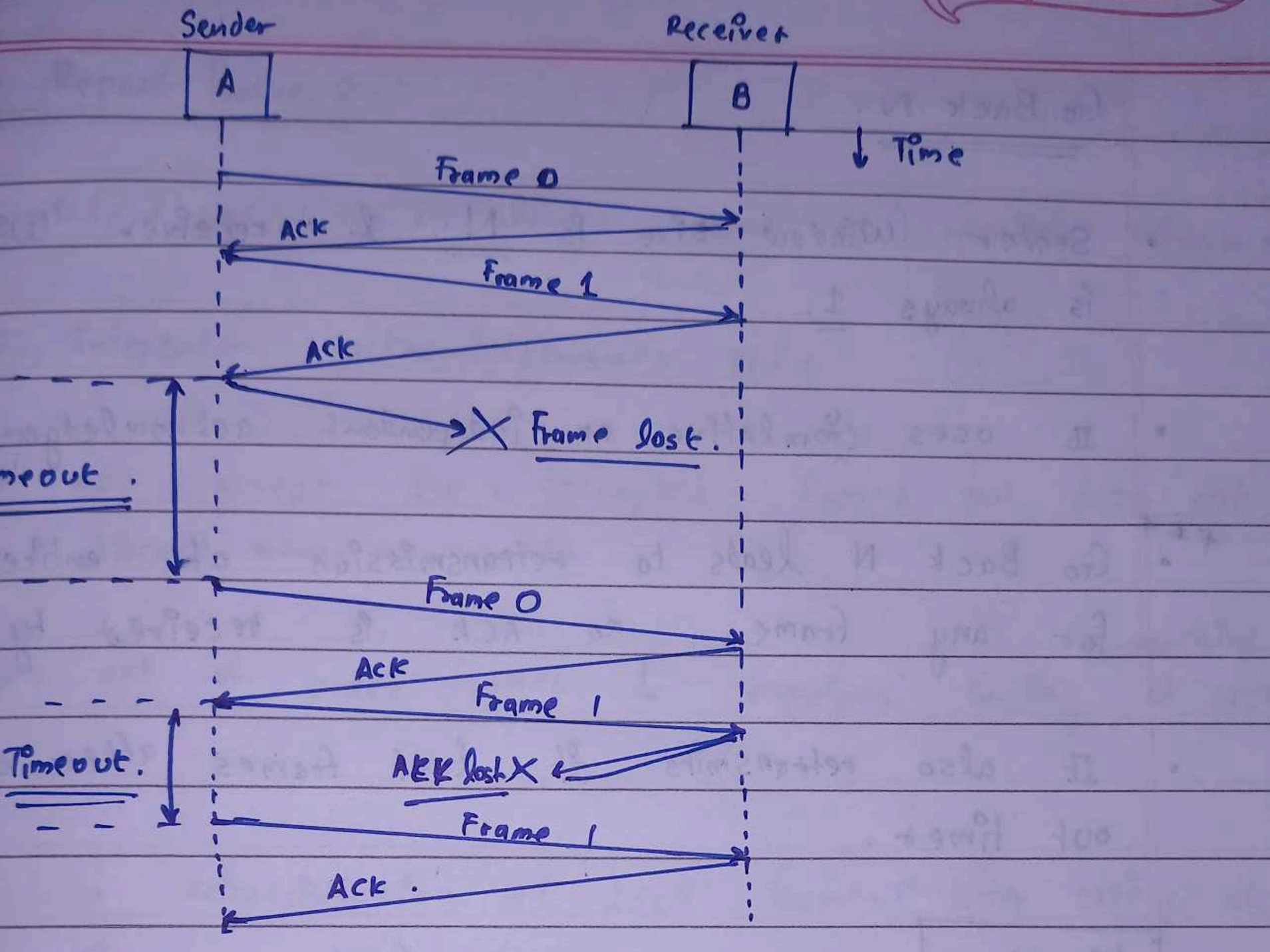
Stop and Wait ARQ → Automatic Repeat Request.

This is the modified version of Stop and Wait. What I wrote in " " is of this only.

Working.

- Similar to Stop and Wait.
- Provides solutions to all limitations of stop & wait.
- Introduces → time out timer
  - ↳ sequence numbers for data packets
  - ↳ sequence numbers for ack.

Minimum number of sequence nos. req. in Stop & wait is 2 which are 0 & 1.



Stop & Wait ARQ has very less effeciancy but it can be improved by improving the window size. For the sliding window protocols like Go Back N & Selective Repeat protols were introduced.

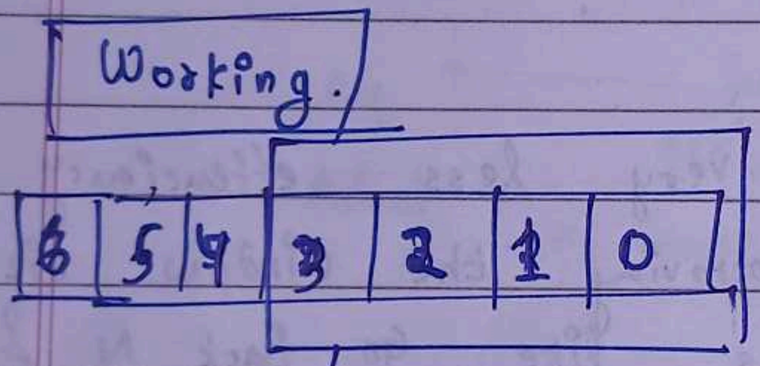
Advantages. Simple, Error Detection, Reliable, Flow control.

Disadvantages. low effiency, High latency, limited bandwidth utilization.

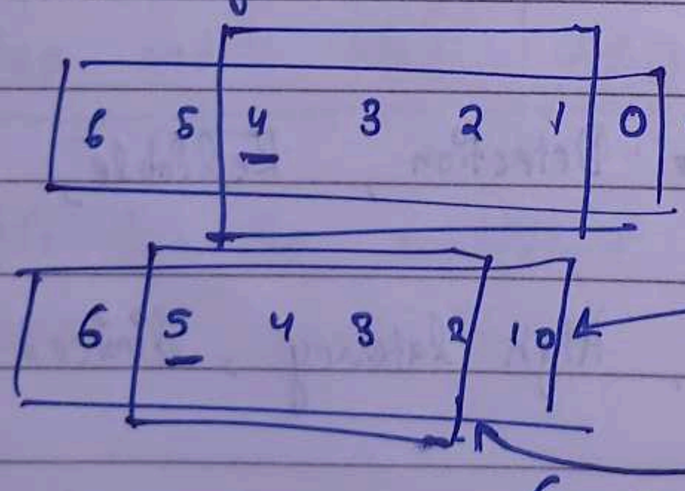
Go Back N.

- Sender window size is N. & receiver window size is always 1.
- It uses cumulative or independent acknowledgements.
- *** Go Back N leads to retransmission of entire window if for any frame, no ACK is received by sender. ***
- It also retransmits of lost frames after expiry of time out timer.

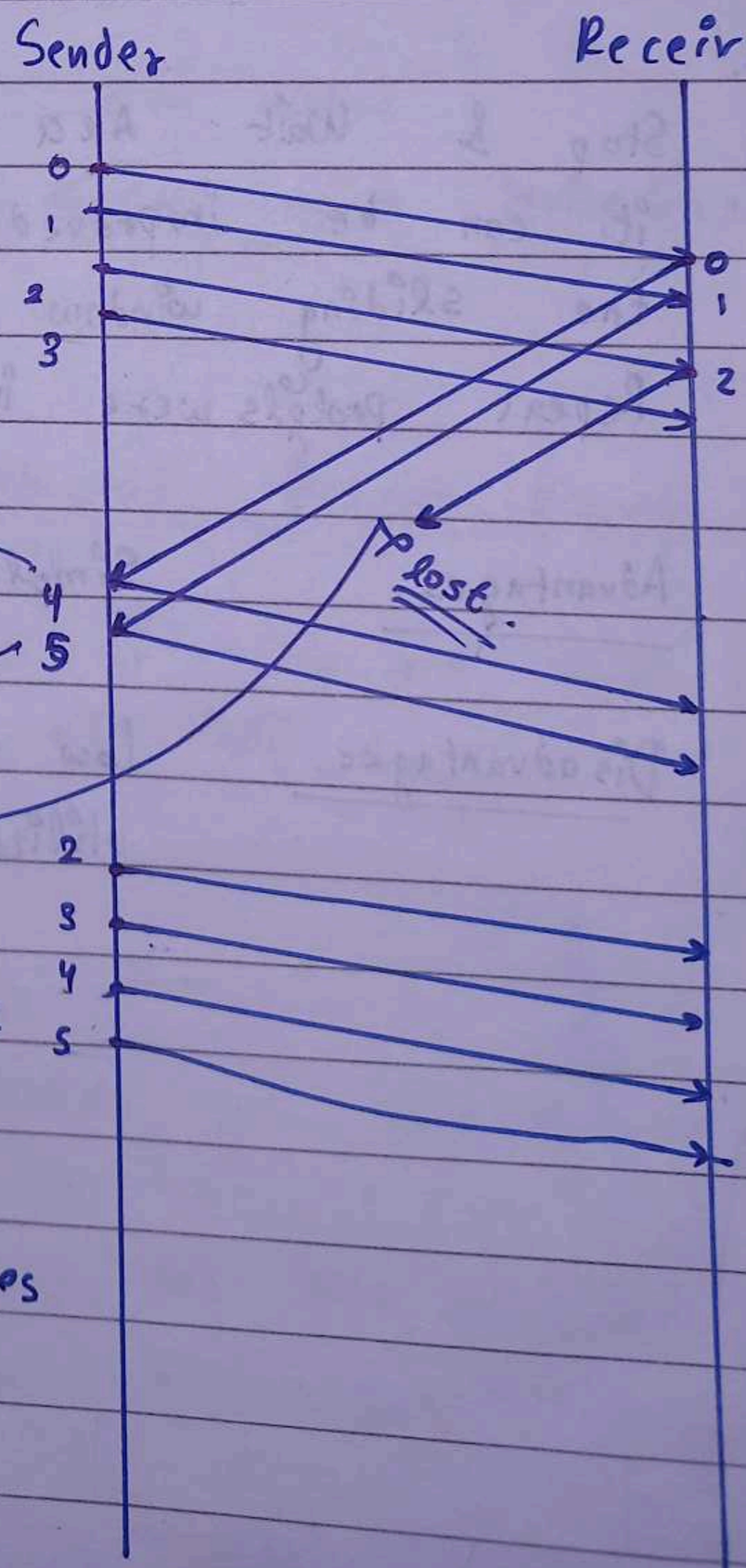
Working.



Sliding window of size: 4



Go Back to 2



- Go Back N does not consider corrupted or out of order frames and simply discards them.

Selective Repeat Protocol.

- Sender window size is always same as received window size.
- It used independent acknowledgements only.
- It does not accept the corrupted frames but does not silently discard them as well.
- It accepts out of order frames & requires sorting & searching at receiver's side.
- It leads to retransmission of lost frames after expiry of time out timer.

Working.

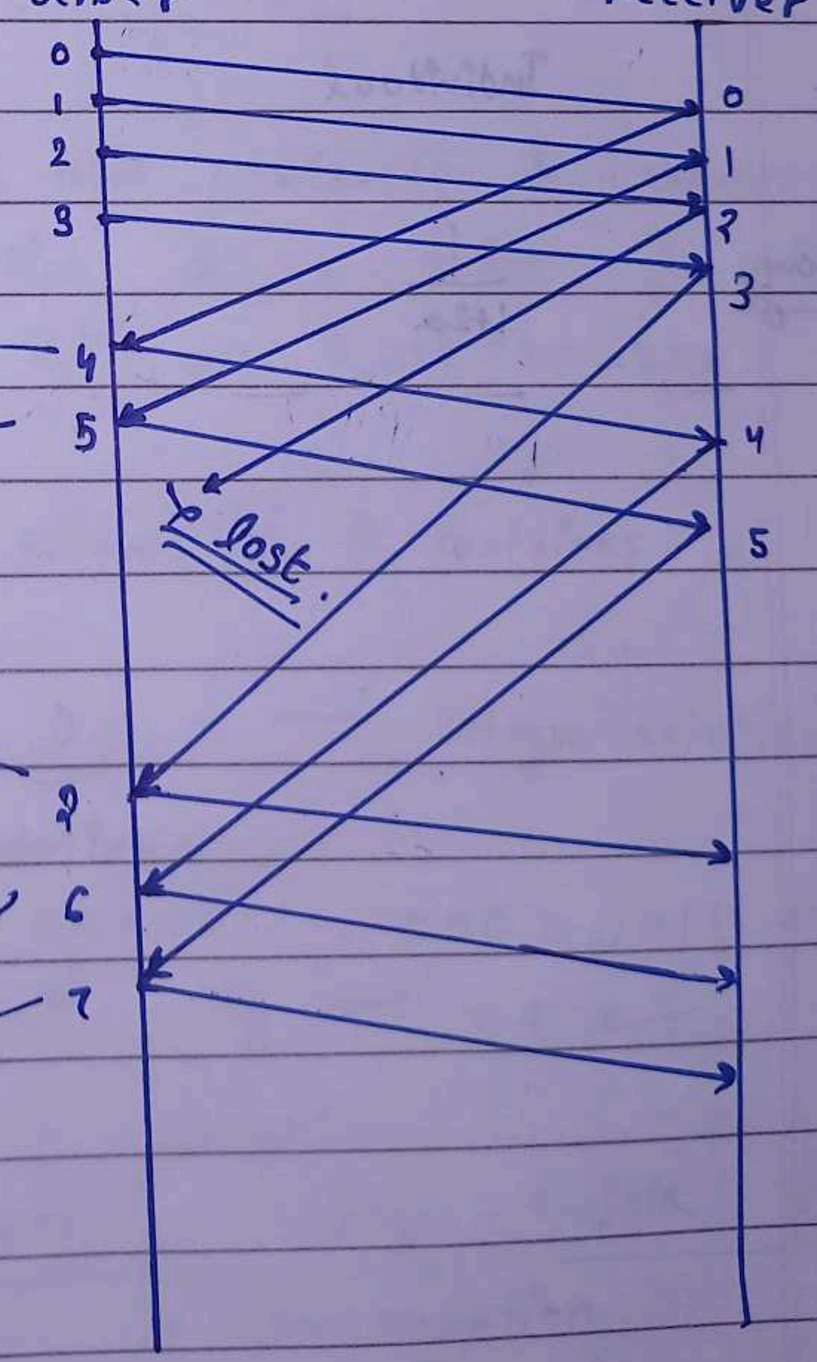
7 6 5 4 | 3 2 1 0

7 6 5 | 4 3 2 1 | 0 &

7 6 | 5 4 3 2 | 1 0 &

7 | 6 5 4 3 | 2 1 0 &

| 7 | 6 5 4 3 | 2 1 0 &



k = no. of. bits.

Differences b/w Stop & wait ARQ, Go Back N, Selective Repeat.

	Stop & Wait ARQ	Go Back N	Selective Repeat
Frames	Single	Multiple	Multiple
Sender window size	1	N $(2^k - 1)$	N (2^{k-1})
Receiver window size	1	1	N 2^{k-1}
Complexity	Least	Moderate	very Complex
Order	Inorder	Inorder	Out of order
Ack.	Individual	Cummulative	Individual.
Efficiency	$\frac{1}{1+2a}$	$N/(1+2*a)$	$N/(1+2*a)$

Concept of IP Packets & Addresses.

An IP address represents a unique address that distinguishes any device on the internet or any network from another. IP defines the set of commands directing the setup of data transferred through the internet or any other local networks.

IP Addresses / ~~Packets~~.

- **Private** → within your own network
- **Public** → What the massive network sees your device as
  - **Dynamic** → changes frequently
  - **Static** → Constant

---

* **IPv4** * → * 32 bits * address that uniquely & universally defines the connection of a device to the internet.

↳ The address space is 2^{32} or 4,294,967,296.

IPv4 address has 4 octaves / 4 containes of 8 bits.

like, 192 . 168 . 1 . 0 → Integer notation

→ Binary notation.

like, 10000000 . 00001011 . 00000011 . 00011111

128 . 11 . 3 . 31

Packet Structure, | IP Header | Layer - 4 Data |

IP Encapsulation.

↓

Payload.

IPv4 format / Datagram. → 20 bytes – 60 bytes.

Version (4)	HLEN (4)	Type of Service (8)	Total length (16)
Identification (16)	Flag (3)	Fragment Offset (13)	
TTL (8)	Protocol (8)	Header Checksum (16)	
Source IP (32)			
Destination IP (32)			
Option 0 - 40 bytes			

Datagram → Header (20-60 bytes) + Payload (0-65515 bytes)

Version → tells the version IPv4.

HLEN → IP header length (4 bits), which is the number of 32 bit words in the header. Min – 5, Max – 15.

Type of Service. → Low delay, high throughput, reliability.

DHCP

↓ Differentiated Service code point

ECN – Explicit congestion network.

Total length → Length of header + Data (16 bits), which has a minimum value of 20 bytes & maximum of 65535 bytes.

Identification → Unique packet id for identifying group of fragments.

Flags → 3 flags of 1 bit each: reserved bit, do not fragment flag, more fragments flag.

Fragment offset → represents the no. of data bytes ahead of the particular fragment in the particular datagram.

Time to live → Datagram's lifetime. Prevents the datagram to loop through the network by restricting the no. of Hops taken by a Packet before delivering to Destination.

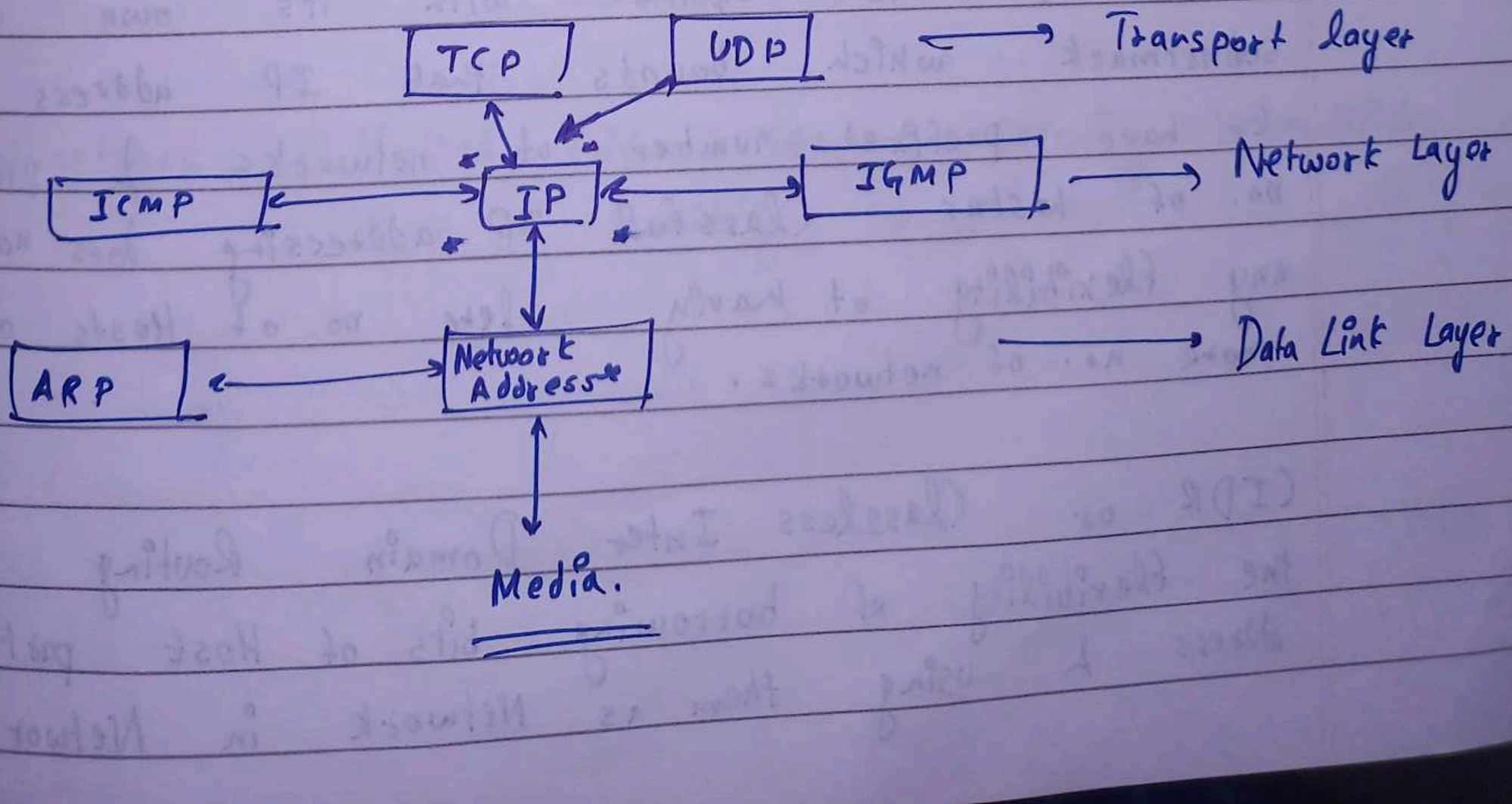
Protocol. → Which Protocol is used. Eg ICMP → 1 TCP → 6 UDP → 17.

Header Checksum → For checking errors.

Option → such as source route, record route .... more.

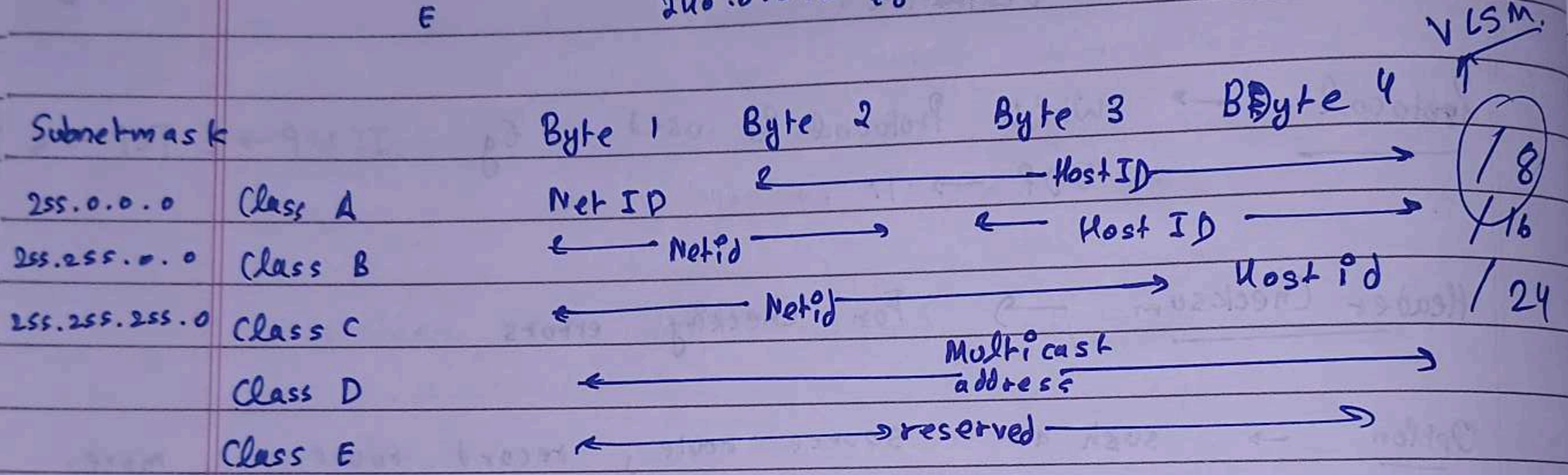
IPv4 Addressing.

- Unicast — one to one
- Broadcast — one to all
- Multicast — one to many.



IPv4 address classes.

Class	Range
A	1.0.0.0 to 127.255.255.255
B	128.0.0.0 to 191.255.255.255
C	192.0.0.0 to 223.255.255.255
D	224.0.0.0 to 239.255.255.255
E	240.0.0.0 to 255.255.255.255



Subnet mask → it is a 32 bit IP address that contains information about its network & host. It is used to distinguish b/w Classes.

Subnetting.

Each IP class is ~~required~~ equipped with its own default subnetmask which bounds that IP address class to have prefixed number of networks & prefixed no. of hostas. Classful IP addressing does not provide any flexibility of having less no. of Hosts of more no. of networks.

CIDR or Classless Inter Domain Routing provides the flexibility of borrowing bits of Host part of IP address & using them as Network in Network, called

Subnet.

IPv4 - VLSM.

Variable Length Subnet Mask is a techique used to design & create subnets with different subnet masks.

This increases efficiency by using smaller subnet masks for subnets with fewer hosts and larger subnet masks for subnets with more hosts.

Reserved IP addresses.

		SNM
Private IP →	10.0.0.0 — 10.255.255.255	255.0.0.0
	172.16.0.0 — 172.31.255.255	255.240.0.0
	192.168.0.0 — 192.168.255.255	255.255.0.0
LoopBack IP →	127.0.0.0 — 127.255.255.255	
Link Local →	169.254.0.0 — 169.254.255.255	

Advantages of IPv4. Widespread Adoption, Simple & Efficient, Robust, Backwards Compatible.

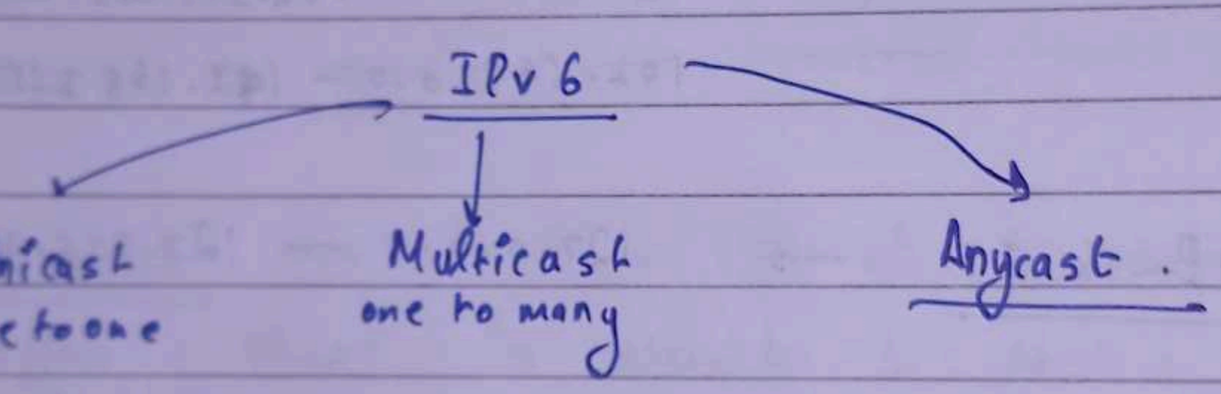
Disadvantages. Limited Address Space, Security Vulnerabilities, Lack of mobility support.

IPv6 → It is a new addressing protocol designed to incorporate whole sort of requirement of future internet known as Internet Version 2.

The problem with IPv4 was its 32 bit length which has about 4.3 billion devices but we have much more devices than 4.3 billion.

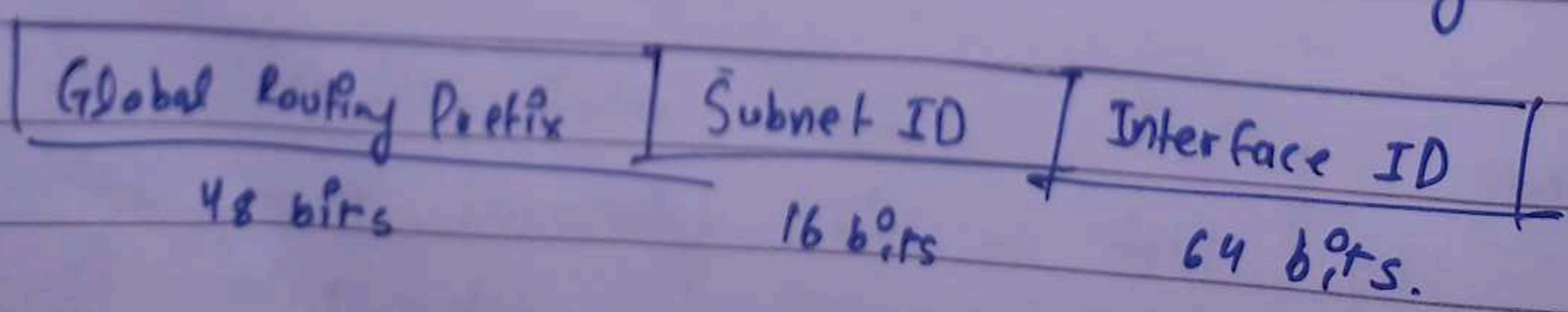
Adv:

IPv6 has a large address space 3.4×10^{38} using a 128 bit address. It has a simplified header. End to End, Connectivity, Auto-configuration, Faster forwarding / routing. IPsec, No broadcast but Anycast support, mobility, extensibility, enhanced priority support.



In Anycast multiple hosts are assigned same Anycast IP address. When a host wishes to communicate with a host equipped with an anycast ip address, it sends a unicast message. With the help of complex routing algorithms, that unicast message is delivered to the host closest to the sender.

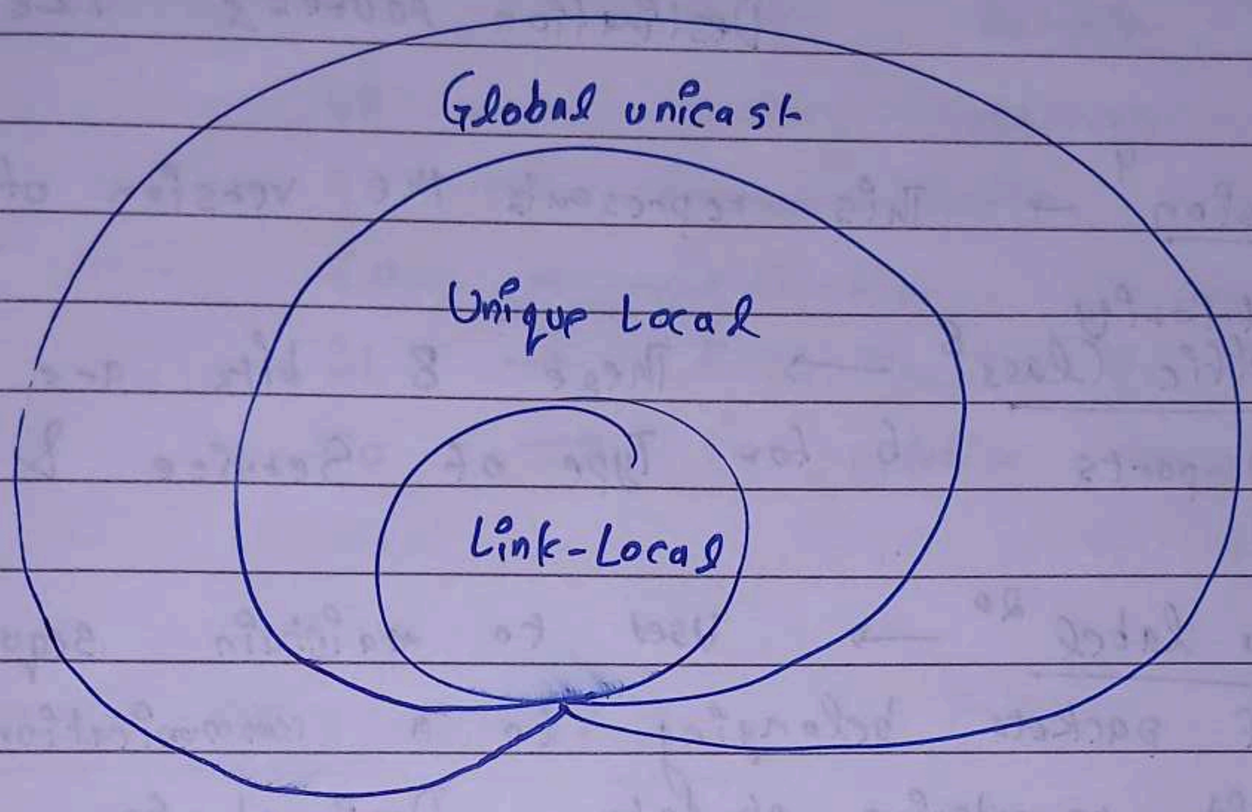
IPv6 addresses are made of 128 bits into eight 16-bits blocks. Each block is converted to a 4 digit Hexa decimal.



Auto configured IPv6 addresses are known as Link-Local address.

Unique-Local address is a IPv6 address which is unique globally but used locally.

Scope :-



Special addresses.

::/128 → Unspecified Address
::/0 → Default Route
::1/128 → Loopback Address.

Reserved for Multicast address for routing protocols.

FF02::5 → OSPFv3
FF02::6 → OSPFv3 Designated Routers
FF02::9 → RIPng
FF02::A → EIGRP

IPv6 Header.

Ver 4 Priority 8 Flow Lable 20
Payload leyth 8 Next Header 8 Hop Limit 16
Source Address 128
Destination Address 128

Version 4 → This represents the version of IP (0110).

Traffic Class 8 → These 8 bits are divided into 2 parts 6 for Type of Service & 2 for ECN.

Flow label 20 → Used to maintain sequential flow of packets belonging to a communication. Helps to avoid re-ordering of data. Designed for streaming/real-time media.

Payload Length. 16 → This field is used to ~~indicate~~ either the type of Extension tell routers how much information this packet contains in its payload. Payload is composed extension headers and Upper layer data.

Next Header 8 → this field is used to ~~tell which~~ indicate either the type of extension header, or if extension header is not present then it indicates the Upper layer PDU.

Hop Limit :- 8 → This field is used to stop packets to loop in the network infinitely. This is same as TTL in IPv4.

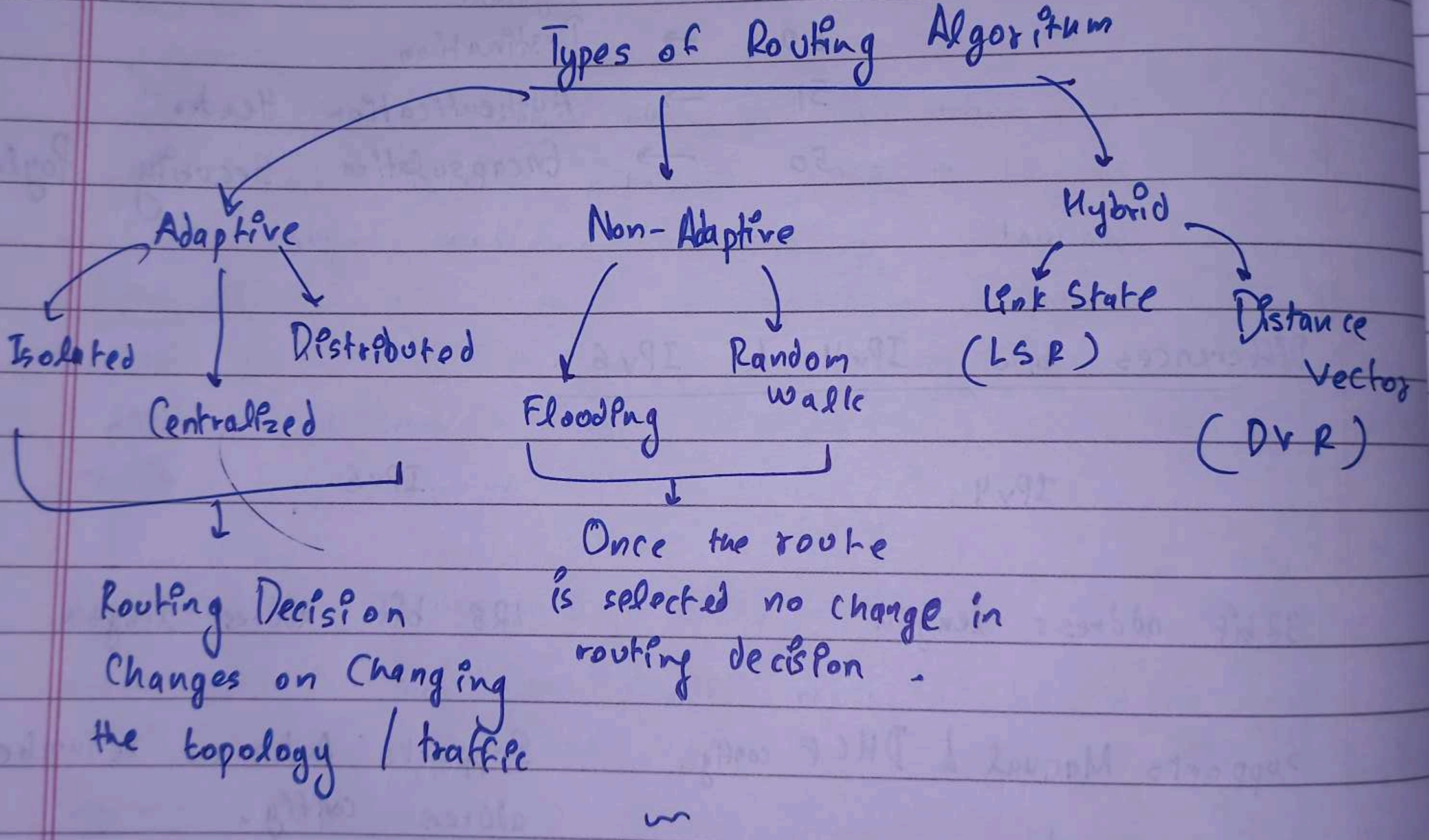
Extensions
0 → Hop by Hop to read all network devices in transit.
43 → Routing Header
44 → Fragment Header
60 → Destination
51 → Authentication Header
50 → Encapsulation Security Payload head

Differences b/w IPv4 & IPv6.

IPv4	IPv6
32 bit address length	128 bit address length
Supports Manual & DHCP config.	Supports Auto and renumbering address config.
4.29×10^9 Address space	3.4×10^{38} address space
Security is dependent on Application	IPSEC is an inbuilt security feature.
Decimal	Hexadecimal.
Fragmentation by Sender & Receiver routers.	Only by sender
20 - 60 bytes header	40 bytes header
Classes (A, B, C, D, E)	No classes.

Routing.

Is the process of establishing the routes that data packets must follow to reach the destination. In this process, a routing table is created which contains information regarding the routes that data packets must follow.



Uses bellman-ford algorithm. Distance measured in Hops.

Each routing Table is known as Vector

Distance Vector Routing. → iterative, asynchronous and distributed

→ **Knowledge about the whole network:** Each router shares its knowledge through the entire network. The router sends its collected knowledge about the network to its neighbors.

→ **Routing only to neighbours:** The router sends it knowledge about the network to only those routers which have direct links. The router sends whatever it has about the network through the ports. The information is received by the router & uses, ~~the information~~ to update its own routing table.

→ **Information sharing at regular intervals:** Within 30 seconds, the routers sends the information to the neighbouring routers.

Advantages → easy to configure & maintain than link state routing.

Disadvantages →
- Slower than link state
- Can just count to infinity
- Creates more traffic
- large routing tables lead to congestion.
- Good news spread fast & bad news spread slowly
- Persistent looping problem.

Link State Routing { OSPF, IS-IS, [redacted] }

It is a dynamic routing algorithm in which ~~Each~~ each router shares knowledge of its neighbours with every other router in the network.

- A router sends its information about its neighbours only to all the routers through flooding.
- Each router in the network maintains a map of entire network topology which is created by flooding Link State Advertisements (LSAs) throughout the network
- Each router then uses Dijkstra's Algorithms to calculate the shortest route.

Problems
- Heavy traffic due to flooding of packets
- Flooding can result in infinite looping which can be solved using TTL field.

RIPv1, IGRP DVR	OSPF, ISIS LSR
• Bandwidth required is less due to local sharing, small packets & no flooding	• Bandwidth required is more due to flooding and sending of large like state packets.
• Based on local knowledge, since it updates table based on info. from neighbours.	• Based on global knowledge, it have knowledge about entire network
• Bellman Ford	• Dijakstra's Algorithm.
• Traffic is less.	• More
• Converges slowly	• Faster
• Count of infinity problem	• No count to infinity.
• Persistent loop.	• Transient loops

Difference b/w Routing & Flooding.

Routing	Flooding
Routing table is required.	No routing table.
May give the shortest path.	Always gives the shortest path.
Less Reliable.	More reliable.
Traffic is less.	Traffic is more / high.
No duplicate packets.	Duplicate packets present.

Connectionless

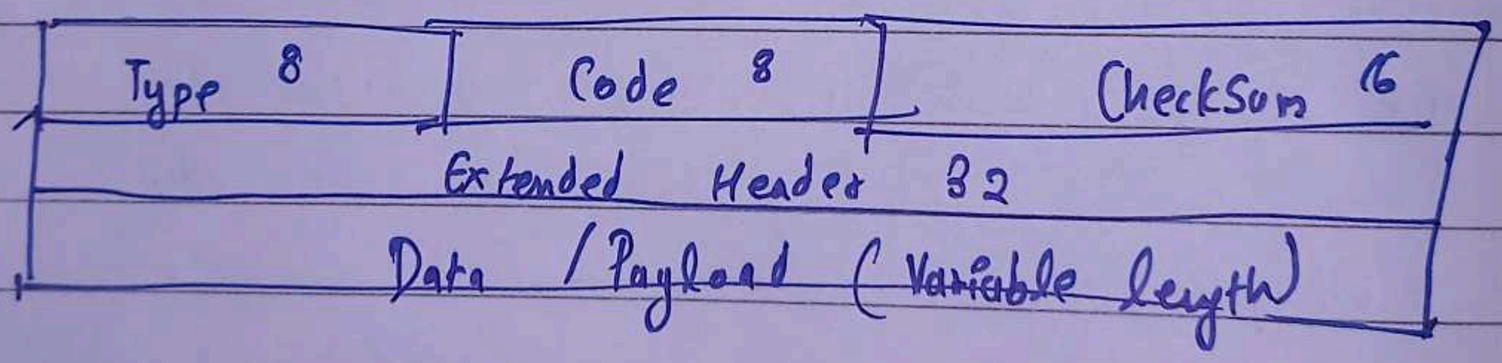
ICMP (Internet Control Message Protocol)

It is a network layer protocol used to diagnose communication errors by performing an error control mechanism. Since IP doesnot have an inbuilt mechanism for sending error and control messages. It depends on ICMP for error control.

- Reporting errors
- Management queries.

Another important use of ICMP protocol is used to perform network diagnosis by making use of traceroute and ping utility.

ICMP Packet format.



Type (8-bit) → is for message type.

0 → Echo reply
3 - Destination unreachable
5 - Redirect message
8 - Echo request
11 - TLE
12 - Parameter problem.

Code (8-bit) → additional info. about the error message and type.

Checksum (16-bit) → Error handling.

IGMP (Internet Group Management Protocol)

used by hosts and adjacent routers for multicasting communication with IP networks and uses the resources efficiently to transmit the message / data packets.

Key roles of IGMP:

- Manage multicast membership
- Notify routers of group membership changes
- Reduce Network traffic

Used in video streaming, gaming and web conferencing.

IGMP is built for IPv4. we use Multicast Listener Discovery Protocol (MLDP) to make IPv6 eligible for multicasting.

IGMPv1 Allows all the supporting hosts to join the multicast groups using membership request and include some basic features. But the host cannot leave the group on their own and have to wait for a timeout to leave the group.

IGMPv2 revised version of IGMPv1. It has added functionality of leaving the multicast group using group membership.

IGMPv3 . revised version of v2 with added source-specific multicast and membership report aggression.

Transition from IPv4 to IPv6

In the current scenario, the IPv4 address is exhausted and IPv6 had come to overcome the limit.

Various organizations are currently working with IPv4 tech and in one day we can't switch directly from IPv4 to IPv6. Instead of only using IPv6, we use combination of both and transition means not replacing IPv4 but co-existing from both.

IPv6 & IPv4 can't communicate with each other

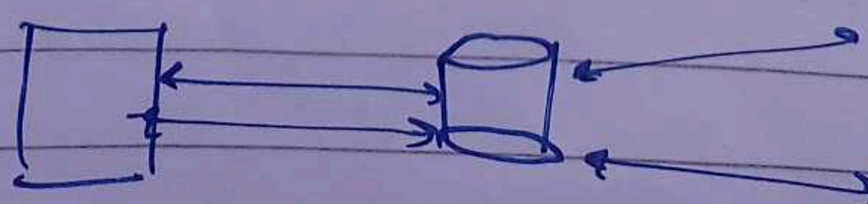
There are 3 main methods for translating b/w IPv4 & IPv6.

1. Dual Stack Transition

This method involves running both IPv4 & IPv6 simultaneously on network interfaces, devices & routers. This allows for coexistance of both.

Advantages → smooth transition, relatively simple to implement.

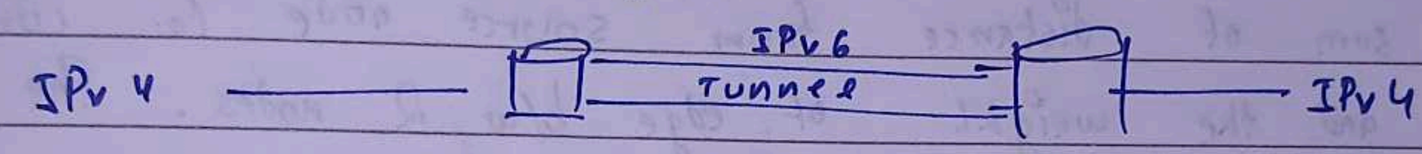
Disadvantages → increases the complexity of network configuration



2. Tunneling.

Encapsulates an IPv6 packet within IPv4 packet. This allows IPv6 devices to communicate with IPv4 devices over an IPv4 network.

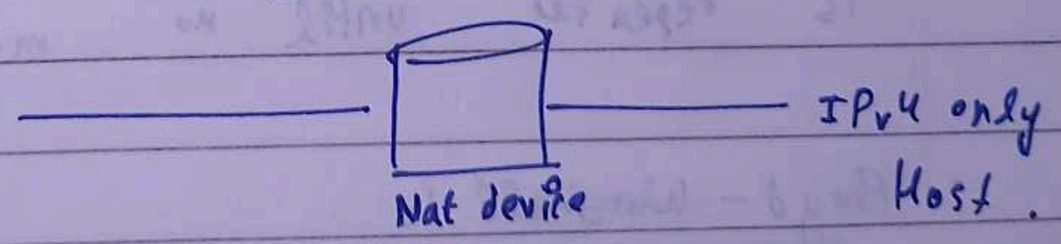
Adv → Enables IPv6 to be sent over IPv4.
Dis → Adds an extra layer of overhead.



3. NAT 64 / DNS 64

Allows IPv6 only devices to communicate with IPv4 by translating IPv6 to IPv4 (Network Address Translation), while DNS64 synthesizes AAAA (IPv6) records for domains that only have IPv4 (A) records.

Adv → In IPv6 only & IPv4 only devices it can be used.
Dis → Complexity in managing translations.



Shortest Path Algorithms

- **Dijkstra's**

Works by maintaining the list of all nodes that have been visited and their distances from source node. It starts with source node & adds its neighbours to the list. It calculates distance to neighbour which is the sum of distance from source node to current node and the weight of edge b/w 2 nodes. The neighbour with shortest distance & this process continues till each visited node.

- **Bellman - Ford's**

Works similarly to Dijkstra's, but it can handle graphs with negative edge weights. It does this by repeatedly relaxing the distance of all nodes, which means updating the distance to a node if it is possible to reach the node with a shorter path. This process is repeated until no more distances can be relaxed.

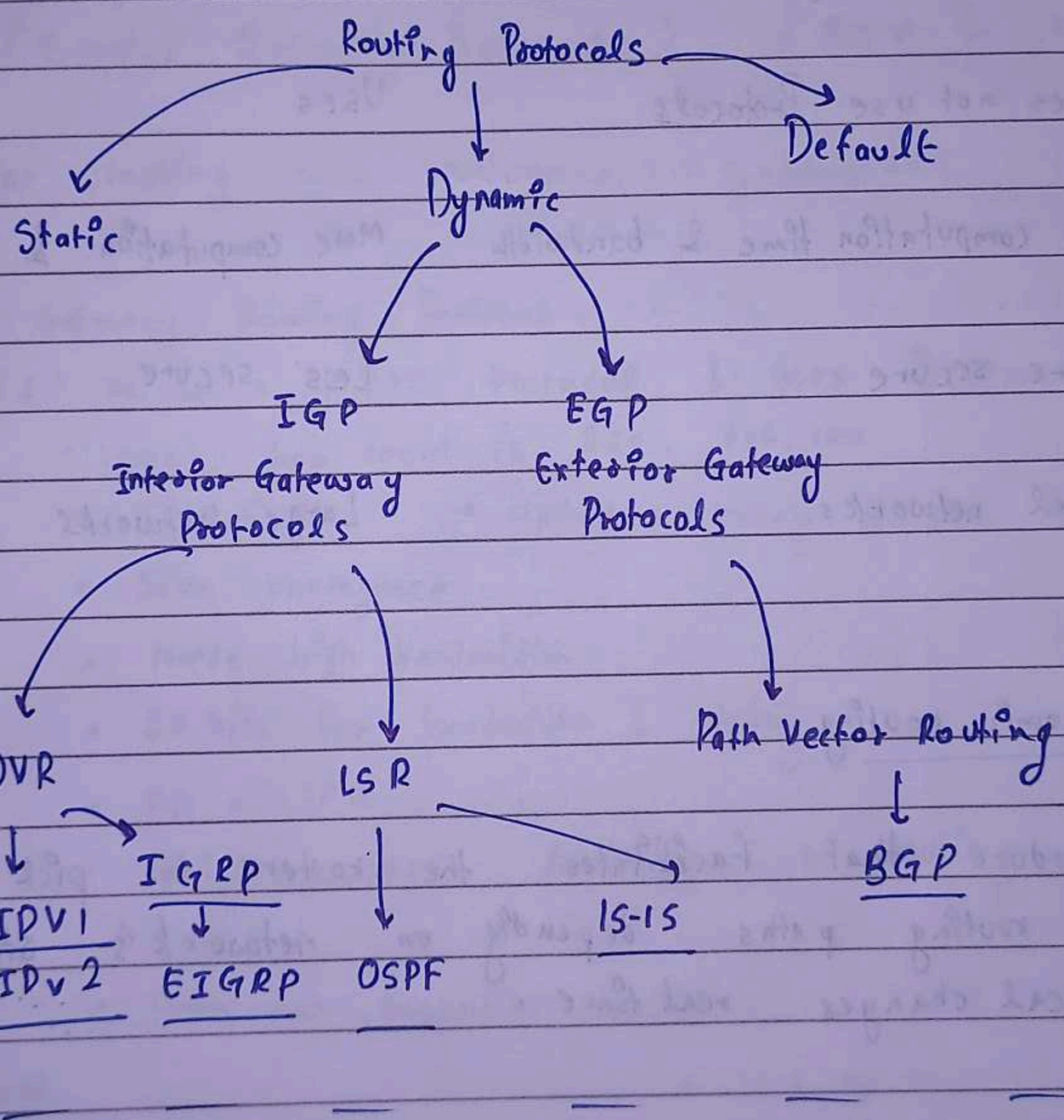
- **Floyd - Warshall's**

Most complex of 3. It can find the shortest paths b/w all pairs of nodes in a graph. It works by creating a pair of nodes. The distances in the table are then updated iteratively until they no longer change.

classmate
Date
Page

Routing Protocols.

Routing is the process of selecting best / optimal path amoung various options.



Static	Dynamic
One preconfigured route	Multiple routes
Manual reconfiguration	Automatic update
Does not use Protocols	Uses
Less computation time & bandwidth	More computation & bandwidth.
More secure	Less secure
usage: Smol networks	Large networks

Dynamic routing

Procedure that facilitates the routers to pick & choose the routing paths dependg on network's structure & logical changes real time.

Autonomous System → group of routers under the contro of a single authority.

IGP (Interior Gateway Protocols) {RIP, EIGRP, OSPF}

Used for routing inside an autonomous system & used to route within the individual networks themselves.

EGP (Exterior Gateway Protocols) {BGP}

Used for routing b/w autonomous systems.

Interior Gateway Routing Protocol.

IGRP (I) is a DVR protocol & uses Bellman Ford.

- Least hop count is 255. Def. 100
- Classful
- Updates every 90 sec. 280 sec session.
- Slow convergence
- Needs high bandwidth
- 24 bits for bandwidth & delay
- AD of 100
- Only supports Multi-part of routing not tables or areas
- Does not support VLSM.

Enhanced
EIGRP (D) is a LSR protocol as well as a DVR protocol. "Hybrid".

- Classless
- Least hop count of 256.
- Fast convergence
- Uses DUAL (Diffusing Update Algorithm)
- Needs low bandwidth
- 32 bits for bandwidth & delay
- AD of 90.
- Neighbor tables and all used.
- VLSM supported.

EIGRP uses some messages to communicate w/t neighbouring devices.

- Hello : Alive / Connection messages every 5 sec.
- Update : To update table after hello message.
- Query : multicasted messeges when device is declared dead.
- Reply : Ack of Query message
- Ack : Acks are Hello packets that contain no data.

Neighbour is declared dead if it is not able to send Hell ~~messa~~ packet after 15 seconds.

RIP Routing Information Protocol.

Routers use it to interchange details on a network.
Its main functions are,

- To find out most effective way to route data
- To avoid routing loops.

The main advantage of RIP is that it's bandwidth utilization is very high and it updates every 30 sec. However, a max of 16 routers can be composed as it supports only 15 max hop counts.
Confluence rate is low.

- Dosen't always get the fastest path.
- Generates lobs of network traffic.

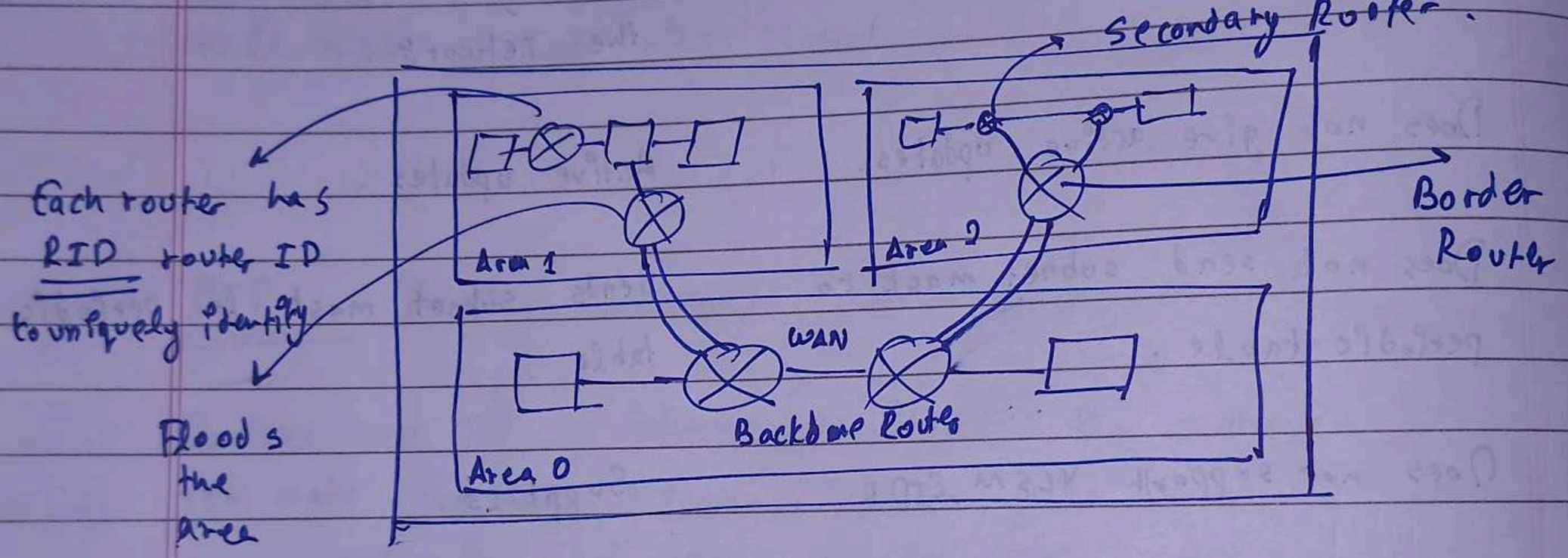
RIPv 1	RIPv 2
DvR	DvR
Classful	Classful & Classless
Hop count limit 15	15
For updates, it uses broadcast traffic.	For updates it uses Multicast traffic, however that weakens the network.
Does not give active updates.	Active updates.
Does not send subnet mask to periodic table.	Sends subnet mask to periodic table.
Does not support VLSM, CIDR	Supports.
Does not support manual route summarization	Supports.
No authentication.	Supports Authentication.

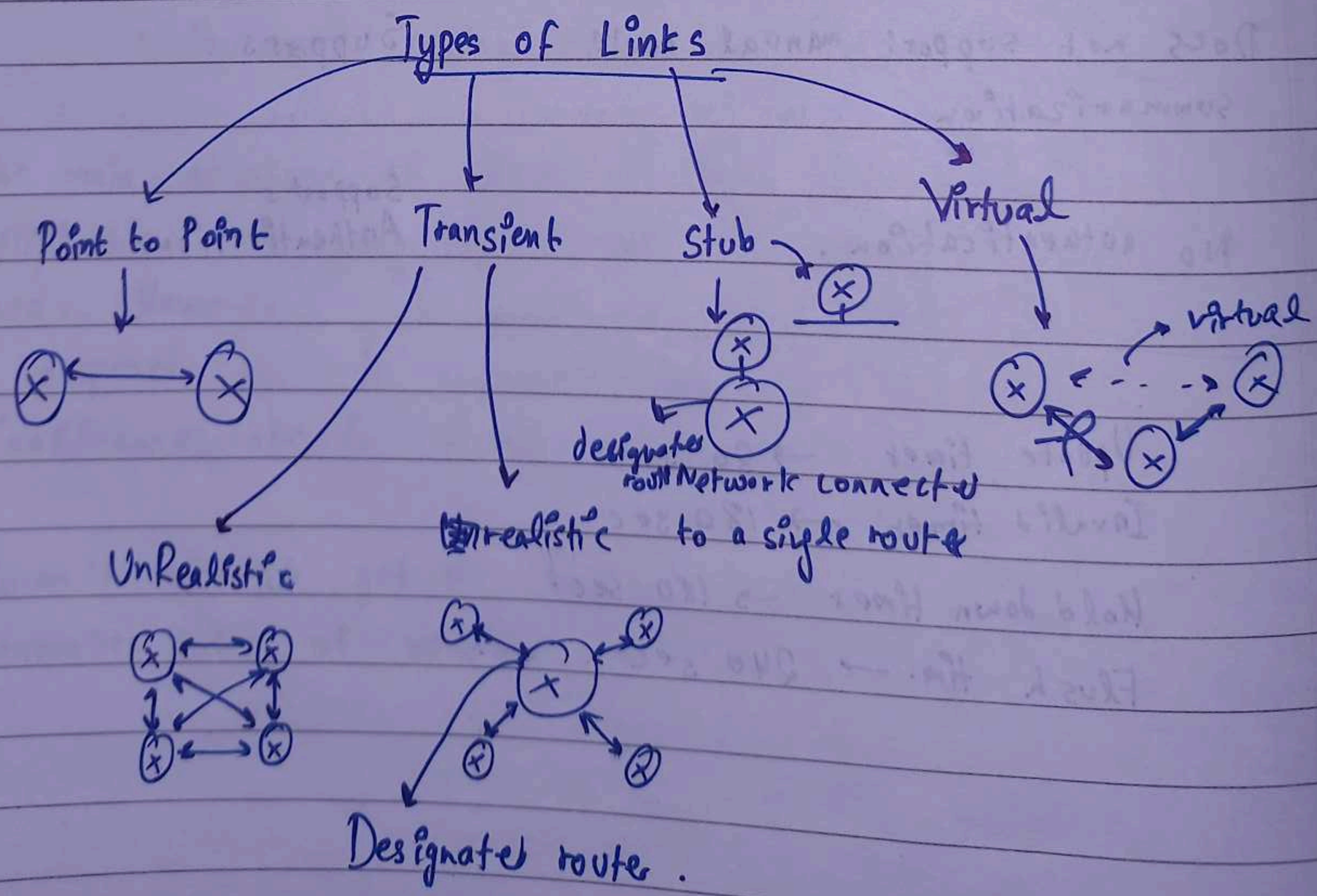
Update timer → 30 sec.
Invalid timer → 180 sec.
Hold down timer → 180 sec.
Flush tim. → 240 sec.

OSPF (Open Shortest Path First)

- Intradomain (IGP) used to exchange routing info...
- it is LSR, i.e, that routers exchange info. about their links with each other.
- used SPF (Shortest Path first) (Dijkastra's)
- Supports multiple paths to a destination.

OSPF areas. → divides Autonomous Systems int areas.
collection of networks, hosts and routers.





Message format.

Version 8 | Type 8 | Message 16

Source Ip

Address Identification → area within routing happening.

Check sum | Auth. Type → 0 / 1 (0 ↓ no auth, 1 ↓ yes)

Authentication 32

Types of message in OSPF.

Hello : To establish connection. ~~5 secs~~

Database Description : after connecting, if the neighbour system is communicating with the system first time, it sends info. about network topology.

Link State Request : when one router wants to find info about the connecting router.

Link State Update : to advertise the state of its links.

Link State ACK : to ack for each link state update.

b/w AS — Internal ↑ ; with diff. AS. — External ↑

BGP (Border Gateway Protocol)

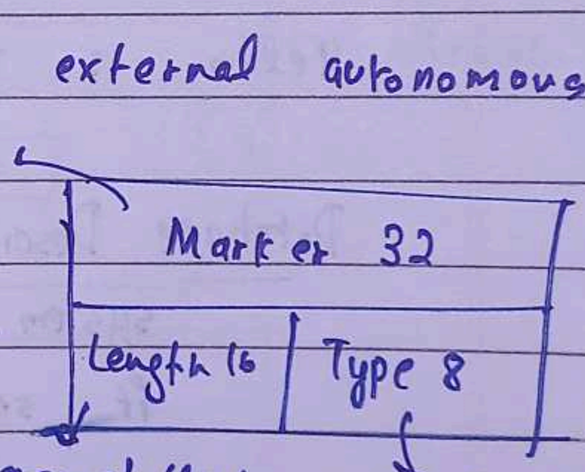
(EGP) b/w multiple AS,

an interdomain routing protocol using path vector routing.

Features of BGP

- Open standard
- EGP
- Supports internet
- Classless
- Incremental & trigger updates
- Path vector routing
- Configure neighbour relationship.
- Rich no. of attributes.
- AD is 20 for information coming from external autonomous system.

200 if from same autonomous system.



BGP chooses the best route based on the attributes of path.

Types of messages.

Open → to create neighbourhood relation

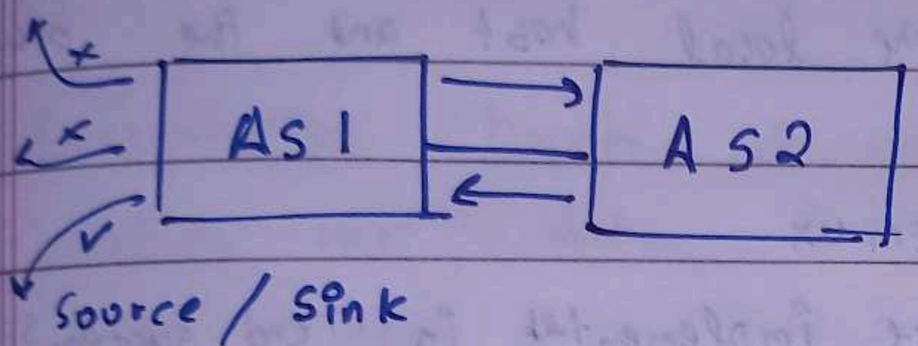
Update → withdraw the destination or announce new destinations..

keep alive → to check b/w neighbours ki jinda hai ki nhi.

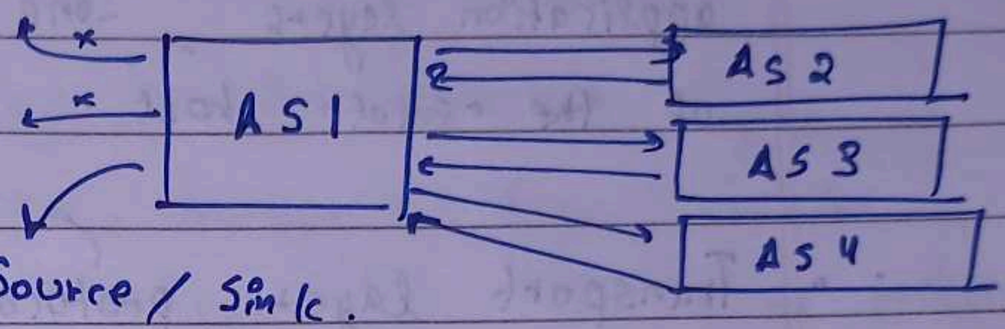
Notification → when route detects error.

Types of Autonomous Systems

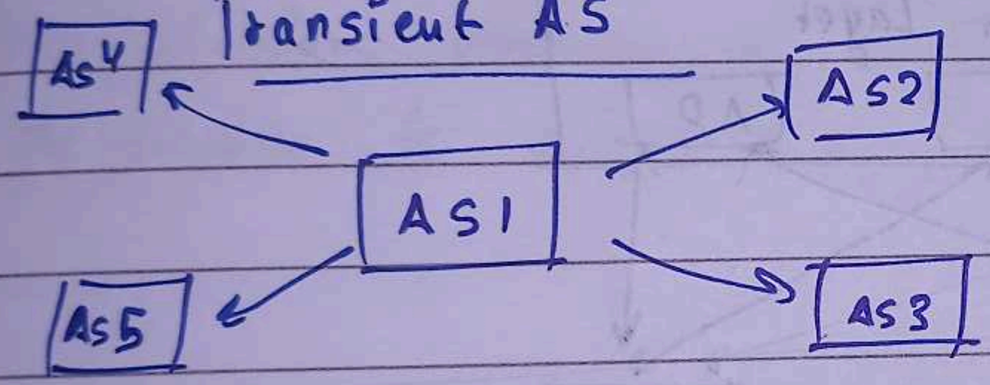
Stub AS



Multihomed AS

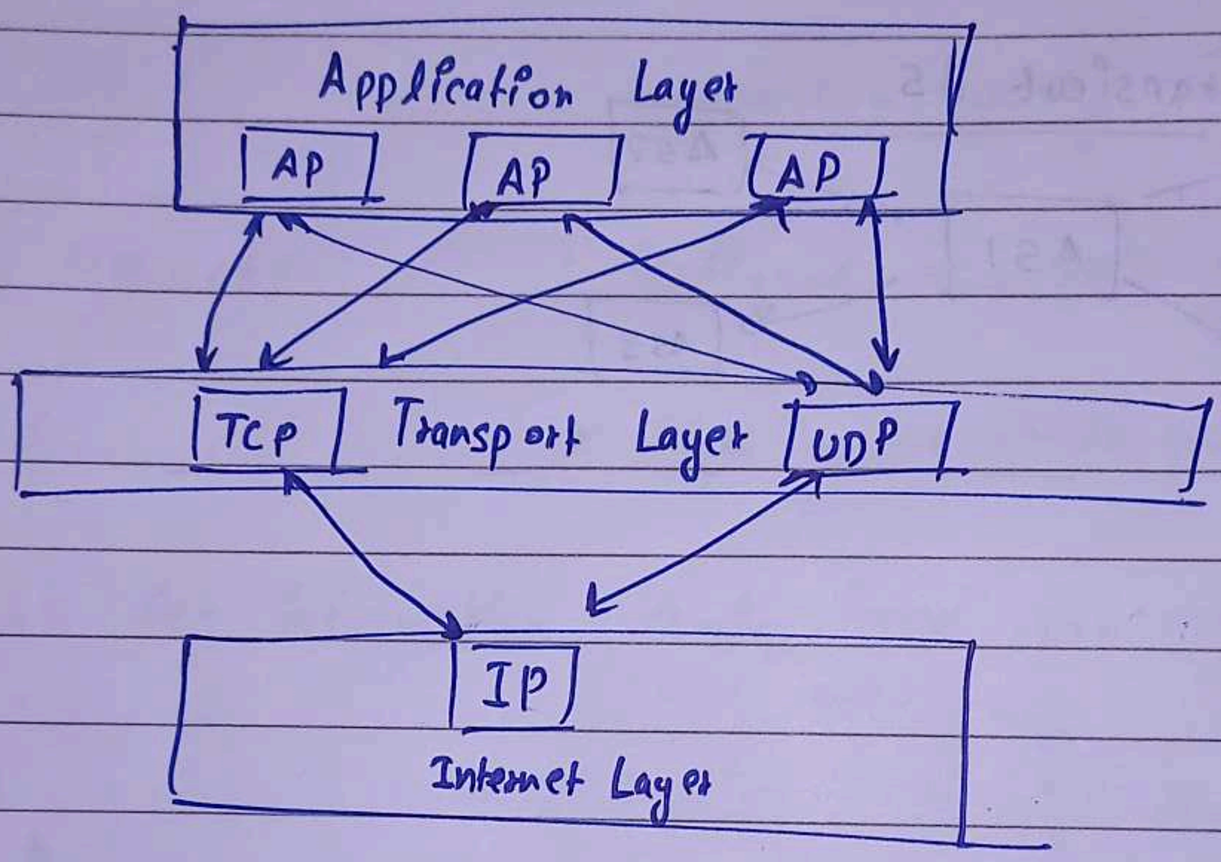


Transient AS



Transport Layer (Heart of OSI)

- Provides a process to process communication b/w 2 application layers, one at the local host and the other at the remote host.
- Transport layer protocols are implemented in end systems. (mux & demux)



Transport Layer Services.

* Process to Process Communication *
or End to End Delivery

Transport Layer is responsible for delivery of message using appropriate processes.

* Addressing. *

- Client serve paradigm to achieve PtPC.
- For communication, the local host and the remote host are defined using IP addresses. To defind processes

port numbers are used.

Encapsulation & Decapsulation

Encapsulation happens at sender site. The transport layer receives the data & adds the transport-layer header. Packets here are called user datagrams, segments or packets.

Decapsulation happens at receivers end. When messages comes, the header is dropped.

Flow Control → balance b/w production & consumption.

Error Control.

To detect & disregard corrupted packets, keeping track of discarded packets & resending. Recognizing duplicate packets. Buffering out of order packets.

Congestion Control → to reduce the load.

2 types of Service

Connection less

- The source process divides message into size acceptable chunks. They are sent one by one with encapsulation.
- Chunks handed over to protocol in order.
- But since there is no dependency b/w packets & transport layer, packets may arrive out of order.

Connection oriented.

- First establish a logical connection & then data exchange will occur.
- End to End service.
- Flow control, Error Control, Congestion control all implemented.

	Connection Oriented	Connection less
Connection Estb.	Yes	No
Gurantee of Delivery	Guaranteed	No
Flow control	Yes	No
Error control	Yes	No
Complexity	More	Less
Overhead	More	less
Sequence	Sequential	No
Resource Allocation	Yes	No
Retransmission	Yes	No
Congestion	Less	More

Transmis

- Full

Flags

Date
Page

Transport Layer Protocols

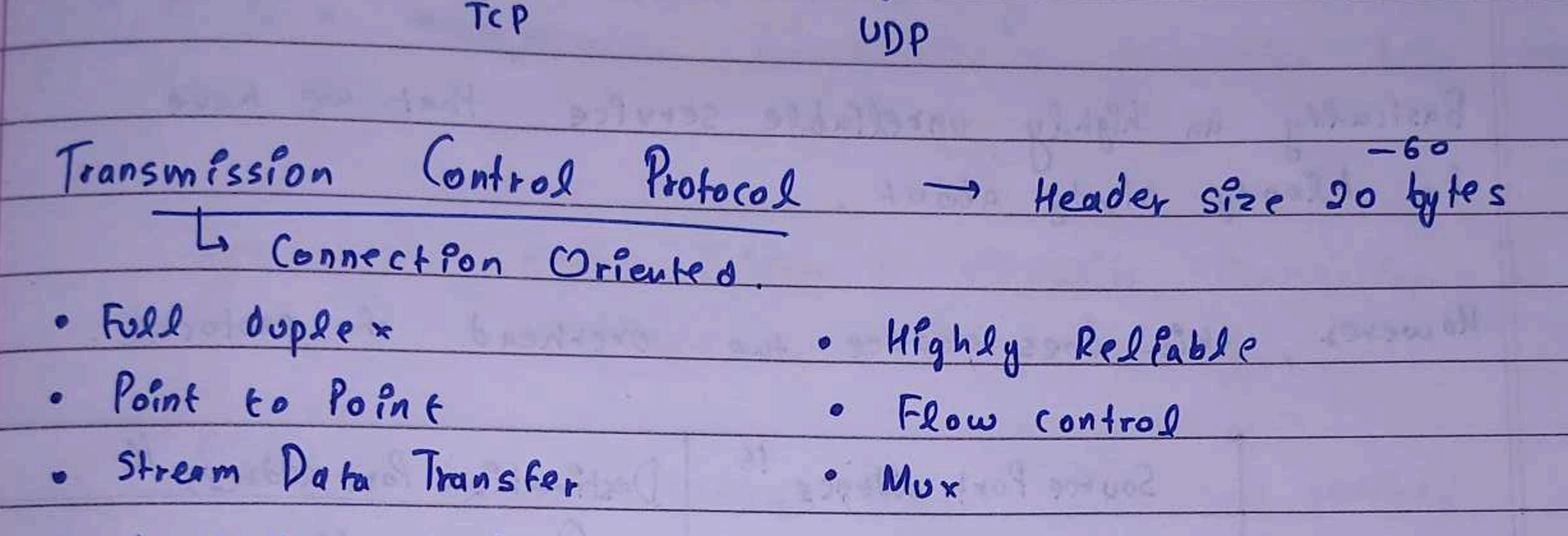
- TCP
- UDP

Transmission Control Protocol → Header size 20 ~~60~~ bytes

↳ Connection Oriented.

- Full duplex
- Point to Point
- Stream Data Transfer
- Logical connection
- Highly Reliable
- Flow control
- Mux

TCP segment format



Flags
URG → Urgent
Ack → Acknowledgement
PSH → Push
Rst → Reset
SYN → Synchronization sequance number
FIN → Finish

UDP (User Datagram Protocol) ⟶ 8 bytes

↳ Connectionless

Basically an highly unreliable service that we have to bloody study about.

However, this does reduce the overhead of protocol.

Source Port address 16	Destination Port address 16
Total length 16	Checksum 16
Data	

	TCP	UDP
Connection Oriented	Yes	No
Guaranteed Delivery	Yes	No
Error checking	Yes	only header
Flow	Yes	No
Overhead	large	small
use cases	File transfer, email	Streaming, VoIP, gaming
Speed	Slow	High

TCP

TCP 3 way Handshake

↳ it is a process before data sharing starts.
to establish a connection b/w server and client.

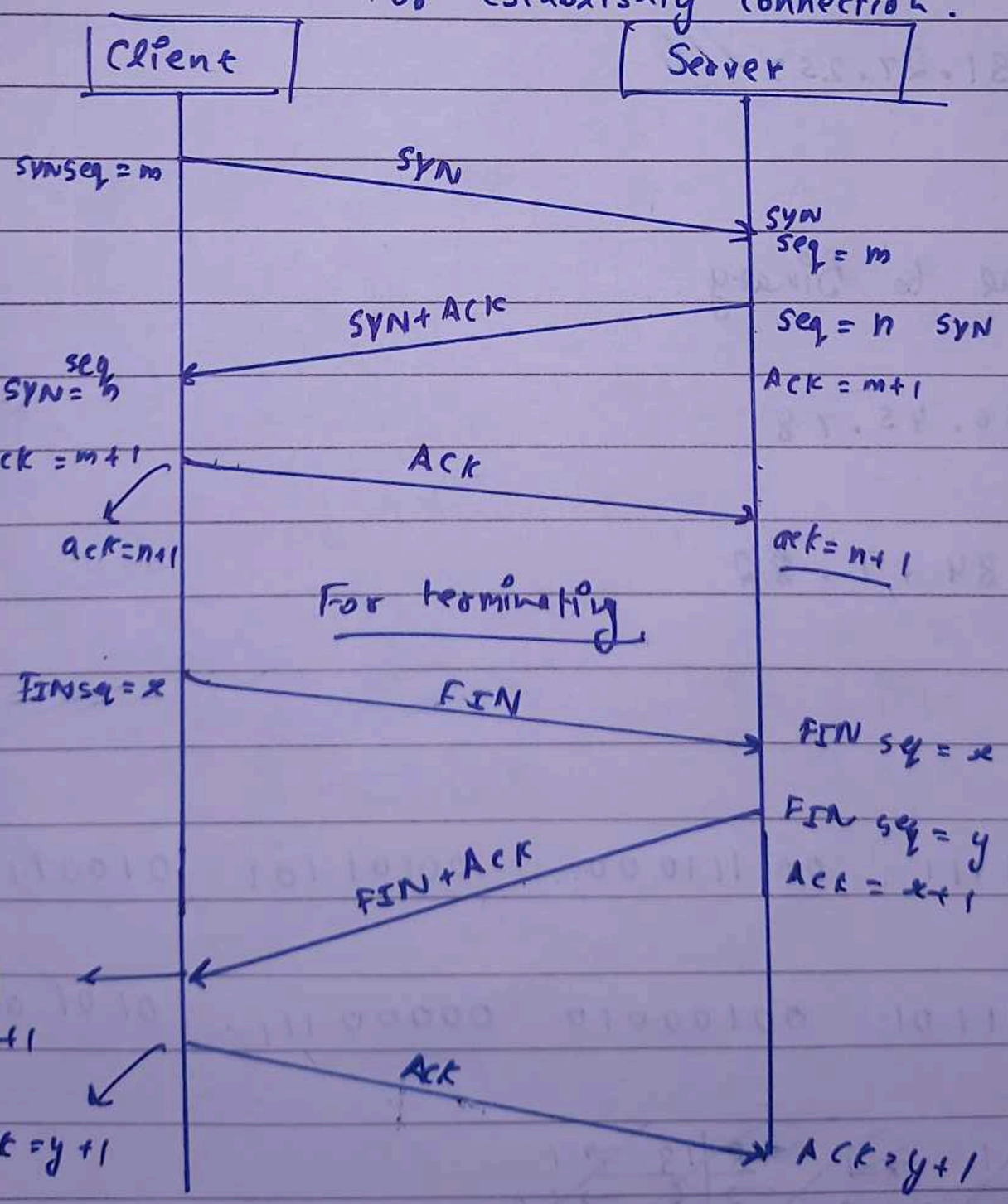
Syn → Helps initiate & establish connection & give sequence no. of syncronization.

Ack → To confirm. that Syn has reached

Syn-Ack → Syn message from local address device & Ack of earlier packet

FIN → to terminate

For establishing connection.



Some Numericals

Q. Binary to Decimal Notation.

a) 10000001 00001011 00001011 11101111

b) 11000001 10000011 00011011 11111111

Soln.

a) 129.11.11.239

10000001
2^7 ... 2^1 2^0

128 + 1 = 129

b) 193.131.27.255

Q. Decimal to Binary.

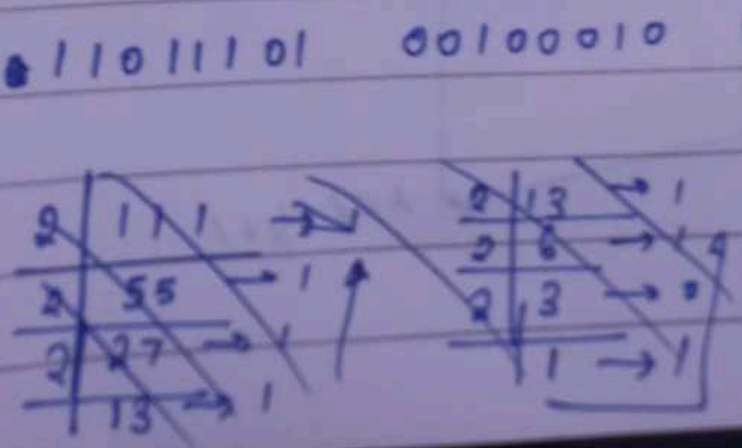
a) 111.56.45.78

b) 221.34.7.82

Soln.

a) 011 01111 00 111000 00101101 01001110

b) 11011101 00100010 00000111 01010010



How to convert eg :-

2	111
2	55 → 1
2	27 → 1
2	13 → 1
2	6 → 1
2	3 → 0
	1 → 1
	0

$(01101111)_2$

Q. Find error if any in following IPv4 addresses.

a) 111.56.045.78. → must be no leading 0's

b) 221.34.7.8.20 → more than 4 nos.

c) 75.45.301.14 → no. greater than equal to 255

d. 11100010.23.14.67 → mixture of binary & decimal.

Q. Find class of each address

a) 00000001 00001011 00001011 11101111 → A class

b) 10000001 10000011 00011011 11111111 → B class

c) A 14.23.120.8

d.) E 252.5.15.111

Class	No. of Blocks	Block Size	Application
A	128	16,777,216	Unicast
B	16,384	65,536	Unicast
C	2,097,152	256	Unicast
D	1	268,435,456	Multicast
E	1	268,435,456	Reserved.

To find first address in a block 32 - n bits to 0s.
To find last address in a block 32 - n bits to 1s.

Q. 205.16.37.39 / 28 What are a first & last block address.

a)

↑ 00100000

32 - 28 to 0s

Last address : 205.16.37.32

32 - 28 to 1s 00101111

First address = 205.16.37.47

Q. If bandwidth of the line is 1.5 mbps, RTT is 45msec and packet size is 1 kb. Find link utilization in Stop and wait.

Ans. (T_t) Transmission delay = Packet Size / Bandwidth

= 1 kB / 1.5 mbps

= 2^{10} x 8 bits / 1.5×10^6 bits/s

= 5.461 msec.

(T_p) Propagation Delay = RTT / 2

= 45 ms / 2

= 22.5 ms

$a = T_p / T_t$

= 22.5 msec / 5.461 msec

= 4.12.

Efficiency = $\frac{1}{1+2a} = \frac{1}{1+2\times4.12} = 10.8\%$

AI Noise-Cancelation
HDMI
WiFi 6 CERTIFIED
DOLBY ATMOS
Hi-Res AUDIO
GEFORCE RTX
*Actual product features and specifications may vary. Check with retailer for details.
15100-2389B000

classmate
Date
Page

Q. Consider the following subnet masks and tell their class and hosts per subnet.

a) 255.0.0.0 Class A no. of subnets = $2^0 = 1$
Net Host

b) 255.128.0.0 Class A. no. of host per subnet = $2^{23} - 2$
Netid = 9 Host = 23 no. of subnet = 9 - 8 = 1
$2^1 = 2$

c) 255.255.0.0 Class B no. of hosts per subnet = $2^{16} - 2$
Net = 16 Host = 16 no. of subnets = $\underbrace{2^8}_{\text{Class A}}$ or $\underbrace{2^0}_{\text{Class B}}$

d) 255.255.254.0 Class B Hosts = $2^9 - 2$
Host = 9 Net=23 no. of subnets = $\underbrace{2^{15}}_{\text{Class A}}$ or $\underbrace{2^7}_{\text{Class B}}$

Harshpreet Singh
2210990393

ST - 3
CN

For any doubts contact my classmate
helpline - 9056299166
5 AM to 10 PM working hours

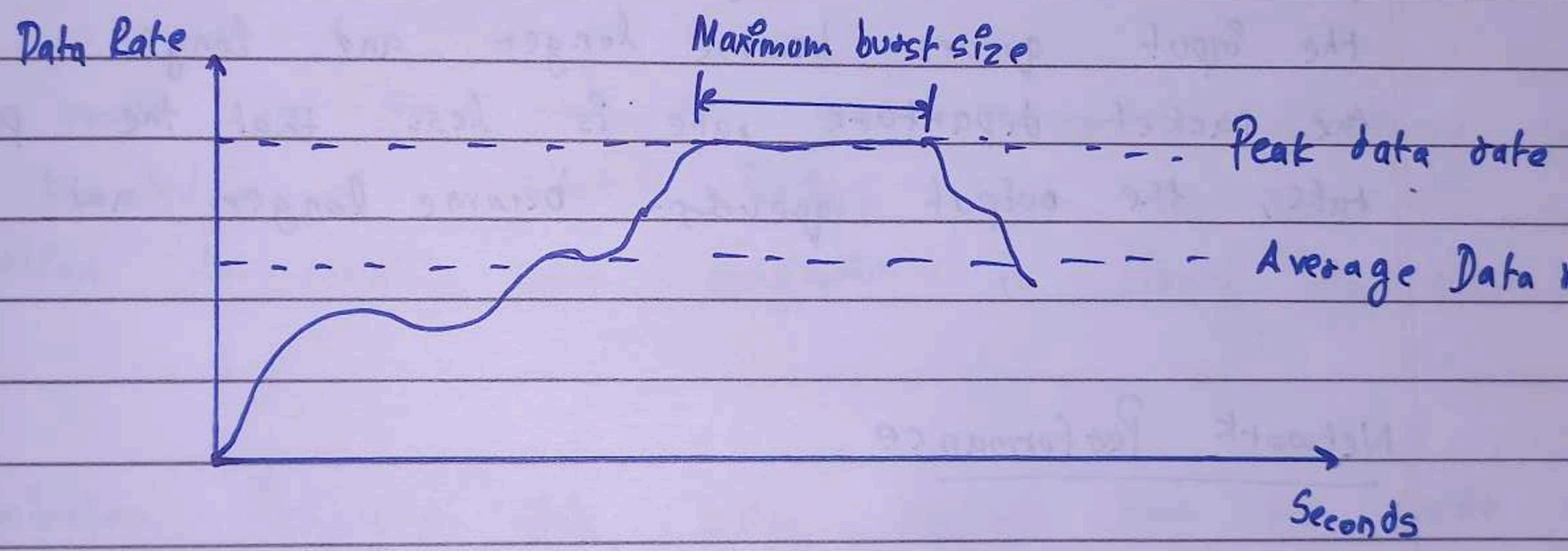
Already explained TCP & UDP message format in ST-2 lol.

Congestion Control & Quality of Service

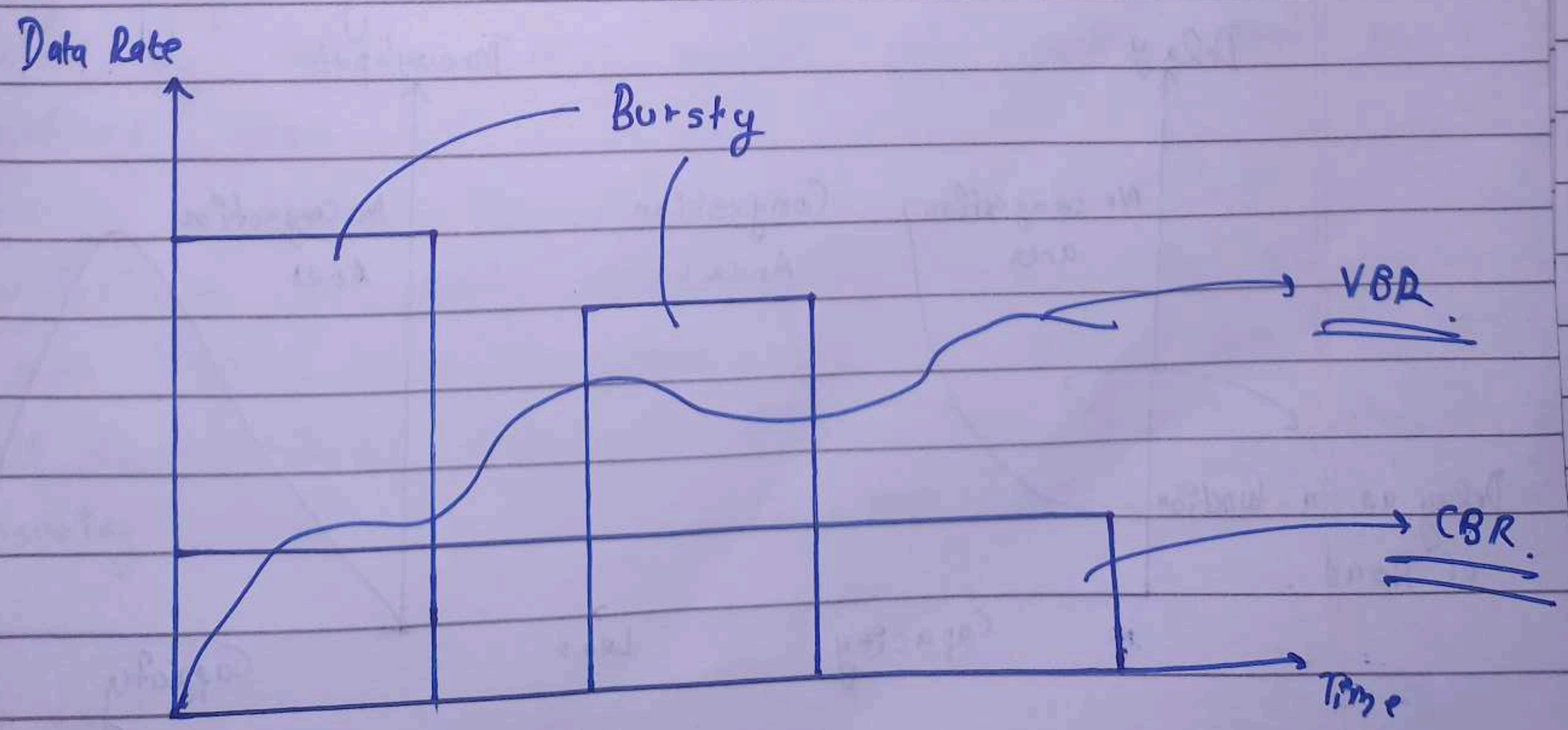
Data Traffic

The main focus of congestion control & quality of service is data traffic. In congestion control we try to avoid traffic congestion. In quality of service, we try to create an appropriate environment for the traffic.

Traffic Descriptor → qualitative values that represent a data flow.



Traffic Profiles → Constant Bit Rate (CBR), Variable Bit Rate (VBR), Bursty



Congestion

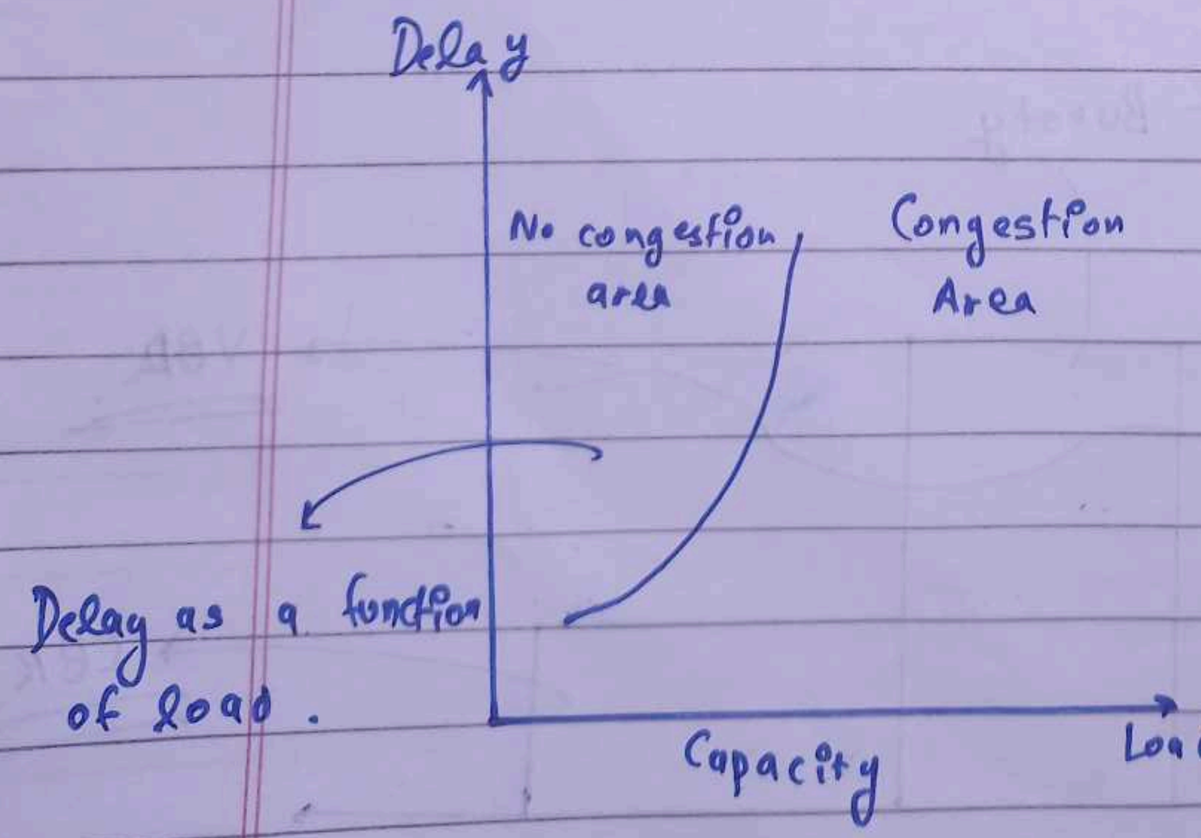
An important issue in packet switched network is congestion. Congestion in a network may occur if the load on the network - the no. of packets sent to the network - is greater that the capacity of the network - the no. of packets a network can handle.

Congestion control refers to the mechanisms and techniques to control the congestion and keep the load below the capacity.

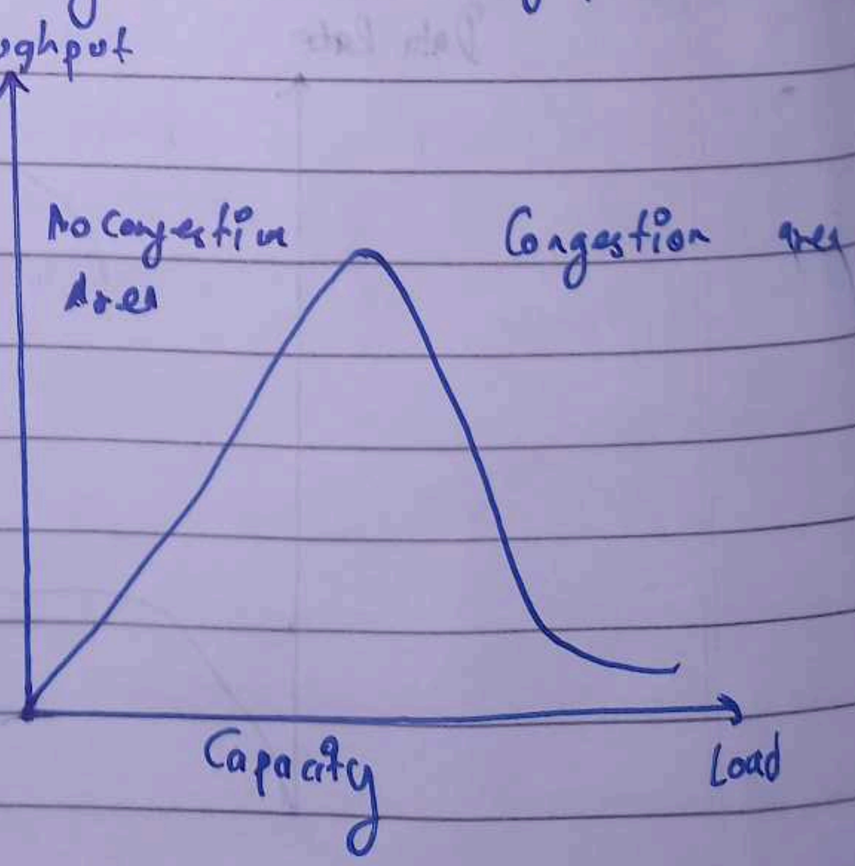
We need to be aware of 2 issues, first, if rate of packet arrival is higher than the packet processing rate, the input queues become longer and longer. Second, if the packet departure rate is less that the packet processing rate, the output queues become longer and longer.

Network Performance

Congestion control involves 2 factors that measure the performance of a network : delay and throughput.



Delay as a function of load.



Congestion Control.

- Open Loop
  - Retransmission Policy
  - Window Policy
  - Acknowledgment Policy
  - Discarding Policy
  - Admission Policy
- Closed Loop
  - Back Pressure
  - Choke Packet
  - Implicit Signaling
  - Explicit Signaling

Open Loop Congestion Control

Set of policies and techniques that are used to prevent congestion in a network before it occurs.

It is based on the assumption that network administrator has some knowledge of network traffic and can use this information to predict when congestion is likely to occur.

Retransmission Policy: This policy controls how the sender of data retransmits packets that have been lost or corrupted. A carefully designed retransmission policy can help to prevent Congestion by reducing the number of packets that are retransmitted unnecessarily.

Window Policy: This policy controls how much data the sender can send before it must wait for an Ack from the receiver. A well chosen window size can help to prevent congestion by ensuring that the sender does not overload the network.

Ack Policy: This policy controls how the receiver acknowledges the receipt of data packets. A carefully designed ack policy can help to prevent congestion by reducing the no. of packets that the sender has to retransmit.

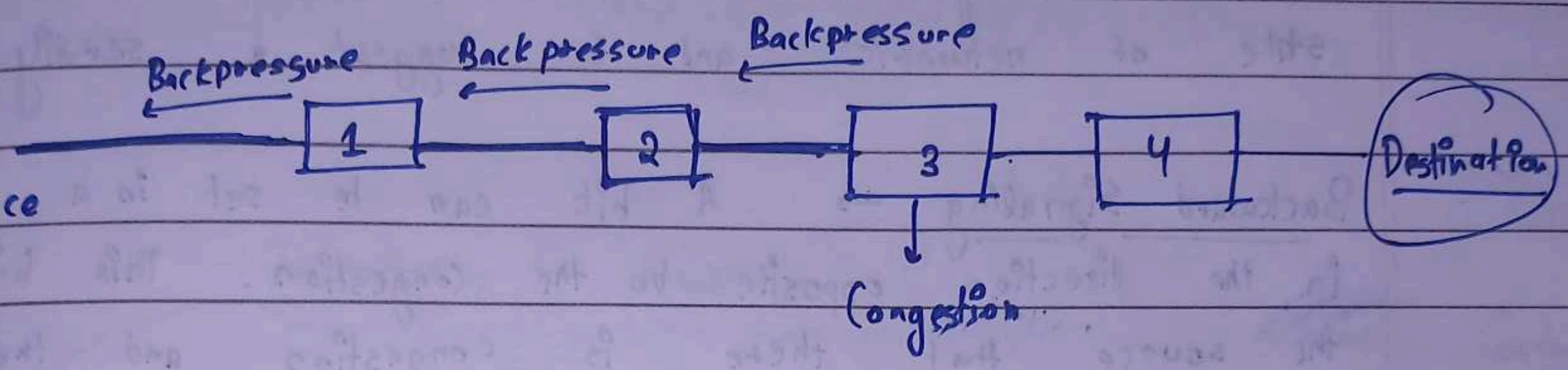
Discarding Policy: This policy controls how the receiver discards packets when it is congested. A well-designed discarding policy can help to prevent congestion by dropping packets in a way that minimizes the impact on the overall performance of the network.

Admission Policy: This policy controls which new connections are allowed to enter the network. A well-designed admission policy can help to prevent congestion by limiting the amount of traffic that enters the network.

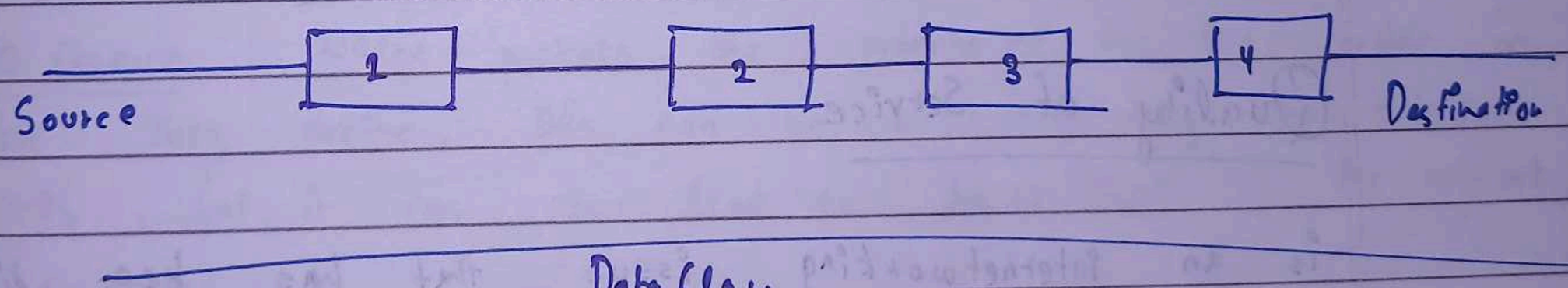
Closed Loop Congestion Control

Set of techniques that use feedback from the network to detect and respond to congestion. This is in contrast to open loop congestion control, which attempts to prevent congestion from occuring in the first place by using predictive methods.

Backpressure : When a router detects congestion, it sends a signal back to the previous router, which then reduces the rate at which it is sending packets. This continues till congestion is resolved.



Choke Packet : When a router detect congestion, it sends a signal back to the previous router, which then reduces it sends a special packets to the sender, signaling it to reduce the rate at which it is sending data.



Implicit Signaling : Protocols such as TCP, use implicit signaling to indicate congestion. For eg, if a TCP sender does not receive an acknowledgment from the receiver for a period of time, it assumes that the network is congested and reduces its sending rate.

Explicit Signaling

Some protocols, such as RSVP, use explicit signaling to indicate congestion. In explicit signaling, router send explicit messages to the sender to indicate the current state of network and to suggest a sending rate.

Backward Signaling → A bit can be set in a packet moving in the direction opposite to the congestion. This bit can warn the source that there is congestion and then it needs to slow down to avoid discarding of packets.

Forward Signaling → A bit can be set in a packet moving in the direction of the congestion. This bit can warn the destination that there is congestion. The receiver in this case can use policies, such as slowing down the ACKs, to alleviate the congestion.

Quality of Service

is an internetworking issue that has been discussed more than defined. We can informally define quality of service as something a flow seeks to attain.

classmate
Date
Page

Flow Characteristics

- Reliability
- Delay
- Jitter
  - variation of delay for packets belonging to same flow
- Bandwidth

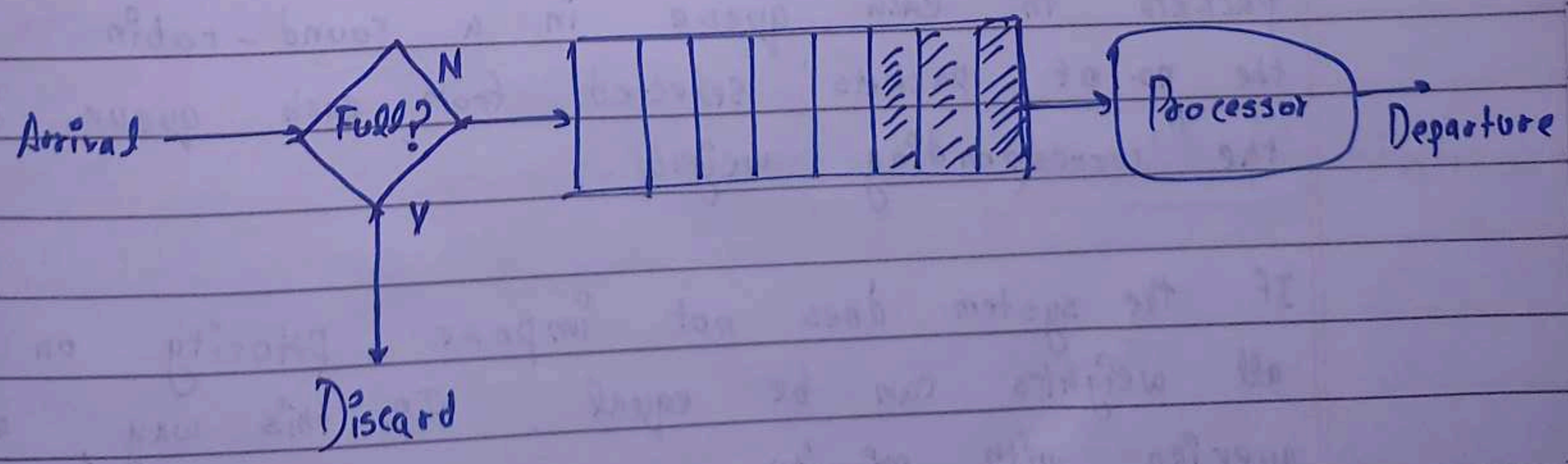
Techniques to improve QoS

- Scheduling
- Traffic Shaping
- Resource Reservation
- Admission Control.

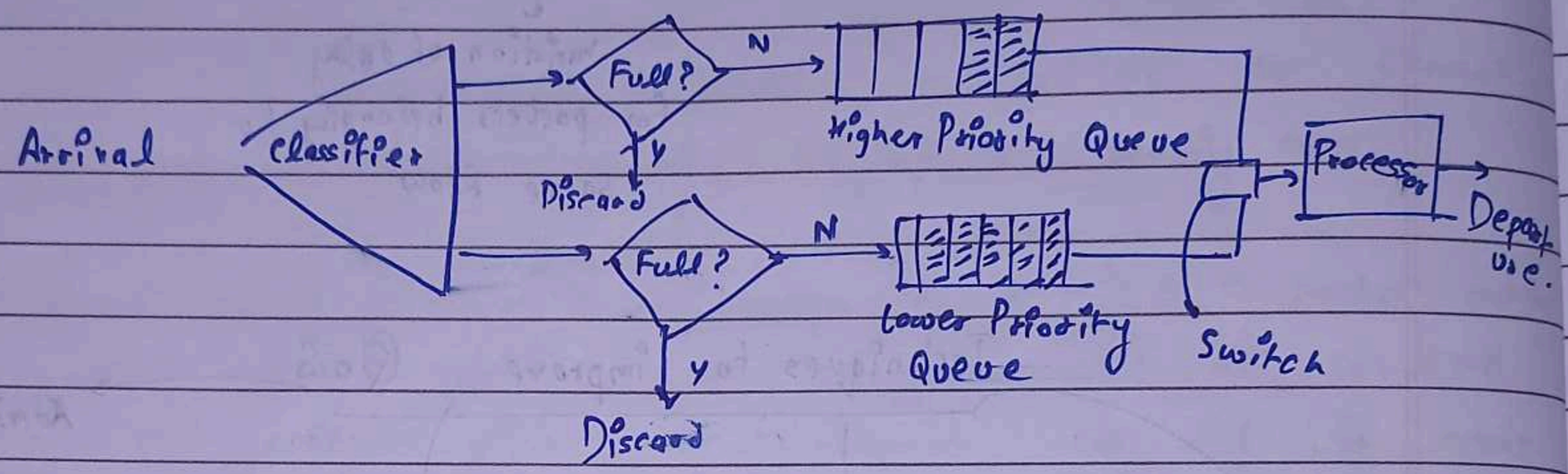
Scheduling

is a technique used to improve QoS by controlling the order in which packets are processed by a router or switch.

FIFO Queue where packets are processed in the order in which they arrive. This can be a fair way to schedule packets, but it can also lead to lower QoS if the network is congested.



Priority Queuing Packets are assigned to a priority class. Each class has their own queue. Higher priorities are processed first & lower at last.

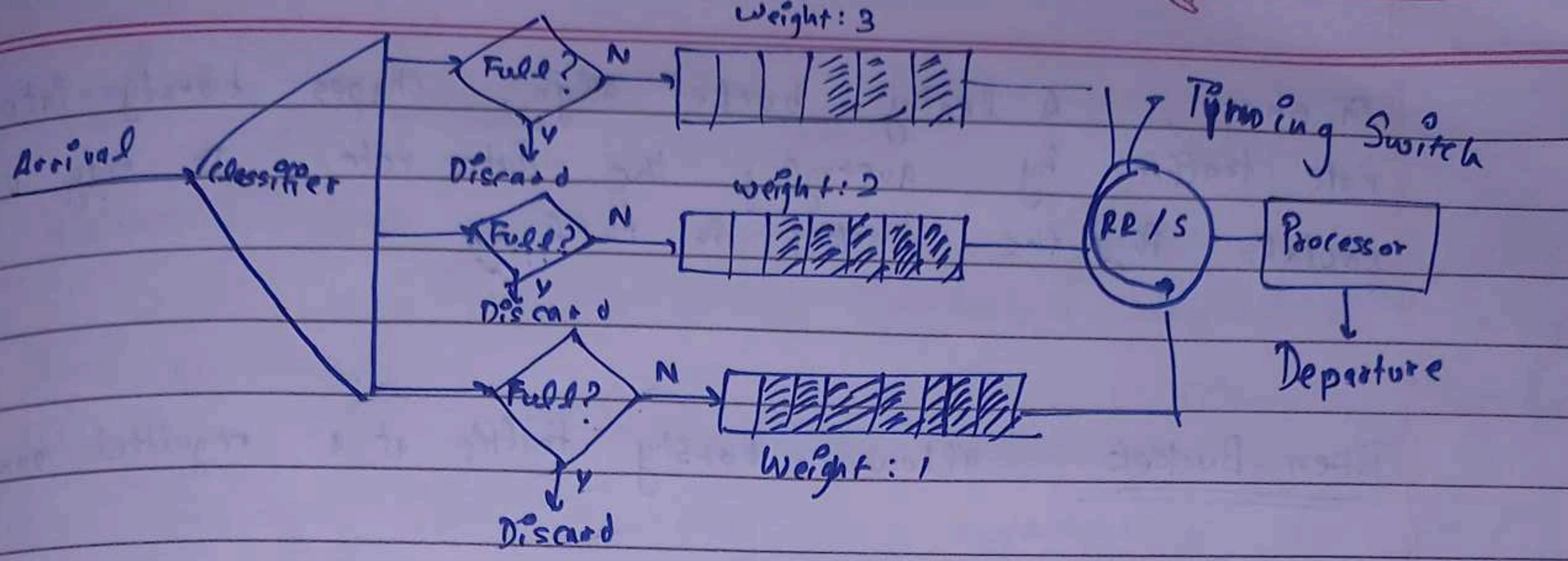


But in Priority Queue Starvation can occur as, if there is a continuous flow in a high-priority queue, the packets in the lower priority queue will never have a chance to be processed.

Weighted Fair Queuing.

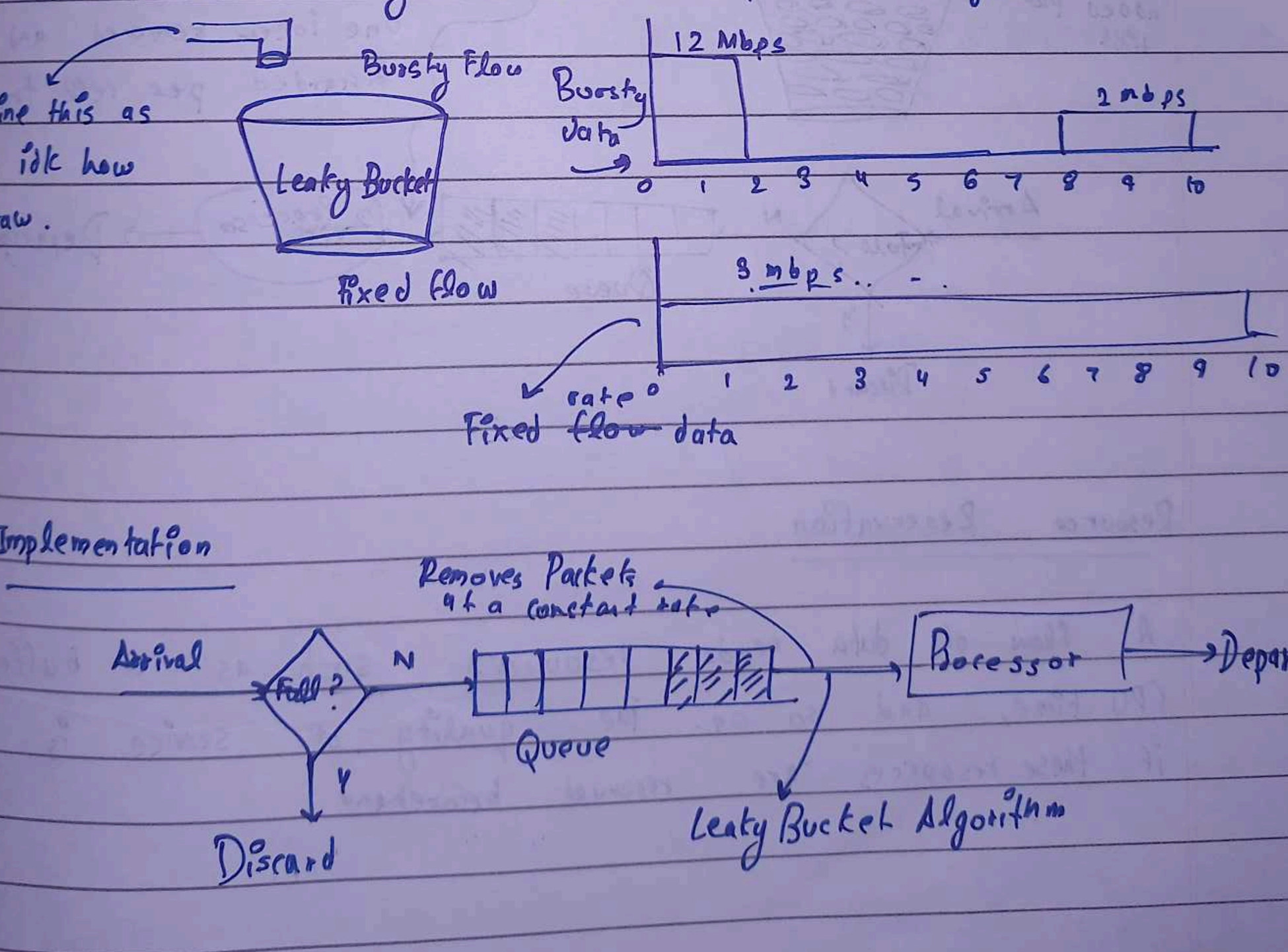
The packets are still assigned to different classes and admitted to different queues. The queues however, are weighted based on the priority of queues; higher priority means a higher weight. The system processes packets in each queue in a round-robin fasion with the no. of packets selected from each queue based on the corresponding weight.

If the system does not impose priority on the classes, all weights can be equal. In this way, we have fair queuing with priority.



Traffic Shaping. Mechanism to control the amount and the rate of the traffic sent to the network.

Leaky Bucket works by storing incoming traffic in a bucket and then releasing it at a constant rate. This helps smooth out bursty traffic and prevent congestion.

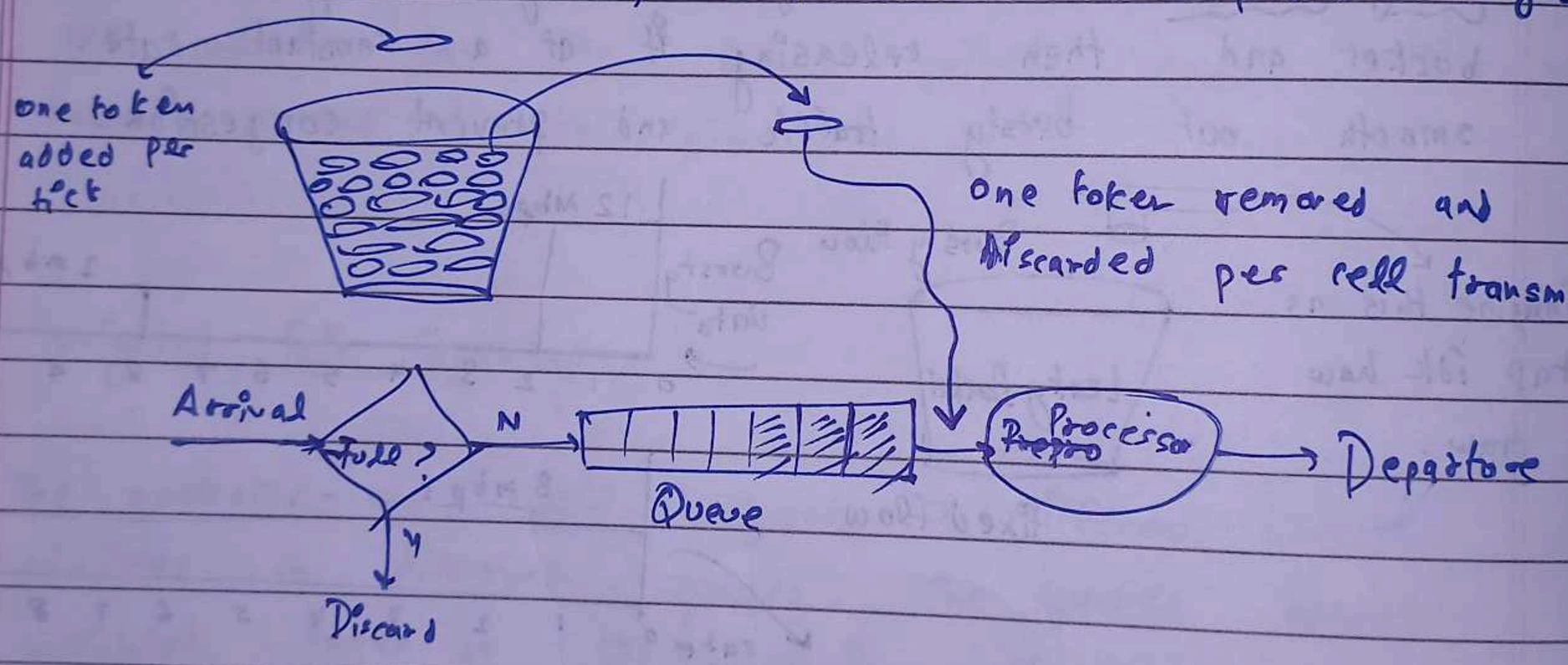


Implementation

In short, A leaky bucket algo. shapes bursty into fixed rate traffic by averaging the data rate. It may drop packets if the bucket is full.

Token Bucket. allows bursty traffic at a regulated maximum rate

If token bucket is used, the network administrator could set the release rate to be lower than the maximum incoming traffic rate for both voice ~~& file transfers,~~ but higher than the maximum incoming traffic rate to file transfers. This would allow voice traffic to burst more than file transfers, but it would still prevent congestion.



Resource Reservation

A flow of data needs resources such as a buffer, bandwidth, CPU time, and so on. The quality of service is improved if these resources are reserved beforehand.

Admission Control.

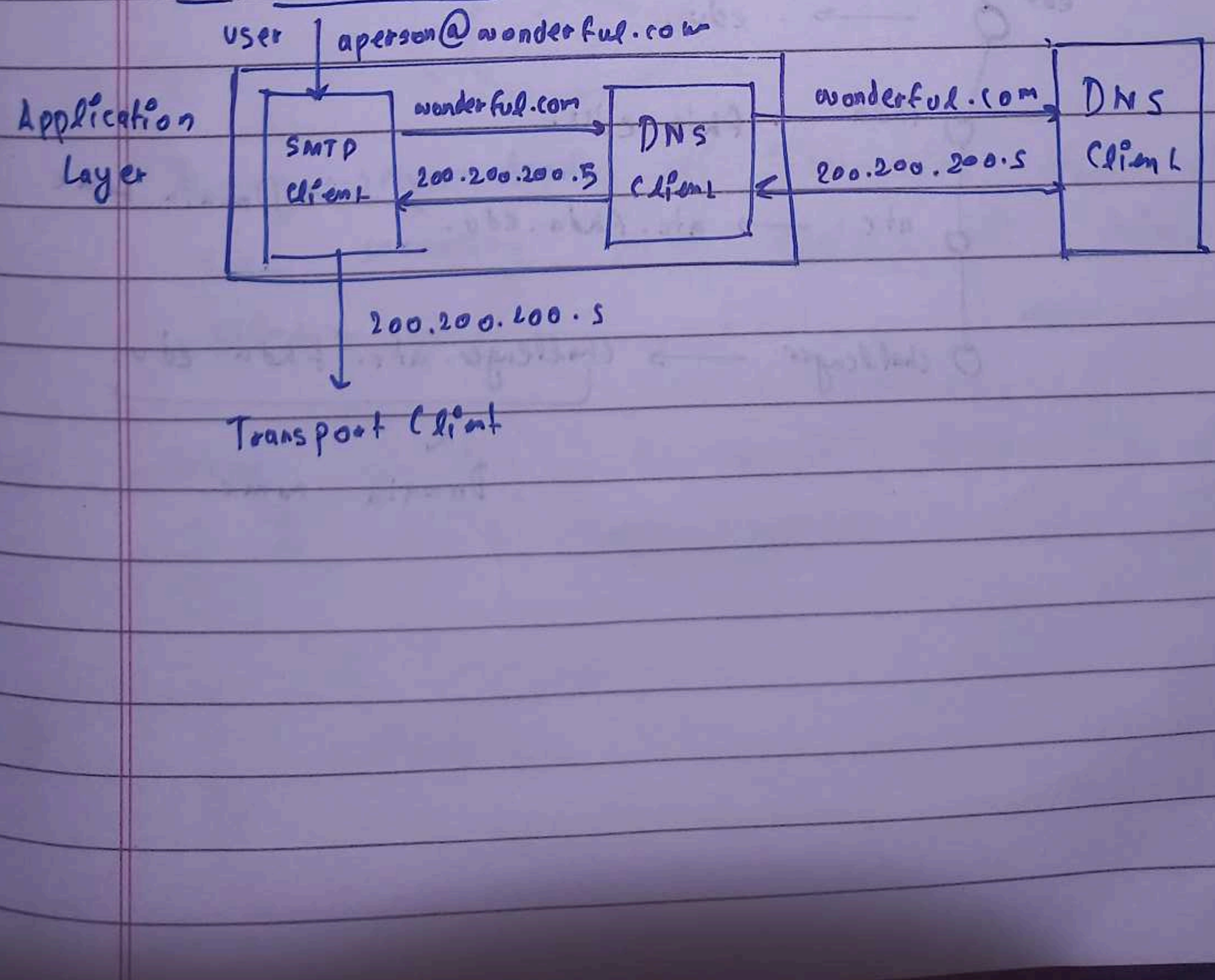
Mechanism used by a router, or a switch, to accept or reject a flow based on predefined parameters called flow specifications. Before a router accepts a flow for processing, it checks the flow specifications to see if its capacity and its previous commitments to other flows can handle the new flow.

Application Layer

Domain Name System (DNS)

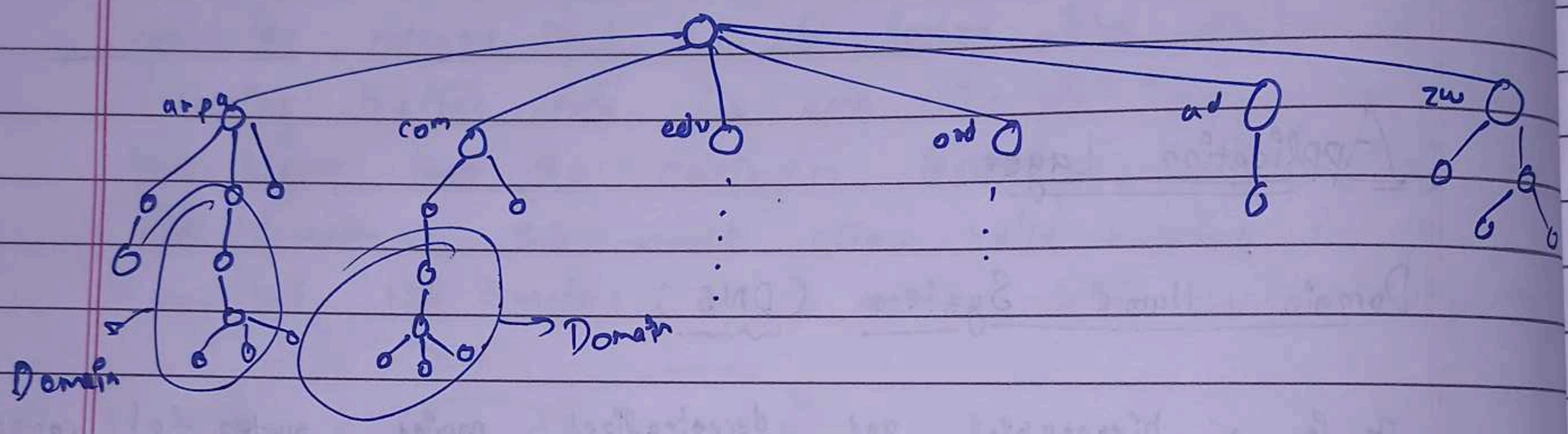
It is a hierarchial and decentralized naming system for computers, services, or any resource connected to the Internet or a private network.

Example of a DNS

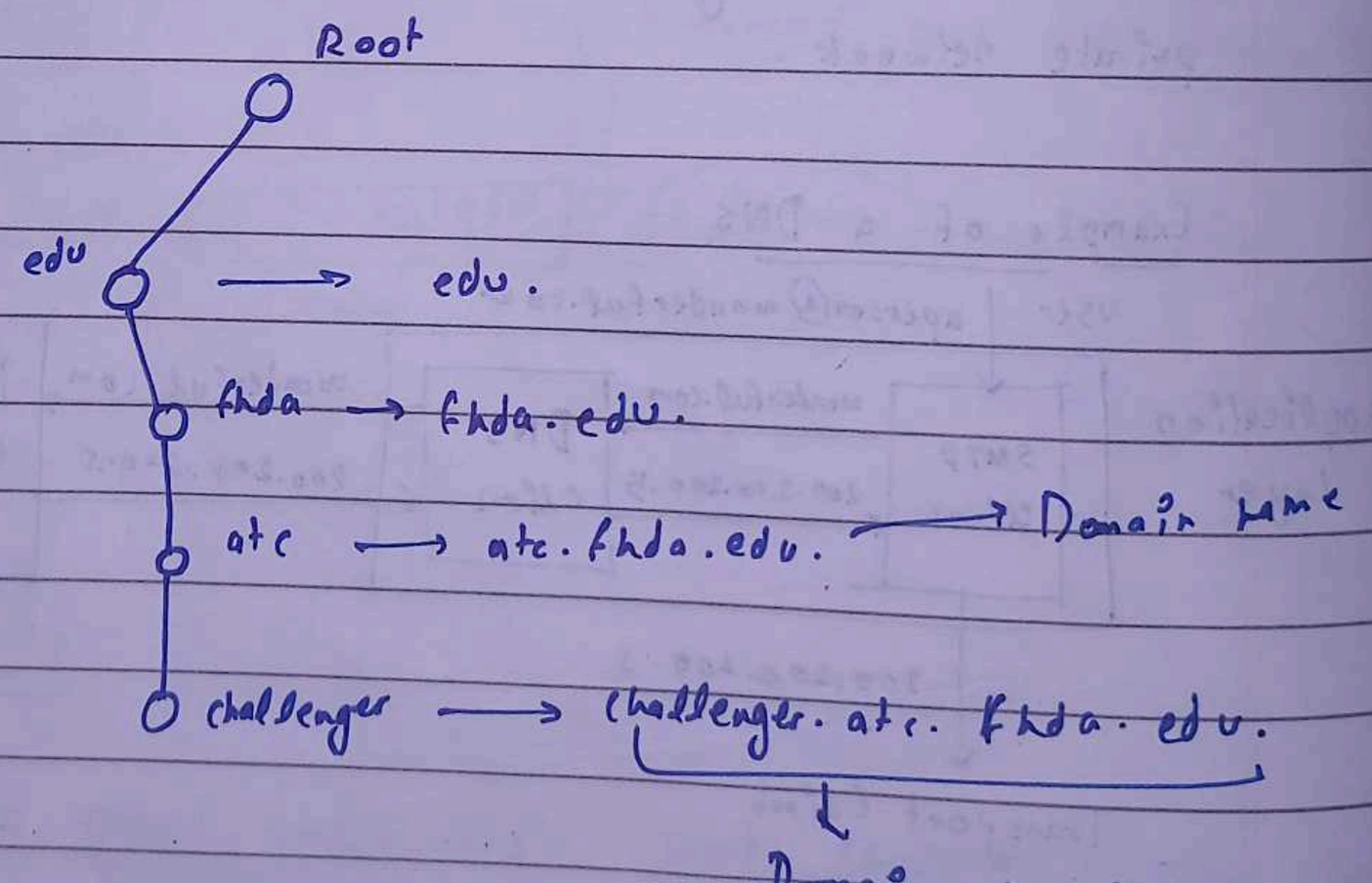


Name Space → To be unambiguous, the names assigned to machines must be carefully selected from a name space with complete control over the binding b/w the names and IP addresses.

Domain Name Space → Inverted tree structure. The tree can only have 128 levels. level 0 to level 127.

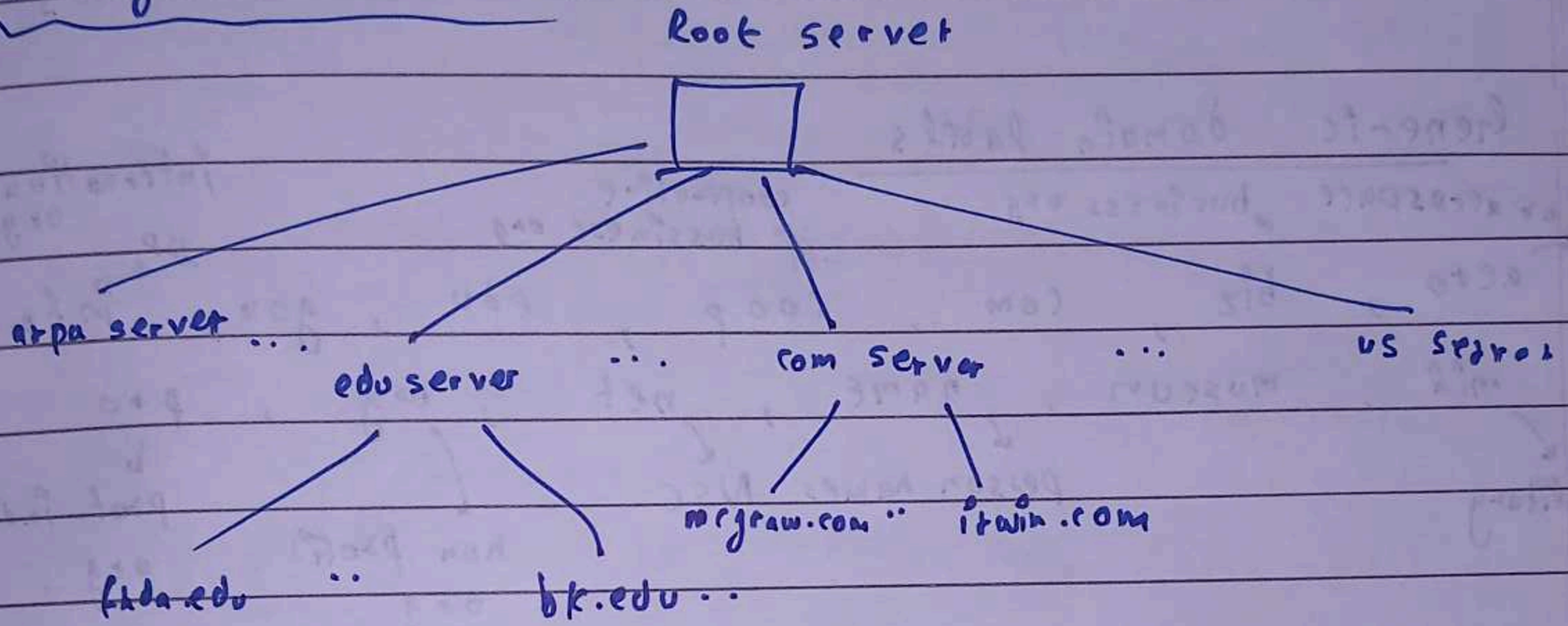


Domain Names and Labels.

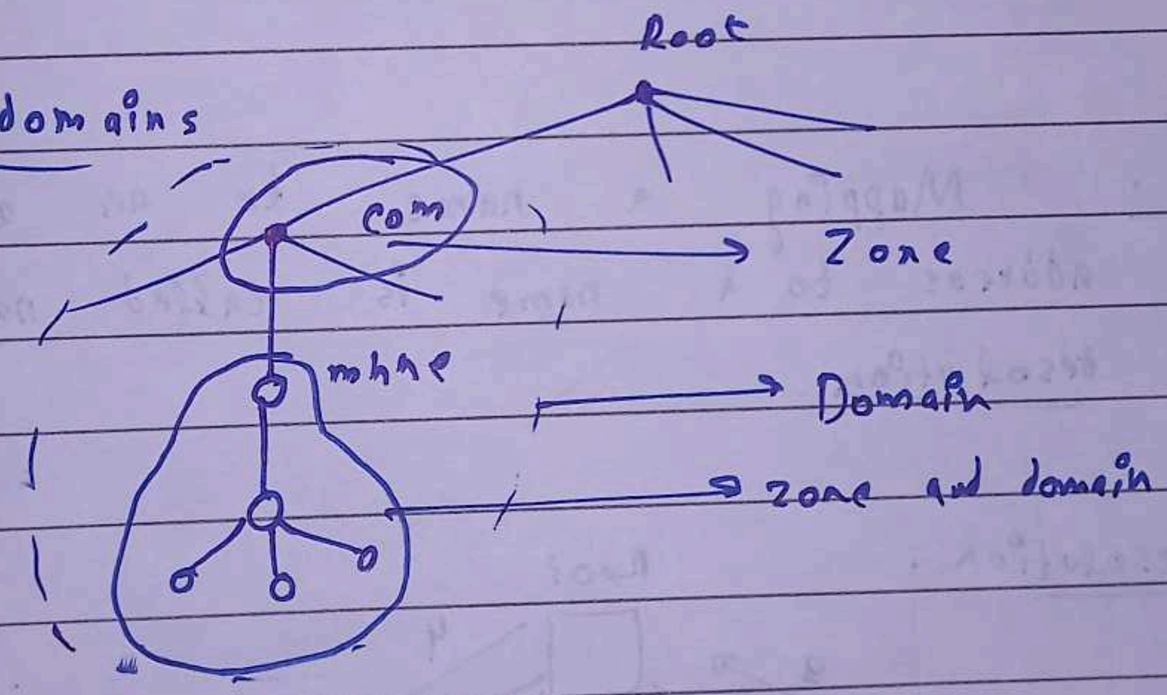


Distribution of Name Space → it is very inefficient and unreliable if we just have one computer store such a huge amount of information.

Hierarchy of name servers



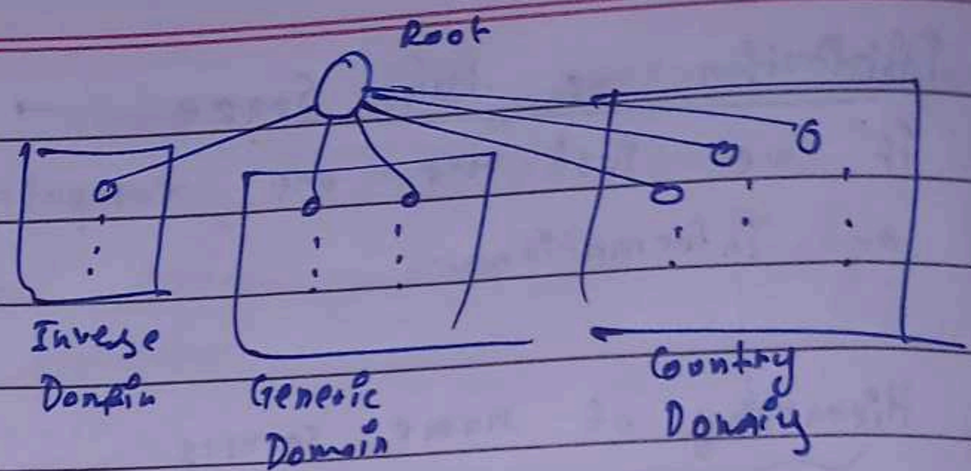
zones and domains



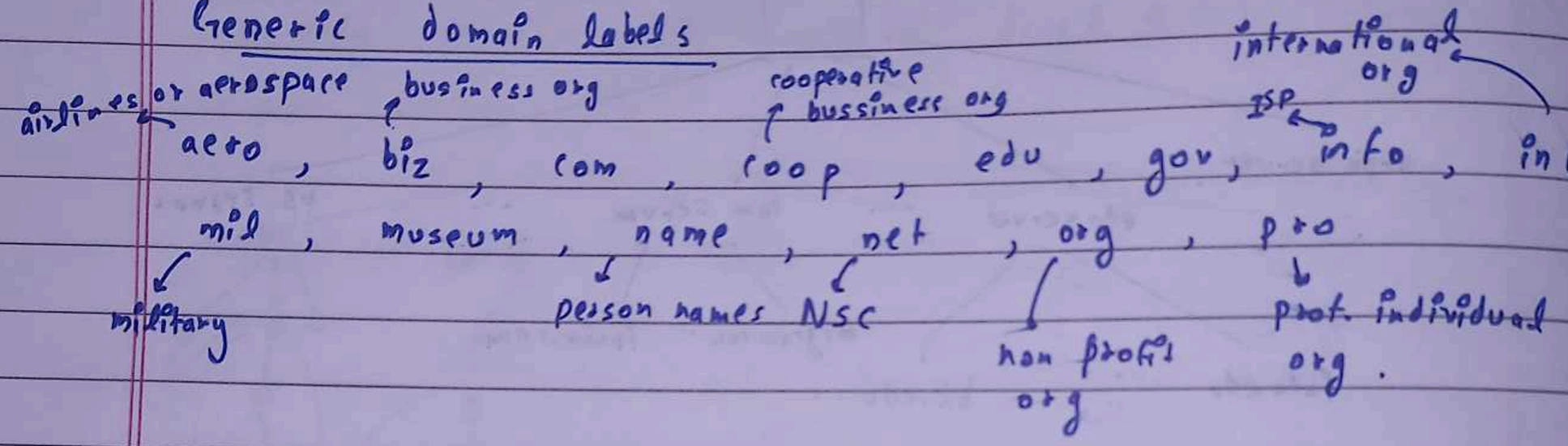
Primary server loads all information from disk file; ~~from~~ the secondary server loads all information from the primary server.

When secondary downloads information from the primary, it is called zone transfer.

DNS in the internet.

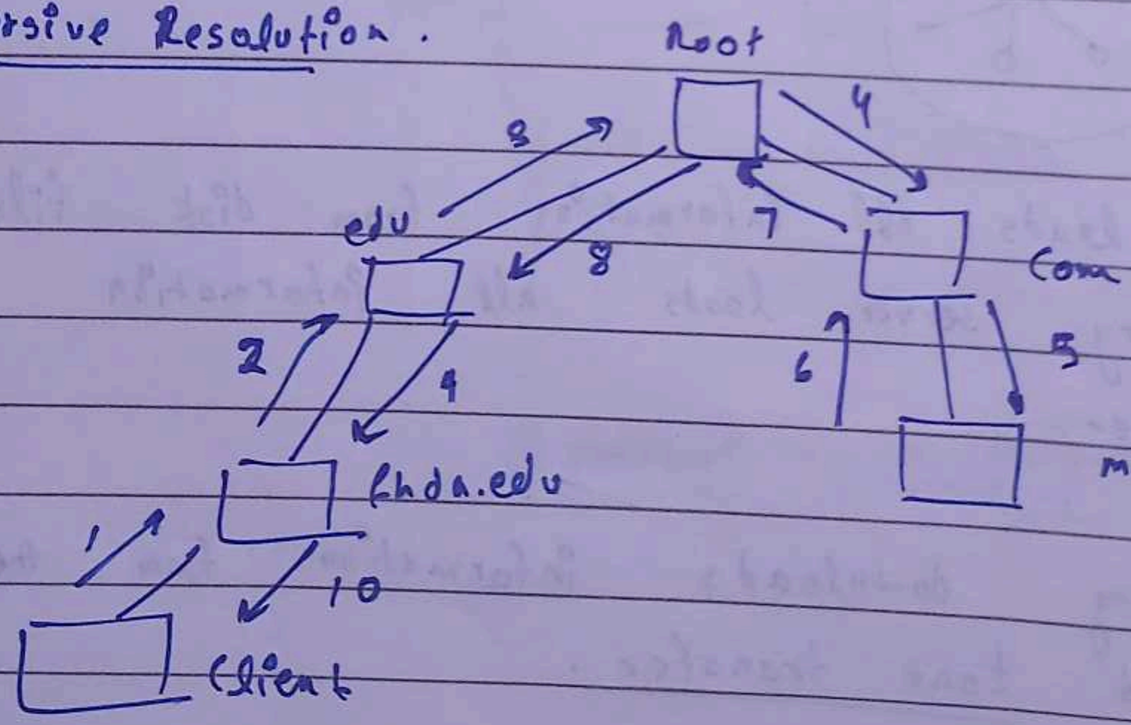


Generic domain labels

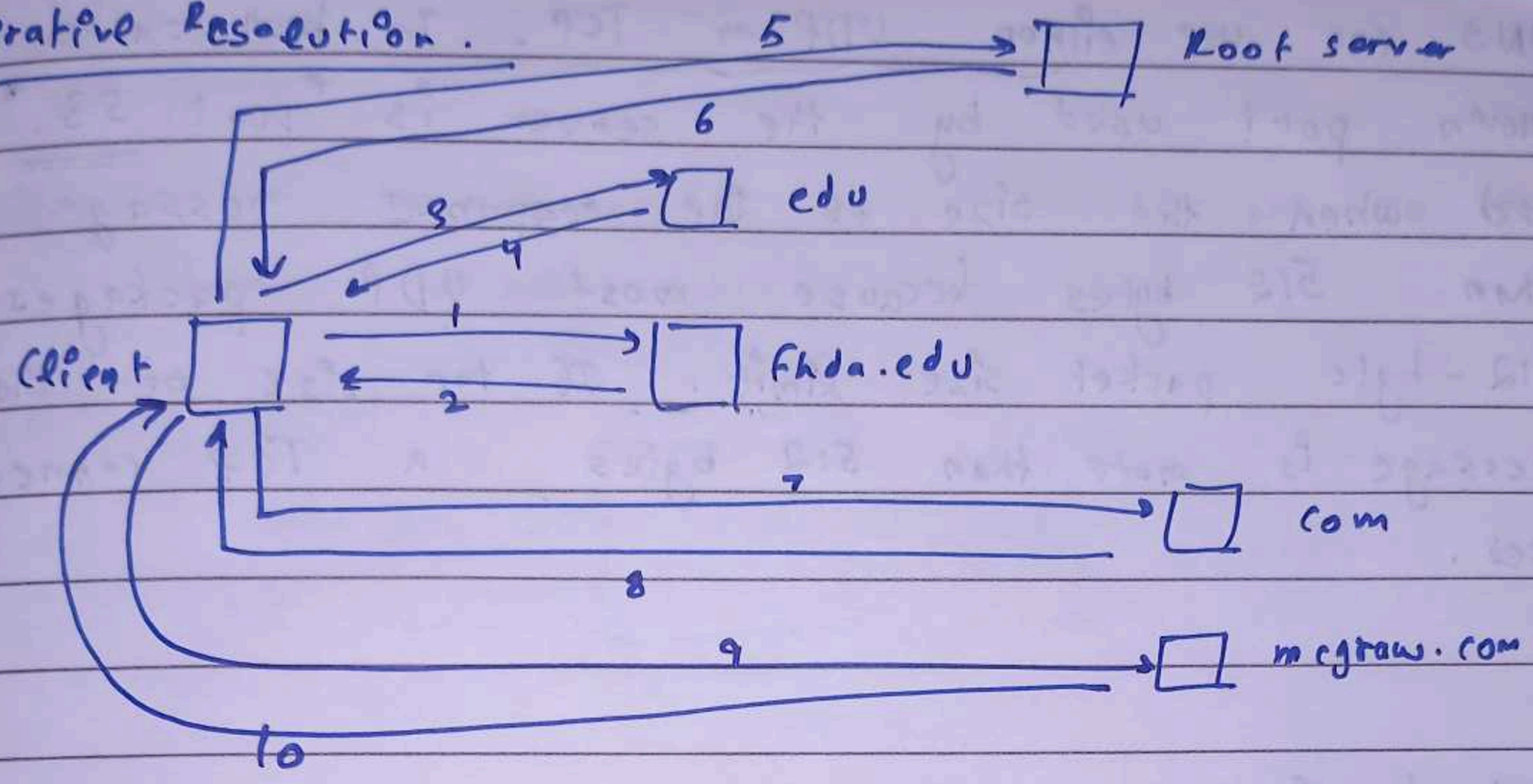


Resolution. Mapping a name to an address or an address to a name is called **name - address resolution.**

Recursive Resolution.



Iterative Resolution.



DNS messages. → query & response.

Query	Response
Header	Header
Question Section	Question Section
0	Answer Section
0	Authoritative Section
0	Additional Section

Both have same format

Header format

Identification	Flags
No. of question records	No. of answer records
No. of authoritative records	No. of additional records

→ all 0s in query message

DNS can use either UDP or TCP. In both cases the well-known port used by the server is port 53. UDP is used when the size of the response message is less than 512 bytes because most UDP packages have a 512-byte packet size limit. If the size of the response message is more than 512 bytes, a TCP connection is used.

Remote Logging.

It would be impossible to write a specific client / server program for each demand. The better solution is a general purpose client / server program that lets a user access any application program on a remote computer.

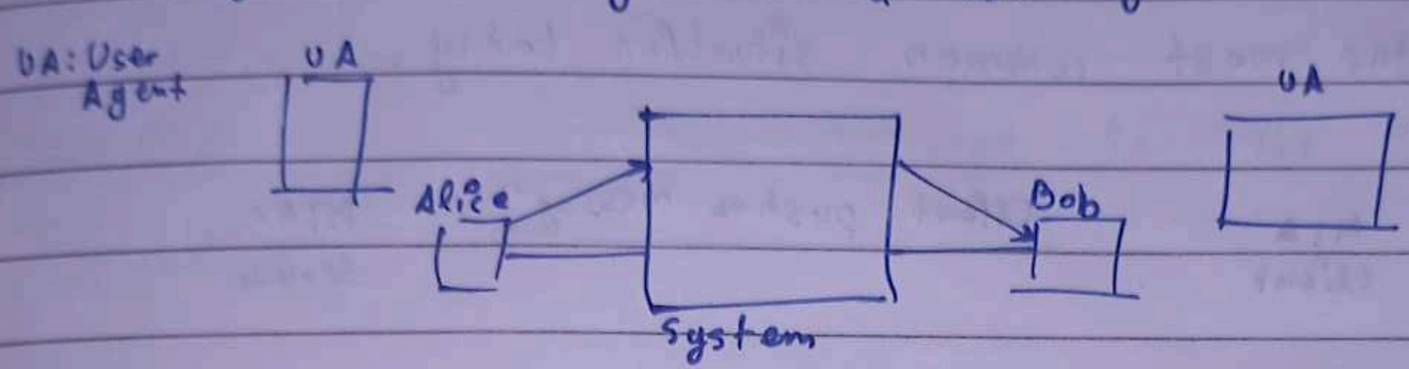
TELNET is a general purpose client / server application program.

Log forwarding: This is where the logs of the remote computer are sent to a central log server. The log server can be accessed from anywhere.

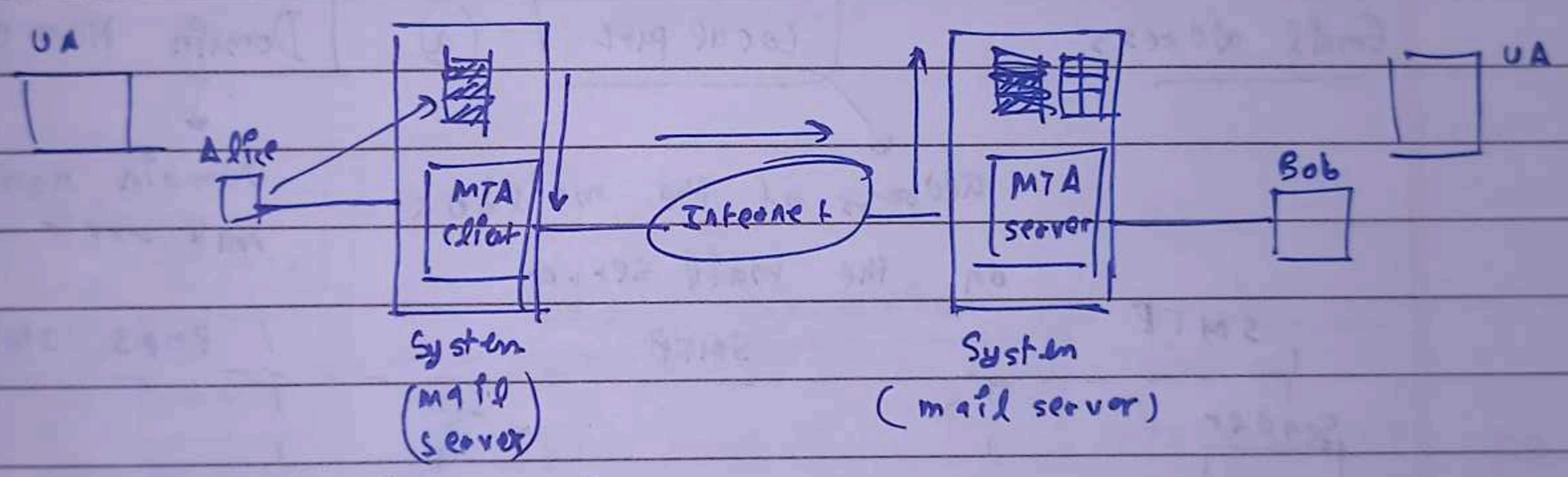
Remote login: This is where you use a remote login protocol to log in to the remote computer and access its logs directly.

Electronic Mail.

Senario 1: sender and receiver of an email are on the same system, we only need 2 user agents

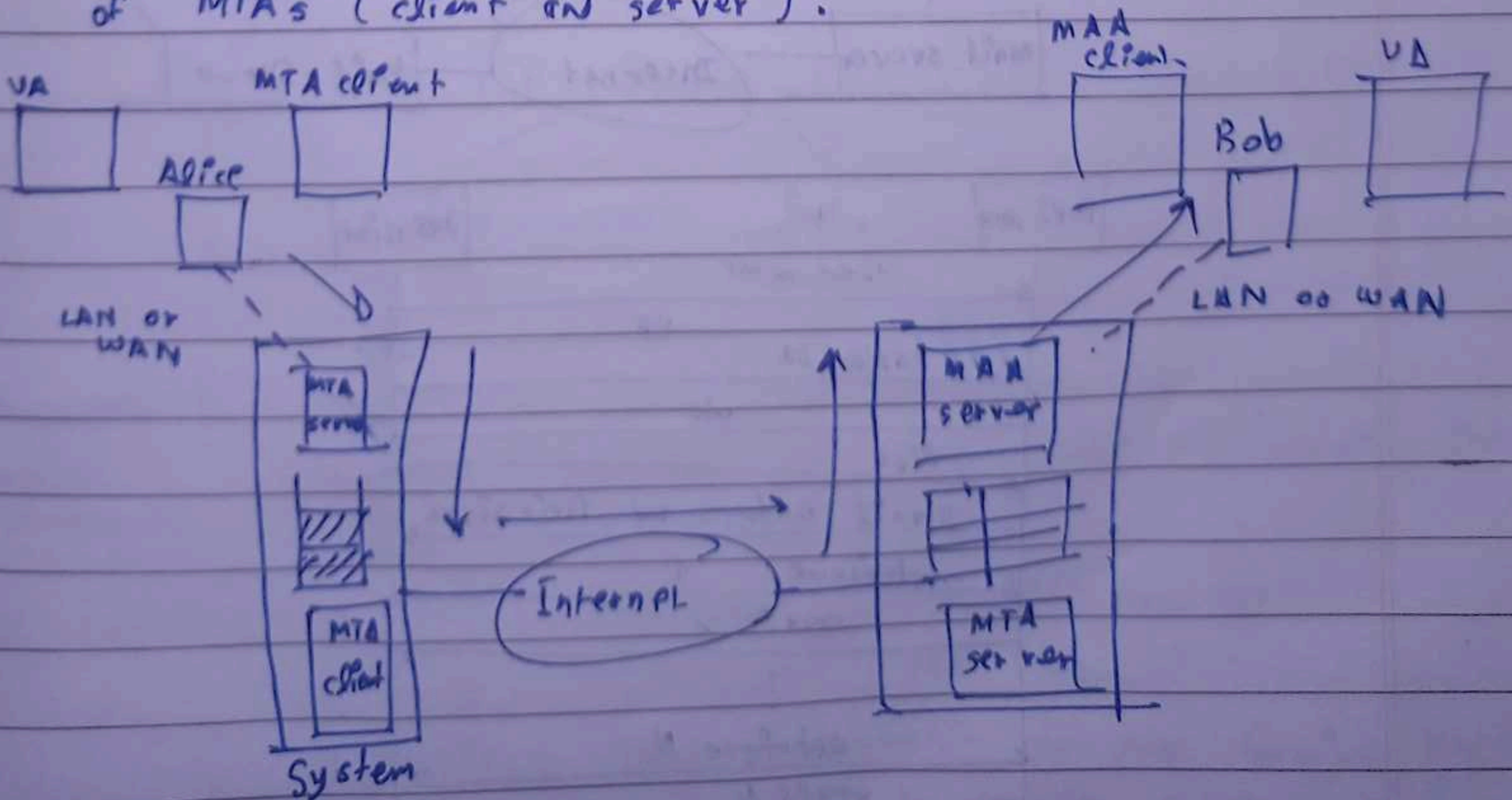


Senario 2: sender and receiver of an email are on different systems we need two UAs and a pair of MTAs (client and server).



MTA: message transfer agent

Senario 3: When sender is connected to the mail server via a LAN or a WAN, we need two UAs and 2 pairs of MTAs (client and server).



MAA: message access agent

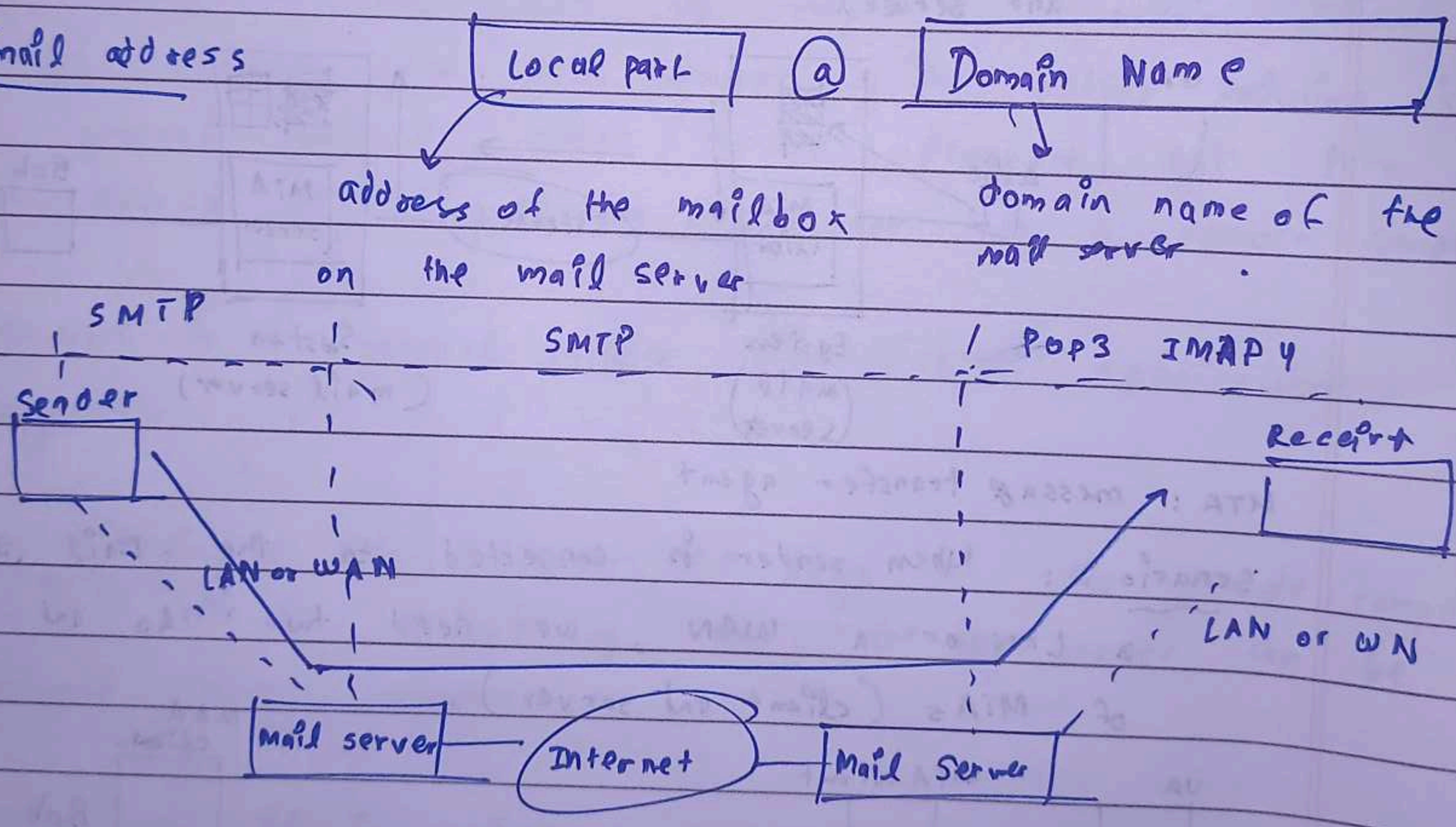
When both sender and receiver are connected to the mail server via a LAN or a WAN, we need two UAs, two pairs of MTAs and a pair of MAAs.

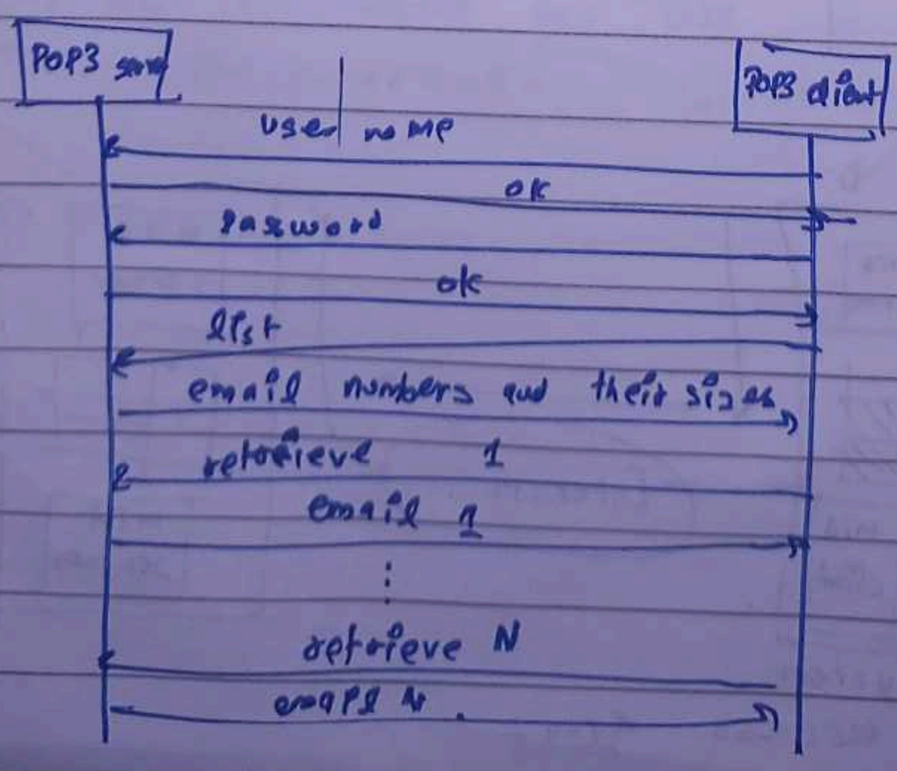
This is the most common situation today.

MTA client — Client pushes message → MTA server

MAA client — Client pulls message ← MAA server

Email address

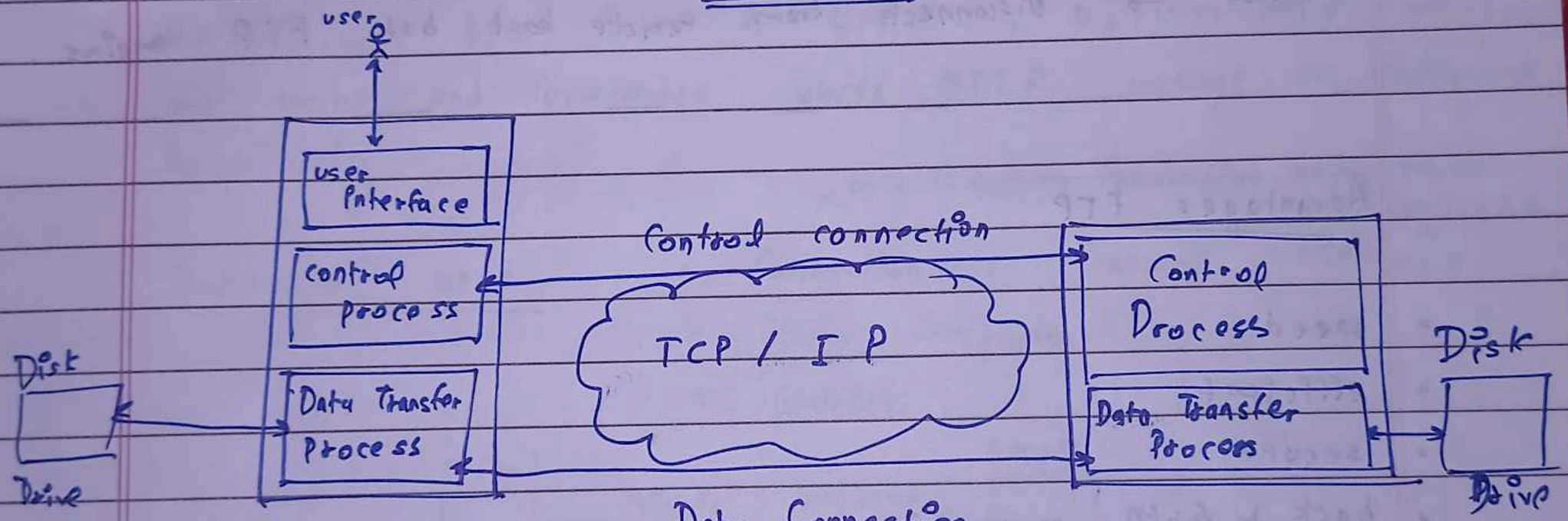




File Transfer Protocol. (FTP)

Transferring files from one computer to another is one of the most common tasks expected from a networking or internetworking environment. As a matter of fact, the greatest volume of data exchange in the Internet today is due to file transfer.

" FTP uses the services of TCP. It needs 2 TCP connections. The well-known port 21 is used for the control connection and the well-known port 20 for the data connection. "



FTP Connections

- Control connection — simple
- Data Connection — complex

Commands

- **open**: selects the remote host and initiates the login session.
- **user**: Identifies the remote user ID.
- **pass**: Authenticates the user.
- **site**: Sends information to the foreign host that is used to provide services specific to that host.

cd → change directory to remote host

lcd → " " to local host.

ls → list on remote directory human readable

dir → same as ls but program readable.

structure → structure of file

type → specifies character sets

mode → Specifies whether the file is treated as having a record structure or a byte stream formal.

get, mget & put, mput also there.

quit → terminates FTP

close → disconnects from remote host but FTP remains.

Advantages FTP

- speed
- efficient
- secure
- back & forth movement.

Disadvantages FTP.

- Not compatible with every systems.
- Unwanted eavesdropping.
- size limit of 2GB

HTTP (HyperText Transfer Protocol)

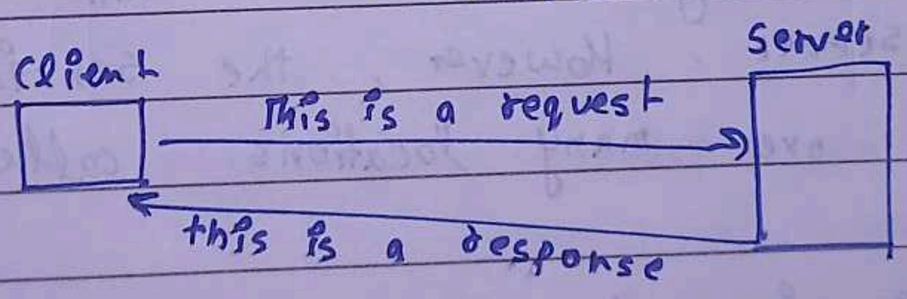
Protocol used to access the data on the www. It can be used to transfer all kinds of data. It allows us to use in a hypertext environment where there are rapid jumps b/w documents. It carries data in MIME-like format.

HTTP is similar to FTP ~~but of~~ as it also transfers files b/w hosts. But, HTTP is simpler than FTP as HTTP uses only one connection i.e. no control connection to transfer the files.

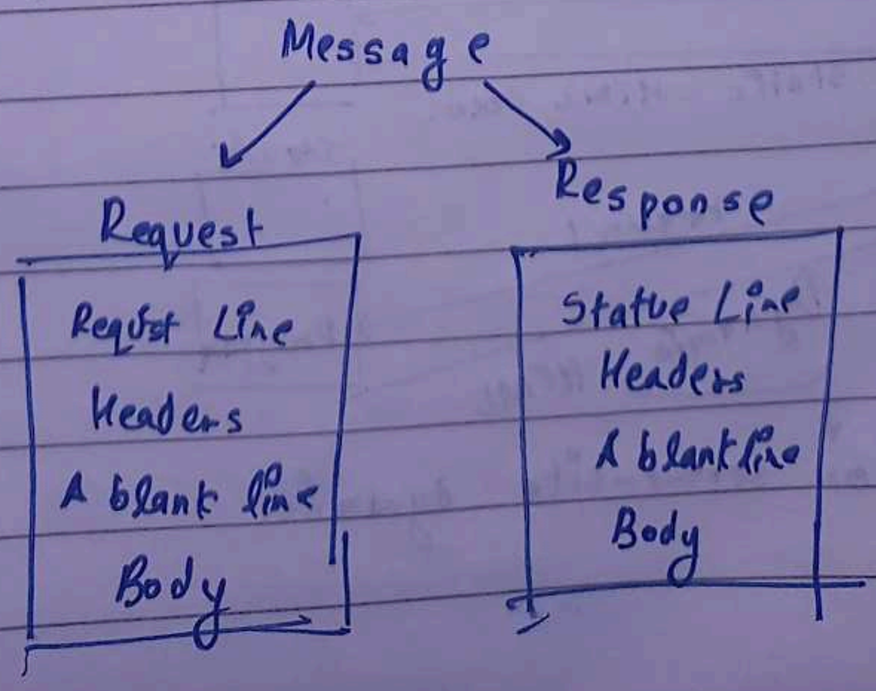
The difference b/w SMTP & HTTP is that SMTP messages are stored and forwarded while HTTP message are delivered immediately.

Features of HTTP

- Connectionless Protocol (uses TCP but. Connection only exists during request & response.)
- Media Independent
- Stateless

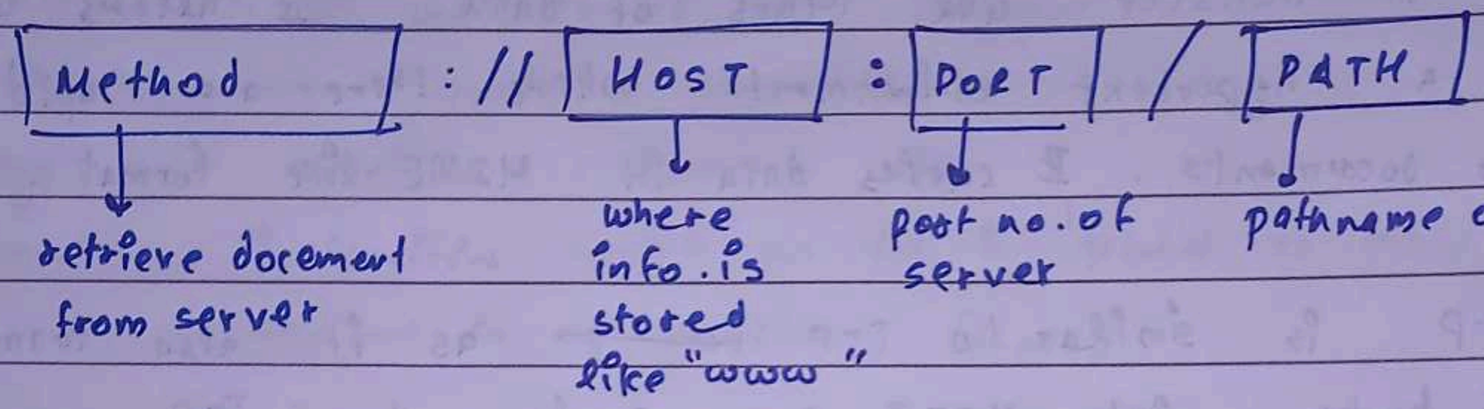


kinda the combination of FTP & SMTP. & uses TCP at transport layer. & is a in-band & stateless protocol.



HTTP uses services of TCP on port 80.
Port 443 for HTTPS

URL Uniform resource locator.

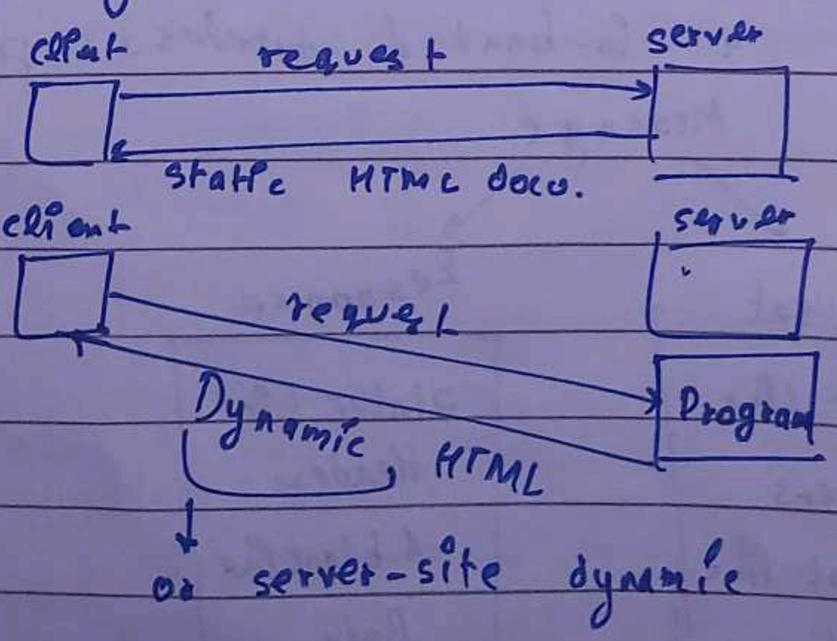


HTTP 1.0 is non-persistent & HTTP 1.1 is Persistent.
connection less

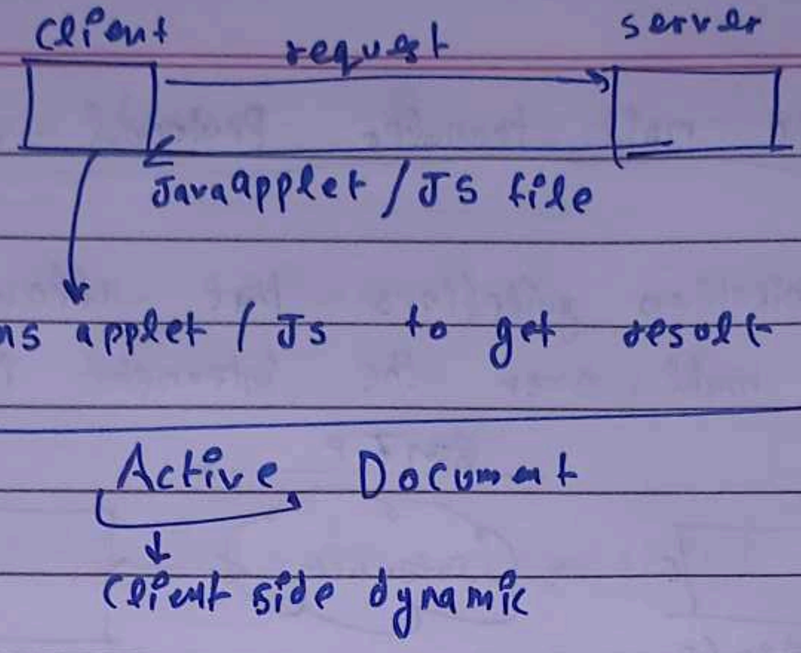
WWW (World Wide Web)

is a distributed client/server service, in which ~~a~~ client using a browser can access a service using a server. However, the service provided is distributed over many locations called sites.

The documents in WWW can be grouped into 3 categories static, dynamic and active.



↓
ex server-site dynamic



HTTP connections

Non-persistent	Persistent
• one that is used for serving exactly one request & one response.	• multiple
• closes TCP connection automatically after sending response.	• closes after a ~~config~~ certain configurable amount of time.
• new connection for every object.	• same.
• HTTP 1.0	• HTTP 1.1

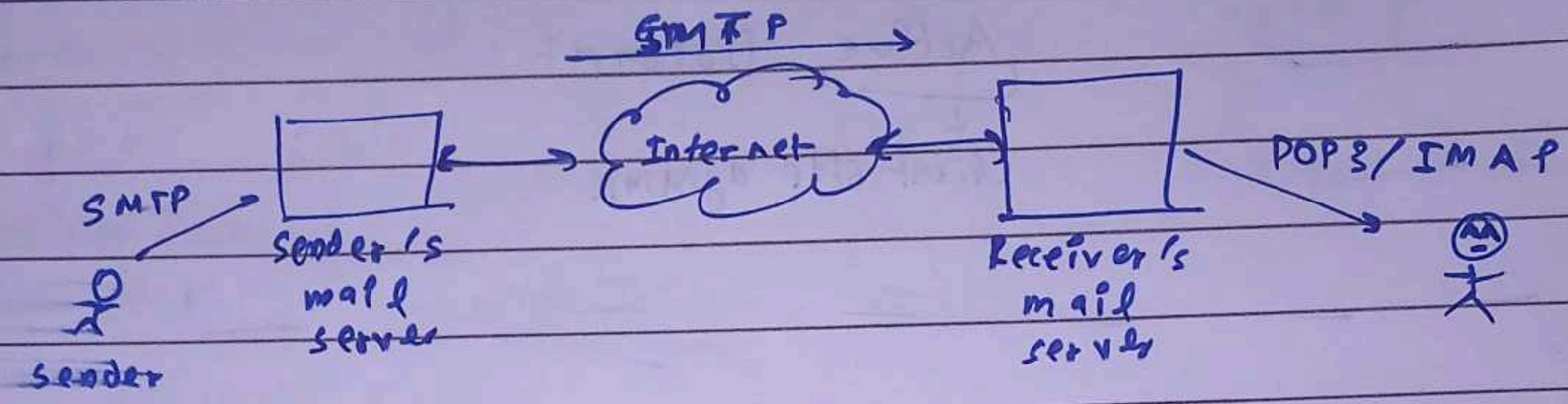
Notizen machen ist so schwer lol.

Pure text based

SMTP Simple mail transfer Protocol.

uses MIME

set of communication guidelines that allow software to transmit an electronic mail over the internet is called SMTP.



SMTP is a push protocol that uses TCP at port no 25. SMTP uses persistent TCP connections, so that it can send multiple emails at once.
It is connection oriented, in-band, stateless.

* We cannot use SMTP at receiver's side. * we need like POP3 or IMAP.

Post office Protocol (POP) or POP3 version of POP.

Message access protocol. Enables clients to receive or download the email from their remote mail server.

POP is a pull protocol uses TCP at port no. 110.
uses Persistent TCP & is connection oriented, in-band.

POP is stateful until mail is downloaded as well as stateless across sessions.

250 → okay 211 → system status 214 → help 220 → service ready 421 → service not available 500 → syntax error 450 → mailbox not available.

Internet Message Access Protocol (IMAP)

Also enables the clients to receive or download the emails from their remote servers.

* POP has been largely superseded by IMAP *

Everything same as POP just port no. 143 & is stateful & distributes mail boxes across multiple servers.

POP	IMAP
• can read only after downloading.	• can read before downloading
• when downloaded copy on serve is delete.	• stores on mail server as well.
• cannot organize mails on server.	• can organize mails on serve.
• can be accessed on a single device.	• multiple devices.
• changes made in client's mail account are not visible on web mail inbox.	• changes made are immediately visible.

SNMP — Simple Network management Protocal.

Before SNMP let's continue SMTP i am just not satisfied lol.

SMTP (Cont.)

Helo → to identify client

Mail From → to identify sender

RCP To → recipient identification

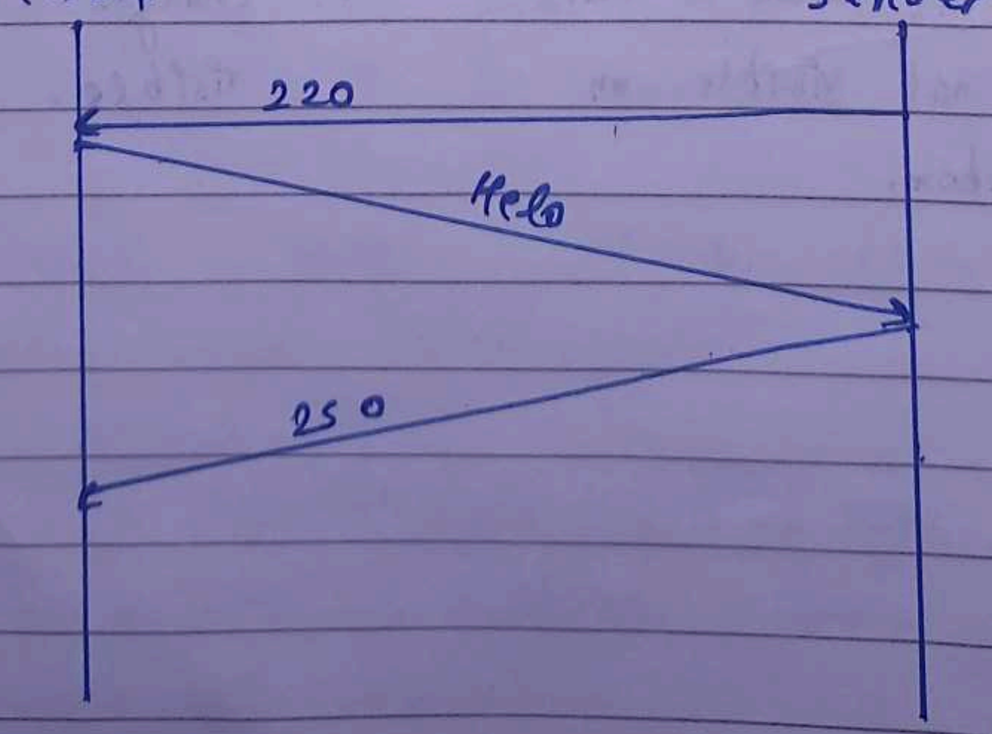
Data → for sending data

Quit → To terminate

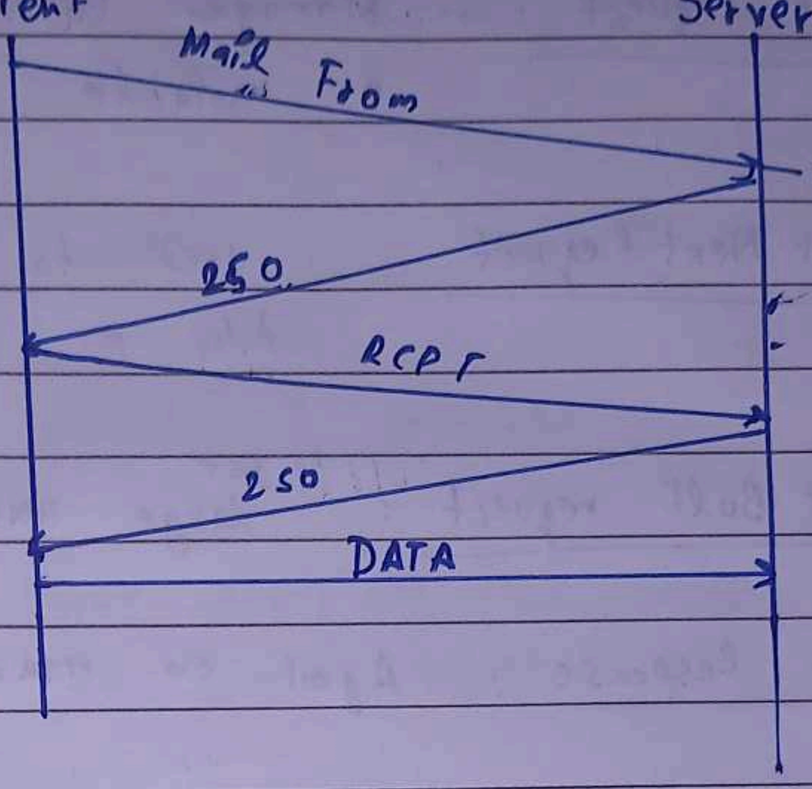
VRFY → verify

RSET → reset

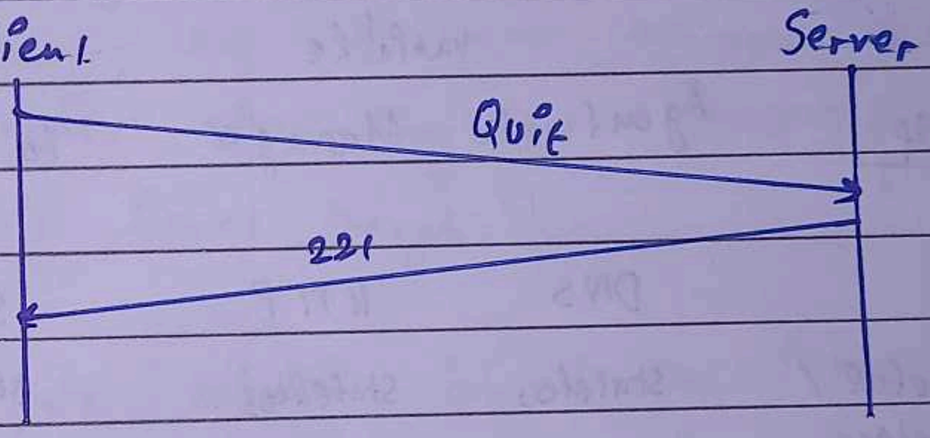
3 stages of SMTP (1) Connection Building



② Data sharing

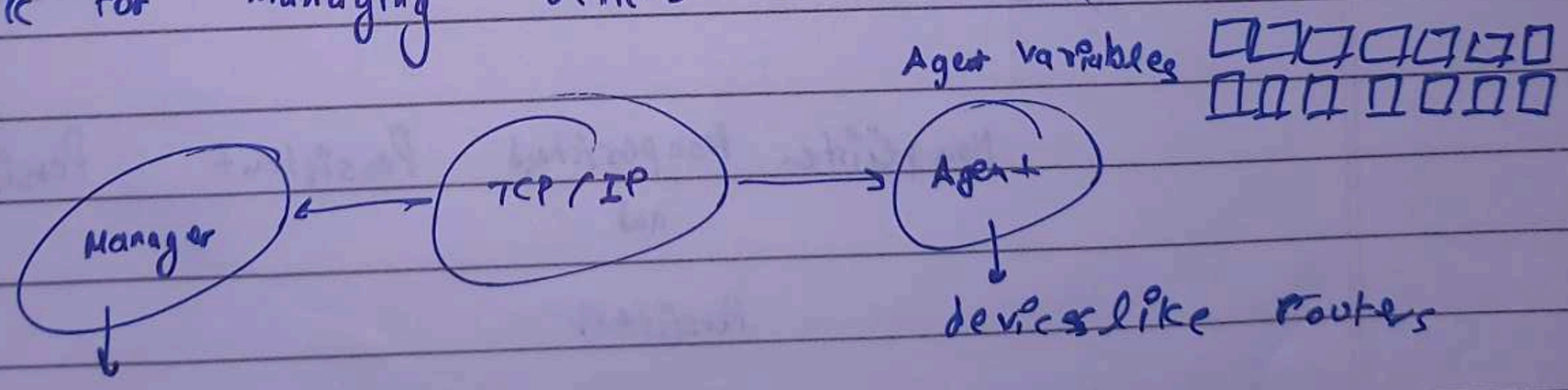


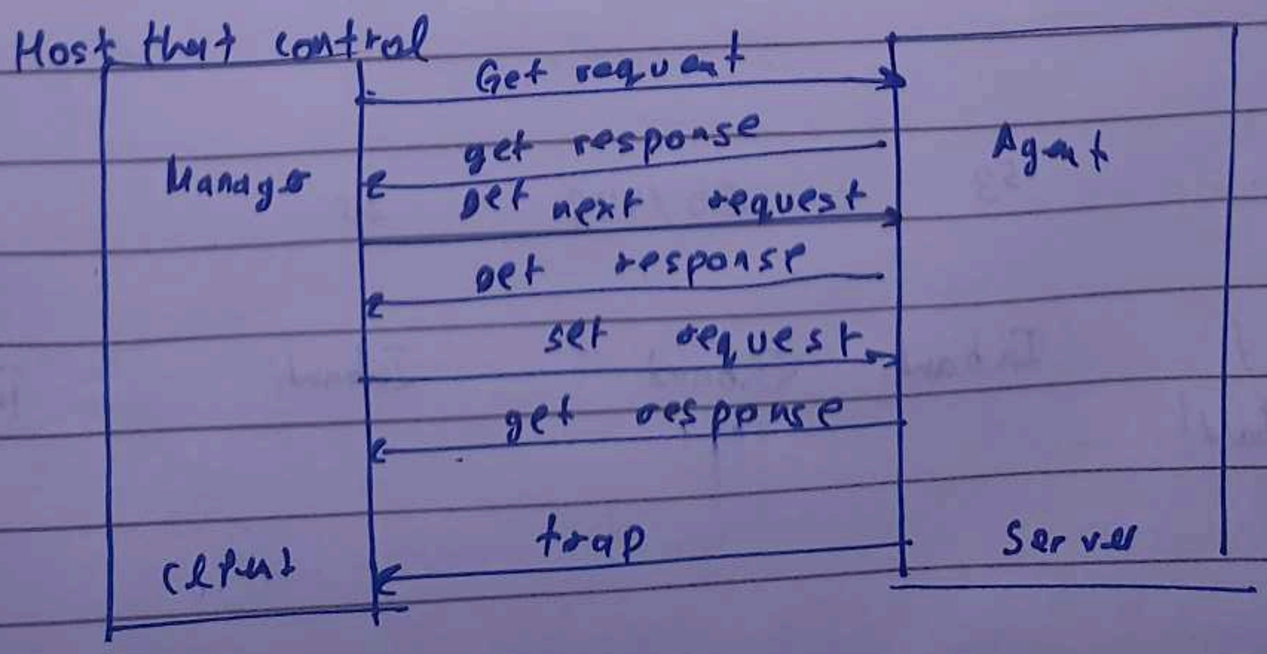
③ Quit



SNMP Simple Network Management Protocol (Again :))

Framework for managing devices on the internet.





Get Request : Manager to Agent to retrieve the value of a variable.

Get Next Request : used to define an object if it data is available or not.

Get Bulk request : to get large amount of data.

Get Response : Agent to manager in response.

Set Request : Manager to Agent to set a value in a variable

Trap : Agent to Manager to report an event.

	DNS	HTTP	SMTP	POP	FTP
Stateful / Stateless	Stateless	Stateless	Stateless	Stateful	Stateful
Transport Protocol used	UDP	TCP	TCP	TCP	TCP
	Connection less	Connection less	Connection oriented	Connection oriented	Connection oriented
	Non persistent	Non persistent and Persistent	Persistent	Persistent	Control is Persistent Data is Non Persistent
Port no.	53	80/443	25	110	20/21
Inband / Out of band	Inband	Inband	Inband	Inband	Out of band

I forgot to tell ya what is Stateful & Stateless & Inband & Out of band.

Stateless protocol : In which each request is independent of previous request. This makes it scalable & reliable.

Stateful : in which server needs to keep track of information about client and previous requests in order to process current request

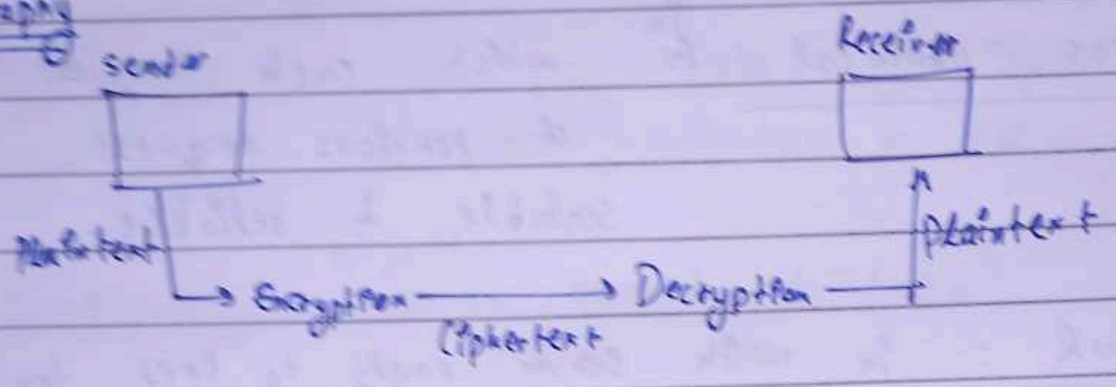
Inband : protocols that use the same network infrastructure as the data traffic that they are transporting. This can lead to network issues though.

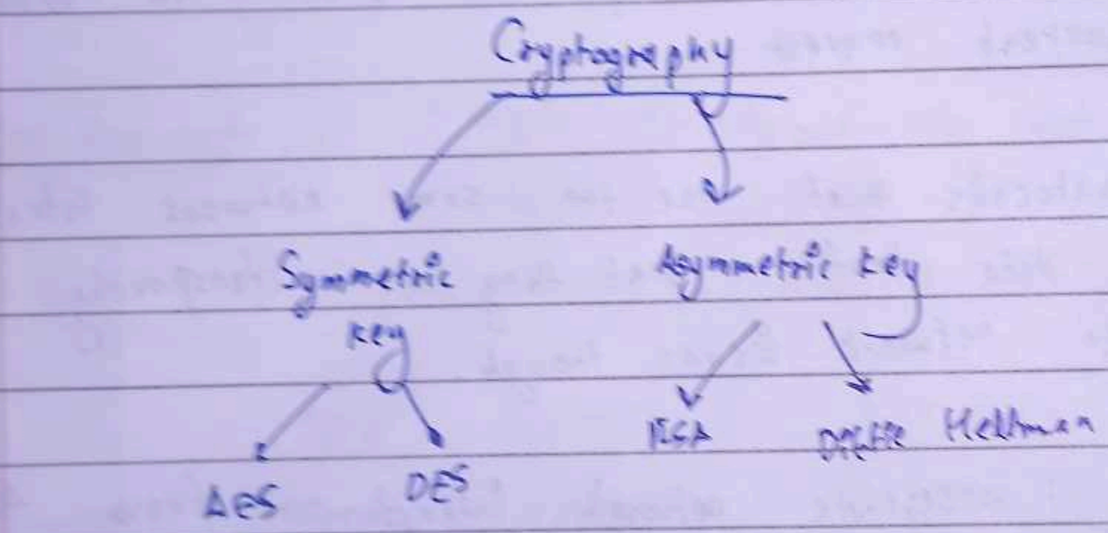
Out of band : use seperate network infrastructure from the data traffic that they are transporting.

classmate
Date
Page

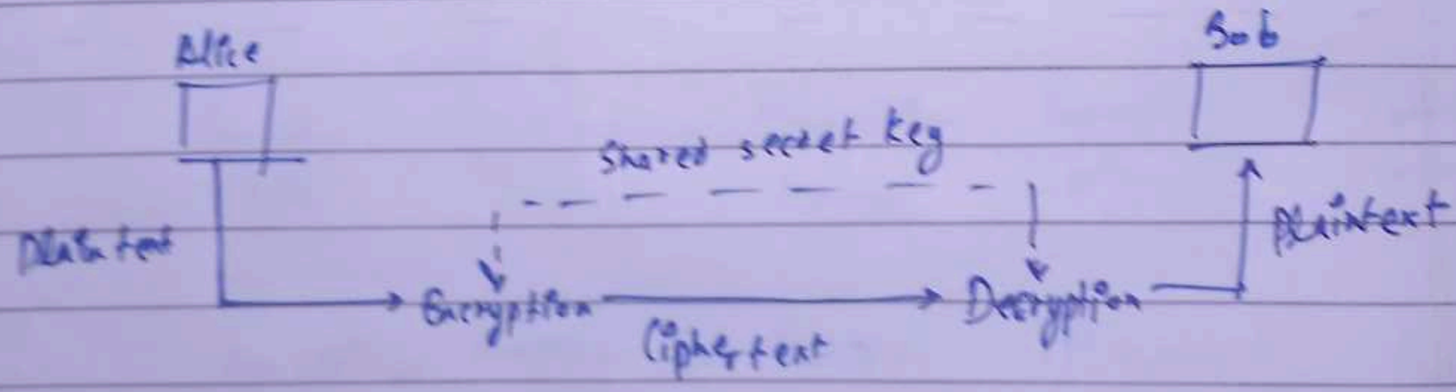
Network Security

Cryptography





Symmetric key Cryptography



In this same key is used by the sender (for encryption) and the receiver (for decryption) the key is shared.

Alice

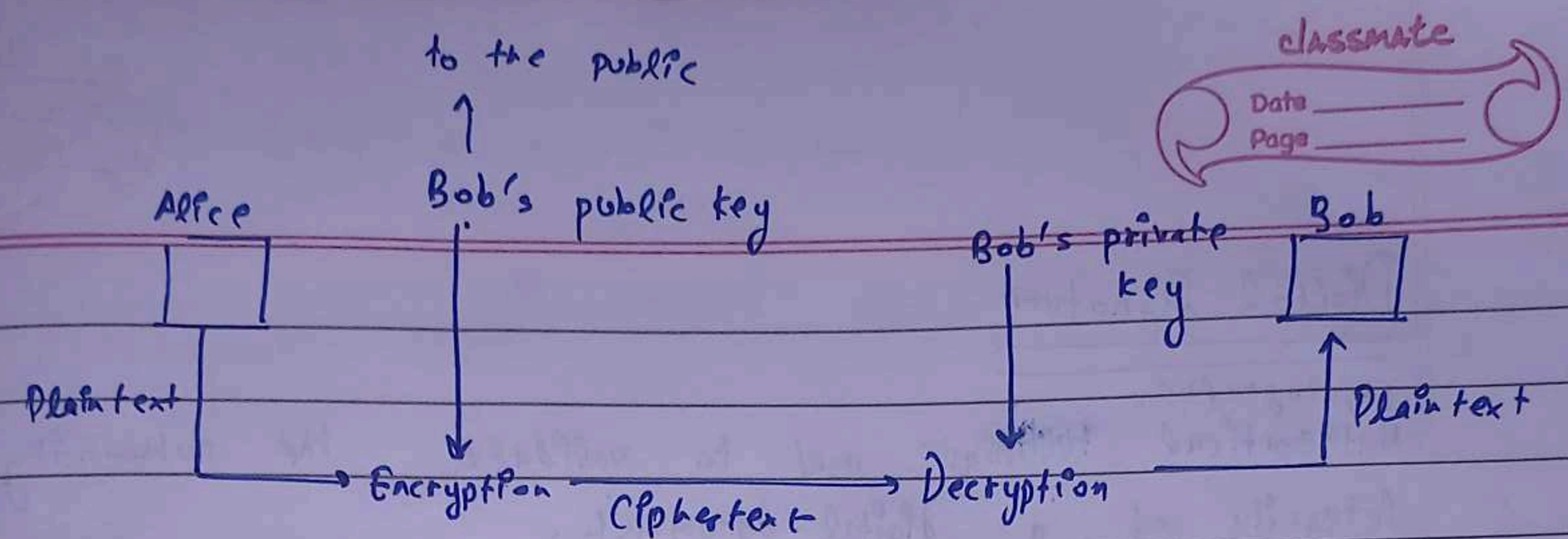
Plaintext

Network sec
by a ne
access,
network

Security

Encryption
is oppos

Services



Network security consists of the provisions and policies adopted by a network administrator to prevent and monitor unauthorized access, misuse, modification, or denial of a computer network and network accessible resources.

Security objectives → identification
→ authentication
→ access control

Encryption is converting plain text to ciphertext. Decryption is opposite of it.

Services → Secracy / Privacy
→ Authentication
→ Non repudiation
→ Data integrity

Digital Signature

Cryptographic Mathematical technique used to validate the authenticity and integrity of a digital document.

It's the digital equivalent of a handwritten signature or stamp but much more secure.

They are used to protect electronic transactions and communications from tampering and impersonation. They also verify identity of user and devices.

How they work?

They rely on public key cryptography, To create a digital signature, the sender uses their private key to encrypt a hash of the message or document they are signing. The hash is a unique digital fingerprint of the message or document.

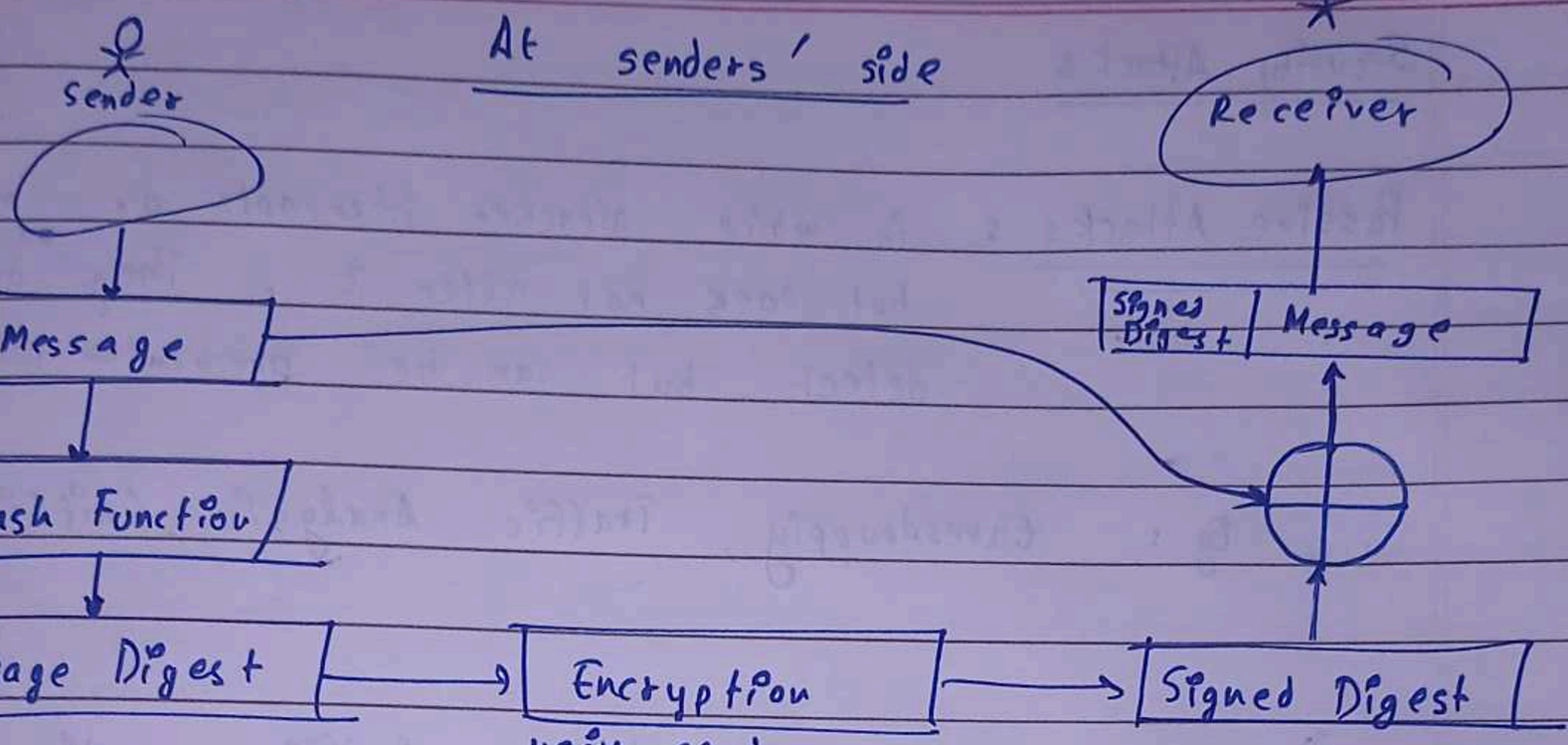
The encrypted hash is then send to the recipient along with the message or document. To verify the digital signature, the recipient uses the sender's public key to decrypt the hash. The recipient then creates a new hash of the message or document and compares it to the decrypted hash. If 2 hashs match then yes the message is clean.

Benefits

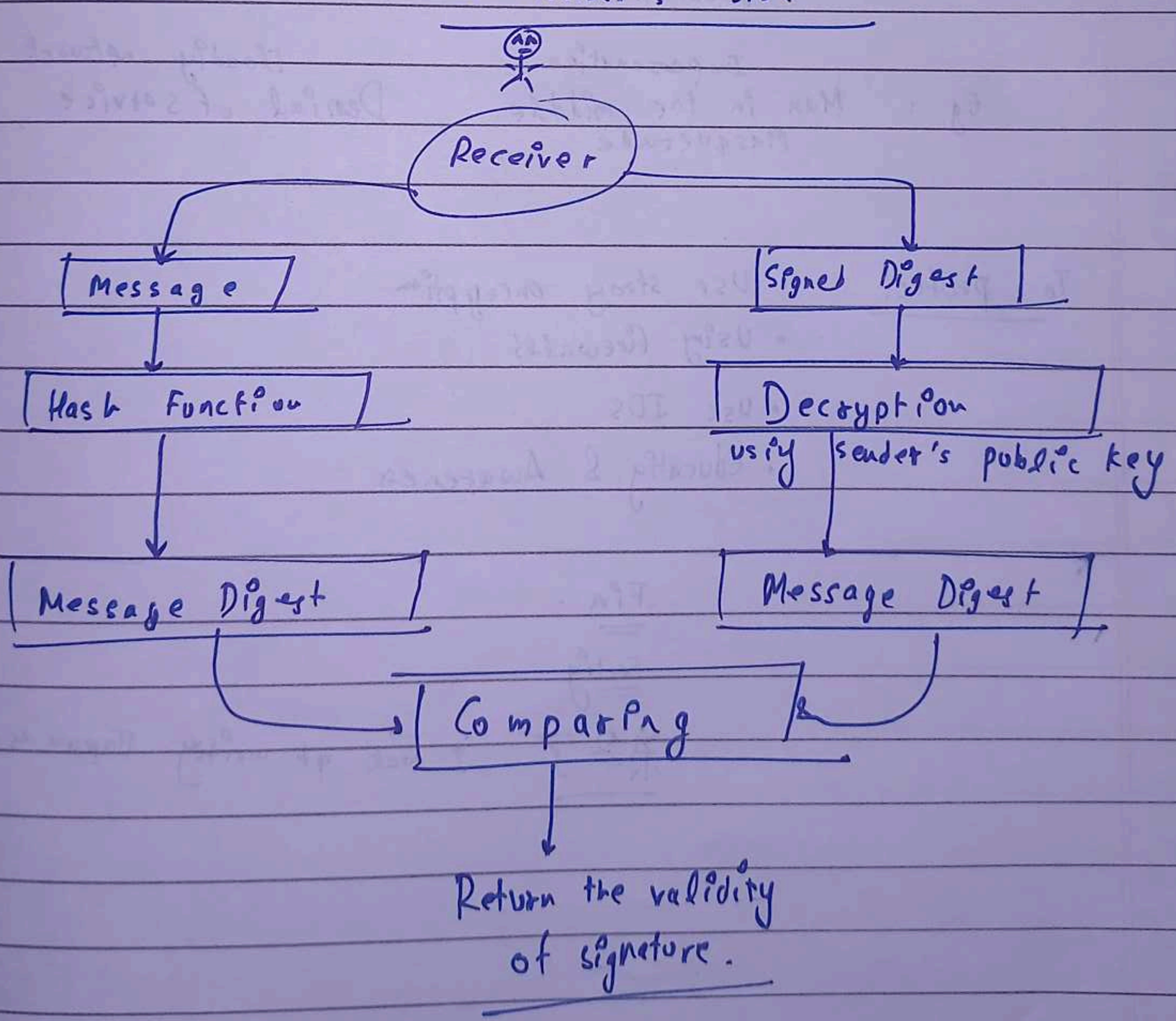
- Authentication
- Integrity
- Non-repudiation

Digital signatures are of 3 classes.

At senders' side



At receiver's side



Security Attacks

Passive Attacks : in which attacker intercepts or monitors data but does not alter it. They are hard to detect but can be prevented.

Eg : Eavesdropping, Traffic Analysis, Sniffing

Active Attacks : in which attacker modifies or disrupts data. There is an integrity & availability loss and DoS they are easily detected but prevention isn't possible.

Eg : Man in the middle (Impersonation / Masquerade), Denial of service (Flooding network), Spoofing, forging.

To protect

- Use strong encryption
- Using firewalls
- Use IDS
- Educating & Awareness

Fin.

Feeling

終了 I suck at writing Japanese lol.